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(54) **MICROCHIP WITH FARADAY CAGES AND
INTERNAL FLEXIBILITY SIPEs**

(56) **References Cited**

U.S. PATENT DOCUMENTS

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D193,914 S 8/1877 Berry
280,791 A 7/1883 Brooks
(Continued)

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FOREIGN PATENT DOCUMENTS

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AU 200963 5/1958
CA 1138194 12/1982
(Continued)

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OTHER PUBLICATIONS

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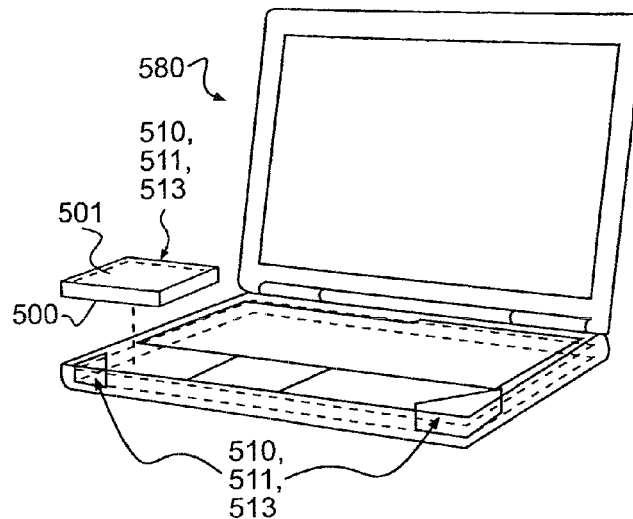
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(58) **Field of Classification Search**
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(57) **ABSTRACT**

A microchip including an outer chamber, compartment, or bladder; an inner chamber, compartment, or bladder inside said outer chamber, compartment, or bladder; an internal sipe separating at least a part of said inner and outer chambers, compartments, or bladders, and a Faraday cage. The microchip is configured to connect to a network of computers and includes an internal hardware firewall configured to deny access a protected portion of the microchip from the network. The sipe is formed by at least a portion of both an inner surface of the outer chamber, compartment, or bladder and an outer surface of the inner chamber, compartment, or bladder. The surface portions forming the sipe oppose each other and can move relative to each other in a sliding motion. At least a portion of an outer surface of the outer chamber, compartment, or bladder is proximate to an outer surface of the microchip.

17 Claims, 31 Drawing Sheets



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continuation of application No. 13/769,292, filed on Feb. 16, 2013, now abandoned, which is a continuation of application No. 12/292,769, filed on Nov. 25, 2008, now Pat. No. 8,125,796, which is a continuation of application No. 12/292,553, filed on Nov. 20, 2008, now Pat. No. 8,164,170.

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(56) **References Cited**

U.S. PATENT DOCUMENTS

288,127 A	11/1883	Shepard	3,186,004 A	6/1965	Carlini
500,385 A	6/1893	Hall	3,204,678 A	9/1965	Worcester
532,429 A	1/1895	Rogers	3,242,500 A	3/1966	Derr
584,373 A	6/1897	Kuhn	3,248,738 A	5/1966	Morgan
1,283,335 A	10/1918	Shillcock	3,253,355 A	5/1966	Menkin
1,289,106 A	12/1918	Bullock	3,305,947 A	2/1967	Kalsoy
D5,515 S	5/1920	Barney	3,308,560 A	3/1967	Jones
D55,115 S	5/1920	Barney	3,407,406 A	10/1968	Werner et al.
1,458,446 A	6/1923	Schaeffer	3,416,174 A	12/1968	Novitske
1,622,860 A	3/1927	Cutler	3,469,576 A	9/1969	Everts et al.
1,639,381 A	8/1927	Manelas	3,512,274 A	5/1970	McGrath
1,701,260 A	2/1929	Fischer	3,535,799 A	10/1970	Onituska
1,735,986 A	11/1929	Wray	3,539,876 A	11/1970	Feinberg et al.
1,853,034 A	4/1932	Bradley	3,600,714 A	8/1971	Cade et al.
2,004,906 A	6/1935	Simister	3,806,950 A	4/1974	Spencer-Foote
2,120,987 A	6/1938	Murray	3,806,974 A	4/1974	DiPaolo
2,147,197 A	2/1939	Glidden	3,824,716 A	7/1974	DiPaolo
2,155,166 A	4/1939	Kraft	3,834,046 A	9/1974	Fowler
2,162,912 A	6/1939	Craver	3,835,530 A	9/1974	Kilby
2,170,652 A	8/1939	Brennan	3,849,801 A	11/1974	Holt et al.
2,179,942 A	11/1939	Lyne	3,863,366 A	2/1975	Auberry et al.
D119,894 S	4/1940	Sherman	3,872,511 A	3/1975	Nichols
2,201,300 A	5/1940	Prue	3,958,291 A	5/1976	Spier
2,206,860 A	7/1940	Sperry	3,964,181 A	6/1976	Holcombe, Jr.
D122,131 S	8/1940	Sannar	3,995,320 A	12/1976	Zafuto
D128,817 S	8/1941	Esterson	3,997,984 A	12/1976	Hayward
2,251,468 A	8/1941	Smith	4,003,145 A	1/1977	Liebscher et al.
2,328,242 A	8/1943	Witherill	4,023,213 A	5/1977	Rovani
2,345,831 A	4/1944	Pierson	4,030,213 A	6/1977	Daswick
2,433,329 A	12/1947	Adler et al.	4,035,846 A	7/1977	Jencks
2,434,770 A	1/1948	Lutey	193,914 A	8/1977	Berry
2,470,200 A	5/1949	Wallach	4,043,058 A	8/1977	Hollister et al.
2,546,827 A	3/1951	Lavinthal	4,059,910 A	11/1977	Bryden
2,627,676 A	2/1953	Hack	4,060,855 A	12/1977	Rappleyea
2,718,715 A	9/1955	Spilman	4,064,565 A	12/1977	Griffiths
2,762,134 A	9/1956	Town	4,067,063 A	1/1978	Ettinger
2,814,133 A	11/1957	Herbst	4,068,395 A	1/1978	Senter
3,005,272 A	10/1961	Shelare et al.	4,083,125 A	4/1978	Benseler et al.
3,087,261 A	4/1963	Russell	4,096,649 A	6/1978	Saurwein
3,100,354 A	8/1963	Lombard et al.	4,098,011 A	7/1978	Bowerman et al.
3,110,971 A	11/1963	Chang	4,128,950 A	12/1978	Bowerman et al.
3,170,250 A	2/1965	Scholl	4,128,951 A	12/1978	Tansill
3,171,133 A	3/1965	Steffen	4,141,158 A	2/1979	Benseler et al.
			4,145,785 A	3/1979	Lacey
			4,149,324 A	4/1979	Lesser et al.
			4,161,828 A	7/1979	Benseler et al.
			4,161,829 A	7/1979	Wayser
			4,170,078 A	10/1979	Moss
			4,172,495 A	10/1979	Zebuhr et al.
			4,183,156 A	1/1980	Rudy
			4,194,310 A	3/1980	Bowerman
			D256,180 S	8/1980	Turner
			D256,400 S	8/1980	Famolare, Jr.
			4,217,705 A	8/1980	Donzis
			4,219,945 A	9/1980	Rudy
			4,223,457 A	9/1980	Borgeas
			4,227,320 A	10/1980	Borgeas
			4,235,026 A	11/1980	Plagenhoef
			4,237,627 A	12/1980	Turner
			4,240,214 A	12/1980	Sigle et al.
			4,241,523 A	12/1980	Daswick
			4,245,306 A	1/1981	Besemer et al.
			4,245,406 A	1/1981	Landay
			4,250,638 A	2/1981	Linnemann
			4,258,480 A	3/1981	Famolare, Jr.
			4,259,792 A	4/1981	Halberstadt
			4,262,433 A	4/1981	Hagg et al.
			4,263,728 A	4/1981	Frecentese
			4,266,349 A	5/1981	Schmohl
			4,268,980 A	5/1981	Gudas
			4,271,606 A	6/1981	Rudy
			4,272,858 A	6/1981	Hlustik
			4,274,211 A	6/1981	Funck
			4,276,594 A	6/1981	Morley
			4,278,837 A	7/1981	Best
			4,297,797 A	11/1981	Meyers
			4,302,892 A	12/1981	Adamik
			4,305,212 A	12/1981	Coomer
			4,308,671 A	1/1982	Bretschneider

(56)

References Cited

U.S. PATENT DOCUMENTS

4,309,832	A	1/1982	Hunt	4,747,220	A	5/1988	Autry et al.
4,316,332	A	2/1982	Giese et al.	D296,149	S	6/1988	Diaz
4,316,335	A	2/1982	Giese et al.	D296,152	S	6/1988	Selbiger
4,319,412	A	3/1982	Muller et al.	4,748,753	A	6/1988	Ju
D264,017	S	4/1982	Turner	4,754,561	A	7/1988	DuFour
4,322,895	A	4/1982	Hockerson	4,756,098	A	7/1988	Boggia
D265,019	S	6/1982	Vernonet	4,757,620	A	7/1988	Tiitola
4,335,529	A	6/1982	Badalamenti	4,759,136	A	7/1988	Stewart et al.
4,340,626	A	7/1982	Rudy	4,768,295	A	9/1988	Ito
4,342,161	A	8/1982	Schmohl	4,769,926	A	9/1988	Meyers
4,345,338	A	8/1982	Frieder, Jr. et al.	4,777,738	A	10/1988	Giese
4,348,821	A	9/1982	Daswick	D298,684	S	11/1988	Pitchford
4,354,284	A	10/1982	Gooding	4,783,910	A	11/1988	Boys
4,354,319	A	10/1982	Block et al.	4,784,462	A	11/1988	Priaroggia
4,361,971	A	12/1982	Bowerman	4,785,557	A	11/1988	Kelley et al.
4,366,634	A	1/1983	Giese et al.	4,802,289	A	2/1989	Guldager
4,370,817	A	2/1983	Ratanangsu	4,817,304	A	4/1989	Parker et al.
4,372,059	A	2/1983	Ambrose	4,827,508	A	5/1989	Shear
4,375,108	A	3/1983	Gooding	4,827,631	A	5/1989	Thornton
4,398,357	A	8/1983	Batra	4,833,795	A	5/1989	Diaz
4,399,620	A	8/1983	Funck	4,837,949	A	6/1989	Dufour
D272,294	S	1/1984	Watanabe	D302,900	S	8/1989	Kolman et al.
4,432,099	A	2/1984	Grick et al.	4,853,980	A	8/1989	Zarotti
4,449,306	A	5/1984	Cavanagh	4,854,057	A	8/1989	Misevich et al.
4,451,994	A	6/1984	Fowler	4,855,903	A	8/1989	Carleton et al.
4,454,662	A	6/1984	Stubblefield	4,858,340	A	8/1989	Pasternak
4,455,765	A	6/1984	Sjöswärd	4,864,737	A	9/1989	Marrello
4,455,767	A	6/1984	Bergmans	4,864,738	A	9/1989	Horovitz
4,467,400	A	8/1984	Stopper	4,866,861	A	9/1989	Noone
4,468,870	A	9/1984	Sternberg	4,876,807	A	10/1989	Tiitola et al.
4,484,364	A	11/1984	Mitchell et al.	4,890,398	A	1/1990	Thomasson
4,484,397	A	11/1984	Curley, Jr.	4,893,174	A	1/1990	Yamada et al.
4,489,397	A	12/1984	Lee	4,894,932	A	1/1990	Harada
4,494,321	A	1/1985	Lawlor	4,894,933	A	1/1990	Tonkel et al.
4,505,055	A	3/1985	Bergmans	4,897,936	A	2/1990	Fuerst
4,506,462	A	3/1985	Cavanagh	4,906,502	A	3/1990	Rudy
4,521,979	A	6/1985	Blaser	4,907,228	A	3/1990	Bruckert et al.
4,527,345	A	7/1985	Lopez	4,918,596	A	4/1990	Nakano
D280,568	S	9/1985	Stubblefield	4,922,631	A	5/1990	Anderie
4,538,301	A	9/1985	Sawatzki et al.	4,934,070	A	6/1990	Mauger
4,542,598	A	9/1985	Misevich et al.	4,934,073	A	6/1990	Robinson
4,546,559	A	10/1985	Dassler	D310,131	S	8/1990	Hase
4,547,979	A	10/1985	Harada et al.	D310,132	S	8/1990	Hase
4,557,059	A	12/1985	Misevich et al.	4,947,560	A	8/1990	Fuerst et al.
4,559,723	A	12/1985	Hamy et al.	4,949,476	A	8/1990	Anderie
4,559,724	A	12/1985	Norton	D310,906	S	10/1990	Hase
4,561,195	A	12/1985	Onoda et al.	4,969,092	A	11/1990	Shorter
4,570,362	A	2/1986	Vernonet	4,972,527	A	11/1990	Wallace
4,577,417	A	3/1986	Cole	4,979,794	A	12/1990	Evans
4,578,882	A	4/1986	Talarico, II	4,982,737	A	1/1991	Guttmann
4,580,359	A	4/1986	Kurrash et al.	4,989,349	A	2/1991	Ellis, III
4,624,061	A	11/1986	Wezel et al.	D315,634	S	3/1991	Yung-Mao
4,624,062	A	11/1986	Autry et al.	5,010,662	A	4/1991	Dabuzhsky et al.
4,638,577	A	1/1987	Riggs	5,012,597	A	5/1991	Thomasson
4,641,438	A	2/1987	Laird et al.	5,014,365	A	5/1991	Schulz
4,642,917	A	2/1987	Ungar	5,014,449	A	5/1991	Richard et al.
4,651,445	A	3/1987	Hannibal	5,024,007	A	6/1991	DuFour
D289,341	S	4/1987	Turner	5,025,369	A	6/1991	Schwartz
4,654,983	A	4/1987	Graham	5,025,573	A	6/1991	Giese et al.
4,658,515	A	4/1987	Oatman	5,027,864	A	7/1991	Conti et al.
4,667,423	A	5/1987	Autry	5,031,089	A	7/1991	Liu et al.
4,670,995	A	6/1987	Huang	5,048,203	A	9/1991	Kling
4,676,010	A	6/1987	Cheskin	D320,302	S	10/1991	Kiyosawa
4,694,591	A	9/1987	Banich et al.	2,052,130	A	10/1991	Barry et al.
4,697,361	A	10/1987	Ganter et al.	5,052,130	A	10/1991	Barry
4,703,436	A	10/1987	Varshney	5,068,780	A	11/1991	Bruckert et al.
D293,275	S	12/1987	Bua	5,077,916	A	1/1992	Beneteau
4,715,133	A	12/1987	Hartjes et al.	5,079,856	A	1/1992	Truelsen
4,724,622	A	2/1988	Mills	5,083,320	A	1/1992	Halstead
D294,425	S	3/1988	Le	5,092,060	A	3/1992	Frachey et al.
4,727,660	A	3/1988	Bernhard	5,095,545	A	3/1992	Lane
4,730,402	A	3/1988	Norton et al.	5,103,393	A	4/1992	Harris et al.
4,731,939	A	3/1988	Parracho et al.	5,109,329	A	4/1992	Strelloff
4,736,317	A	4/1988	Hu et al.	5,109,512	A	4/1992	Bahr et al.
4,747,139	A	5/1988	Taaffe	D327,164	S	6/1992	Hatfield
				D327,165	S	6/1992	Hatfield
				5,131,173	A	7/1992	Anderie
				5,136,708	A	8/1992	LaPourtre et al.
				D328,968	S	9/1992	Tinker

(56)

References Cited

U.S. PATENT DOCUMENTS

D329,528	S	9/1992	Hatfield	5,606,615	A	2/1997	Lapointe et al.
D329,739	S	9/1992	Hatfield	5,608,448	A	3/1997	Smoral et al.
5,153,945	A	10/1992	Kobayashi et al.	5,615,127	A	3/1997	Beatty et al.
5,155,808	A	10/1992	Shimizu	5,627,879	A	5/1997	Russell et al.
D330,972	S	11/1992	Hatfield et al.	5,666,484	A	9/1997	Orimo et al.
D332,344	S	1/1993	Hatfield et al.	5,675,844	A	10/1997	Guyton et al.
D332,692	S	1/1993	Hatfield et al.	5,678,028	A	10/1997	Bershteyn et al.
5,175,889	A	1/1993	Infusino	5,678,327	A	10/1997	Halberstadt
5,191,727	A	3/1993	Barry et al.	5,680,461	A	10/1997	McManis
5,195,031	A	3/1993	Ordish	5,680,548	A	10/1997	Trugman
5,204,998	A	4/1993	Liu	5,696,902	A	12/1997	Leclercq et al.
5,212,780	A	5/1993	Padgaonkar et al.	5,699,528	A	12/1997	Hogan
5,214,657	A	5/1993	Farnworth et al.	5,701,507	A	12/1997	Bonneau, Jr. et al.
5,224,280	A	7/1993	Preman et al.	D388,594	S	1/1998	Turner et al.
5,224,810	A	7/1993	Pitkin	5,710,884	A	1/1998	Dedrick
5,237,507	A	8/1993	Chasek	5,734,913	A	3/1998	Iwamura et al.
5,237,758	A	8/1993	Zachman	5,748,489	A	5/1998	Beatty et al.
5,247,742	A	9/1993	Kilgore et al.	5,751,881	A	5/1998	Konda et al.
5,259,070	A	11/1993	De Roza	5,752,067	A	5/1998	Wilkinson et al.
5,259,071	A	11/1993	Scott et al.	5,754,766	A	5/1998	Shaw et al.
5,260,943	A	11/1993	Comroe et al.	5,758,077	A	5/1998	Danahy et al.
5,263,203	A	11/1993	Kraemer et al.	5,758,345	A	5/1998	Wang
5,282,272	A	1/1994	Guy et al.	5,761,507	A	6/1998	Govett
5,283,819	A	2/1994	Glick et al.	5,764,889	A	6/1998	Ault et al.
D345,249	S	3/1994	Sell, Jr.	5,774,337	A	6/1998	Lee et al.
5,291,494	A	3/1994	Bruckert et al.	5,774,668	A	6/1998	Choquier et al.
5,291,502	A	3/1994	Pezeshki et al.	5,774,721	A	6/1998	Robinson
5,291,505	A	3/1994	Nielsen	5,784,551	A	7/1998	De Leva et al.
D347,105	S	5/1994	Johnson	5,784,628	A	7/1998	Reneris
5,317,819	A	6/1994	Ellis, III	5,790,431	A	8/1998	Ahrens, Jr. et al.
5,341,477	A	8/1994	Pitkin et al.	5,793,968	A	8/1998	Gregerson et al.
5,343,639	A	9/1994	Kilgore	5,794,059	A	8/1998	Barker et al.
5,349,682	A	9/1994	Rosenberry	5,802,320	A	9/1998	Baehr et al.
5,353,459	A	10/1994	Potter	5,809,190	A	9/1998	Chen
5,353,523	A	10/1994	Kilgore	5,813,141	A	9/1998	Cho
5,357,404	A	10/1994	Bright et al.	5,813,142	A	9/1998	Demon
5,357,632	A	10/1994	Pian et al.	5,815,665	A	9/1998	Teper et al.
5,361,362	A	11/1994	Benkeser et al.	5,815,793	A	9/1998	Ferguson
5,361,410	A	11/1994	Sigl	5,823,852	A	10/1998	Chu
5,369,896	A	12/1994	Frachey et al.	5,826,014	A	10/1998	Coley et al.
5,381,534	A	1/1995	Shi	5,826,029	A	10/1998	Gore, Jr. et al.
5,388,211	A	2/1995	Hornbuckle	5,828,833	A	10/1998	Belville et al.
5,392,400	A	2/1995	Berkowitz et al.	5,835,726	A	11/1998	Shwed et al.
5,410,651	A	4/1995	Sekizawa et al.	5,838,436	A	11/1998	Hotaling et al.
5,426,741	A	6/1995	Butts, Jr. et al.	5,838,542	A	11/1998	Nelson et al.
5,428,783	A	6/1995	Lake	5,843,799	A	12/1998	Hsu et al.
5,434,998	A	7/1995	Akai et al.	5,844,594	A	12/1998	Ferguson
5,446,843	A	8/1995	Fucito et al.	5,845,074	A	12/1998	Kobata
5,457,797	A	10/1995	Butterworth et al.	5,846,063	A	12/1998	Lakic
5,475,606	A	12/1995	Muyshondt et al.	5,850,449	A	12/1998	McManis
5,497,465	A	3/1996	Chin et al.	5,861,817	A	1/1999	Palmer et al.
5,515,511	A	5/1996	Nguyen et al.	5,862,357	A	1/1999	Hagersten et al.
5,522,070	A	5/1996	Sumimoto	5,864,738	A	1/1999	Kessler et al.
D372,114	S	7/1996	Turner et al.	5,870,721	A	2/1999	Norris
5,535,408	A	7/1996	Hillis	5,872,987	A	2/1999	Wade et al.
5,539,934	A	7/1996	Ponder	5,881,284	A	3/1999	Kubo
5,543,194	A	8/1996	Rudy	5,889,989	A	3/1999	Robertazzi et al.
5,544,429	A	8/1996	Ellis, III	5,896,499	A	4/1999	McKelvey
5,546,594	A	8/1996	Wazumi	D409,362	S	5/1999	Turner et al.
5,550,984	A	8/1996	Gelb	D409,826	S	5/1999	Turner et al.
5,555,584	A	9/1996	Moore, III et al.	D410,138	S	5/1999	Turner et al.
5,568,375	A	10/1996	Rausch	5,905,429	A	5/1999	Hornstein et al.
5,570,270	A	10/1996	Naedel et al.	5,909,052	A	6/1999	Ohta et al.
5,572,643	A	11/1996	Judson	5,909,681	A	6/1999	Passera et al.
5,572,805	A	11/1996	Giese et al.	5,909,948	A	6/1999	Ellis, III
5,576,554	A	11/1996	Hsu	5,913,412	A	6/1999	Huber et al.
5,586,121	A	12/1996	Moura et al.	5,917,629	A	6/1999	Hortensius et al.
5,587,928	A	12/1996	Jones et al.	5,919,247	A	7/1999	Van Hoff et al.
5,588,003	A	12/1996	Ohba et al.	5,930,511	A	7/1999	Hinsley
5,590,284	A	12/1996	Crosetto	5,930,918	A	8/1999	Healy et al.
5,592,376	A	1/1997	Hodroff	5,933,984	A	8/1999	Carlson et al.
5,592,632	A	1/1997	Leung et al.	5,943,421	A	8/1999	Grabon
5,594,491	A	1/1997	Hodge et al.	5,956,777	A	9/1999	Popovich
5,600,597	A	2/1997	Kean et al.	5,964,832	A	10/1999	Kisor
5,604,882	A	2/1997	Hoover et al.	5,976,451	A	11/1999	Skaja et al.
				5,978,829	A	11/1999	Chung et al.
				5,979,078	A	11/1999	McLaughlin
				5,987,781	A	11/1999	Pavesi et al.
				6,003,133	A	12/1999	Moughanni et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,023,857	A	2/2000	Vizy et al.	6,877,254	B2	4/2005	Ellis
6,052,555	A	4/2000	Ferguson	6,918,197	B2	7/2005	Ellis
6,065,118	A	5/2000	Bull et al.	6,946,050	B2	9/2005	Dojan et al.
6,067,082	A	5/2000	Enmei	6,950,947	B1	9/2005	Purtell et al.
6,073,209	A	6/2000	Bergsten	6,957,000	B2	10/2005	McAlpine et al.
6,073,271	A	6/2000	Alexander et al.	6,971,193	B1	12/2005	Potter et al.
6,093,933	A	7/2000	Farnworth et al.	6,976,321	B1	12/2005	Lakic
6,098,091	A	8/2000	Kisor	6,990,755	B2	1/2006	Hatfield et al.
6,098,209	A	8/2000	Bainbridge et al.	7,010,869	B1	3/2006	Ellis
6,112,225	A	8/2000	Kraft et al.	7,024,449	B1	4/2006	Ellis
6,112,243	A	8/2000	Downs et al.	7,035,906	B1	4/2006	Ellis
6,115,698	A	9/2000	Tuck et al.	7,037,571	B2	5/2006	Fish et al.
6,115,941	A	9/2000	Ellis, III	7,047,275	B1	5/2006	Ellis
6,115,945	A	9/2000	Ellis, III	7,076,890	B2	7/2006	Grove et al.
6,163,982	A	12/2000	Ellis, III	7,082,697	B2	8/2006	Ellis
6,178,560	B1	1/2001	Halstead et al.	7,086,179	B2	8/2006	Dojan et al.
6,202,153	B1	3/2001	Diamant et al.	7,093,379	B2	8/2006	Ellis
6,208,634	B1	3/2001	Boulos et al.	7,123,801	B2	10/2006	Fitz
6,219,627	B1	4/2001	Bonneau et al.	7,127,834	B2	10/2006	Ellis
6,226,801	B1	5/2001	Alexander et al.	7,132,032	B2	11/2006	Tawney et al.
D444,293	S	7/2001	Turner et al.	7,140,129	B2	11/2006	Newson et al.
6,253,466	B1	7/2001	Harmon-Weiss et al.	7,148,565	B2	12/2006	Kim et al.
6,268,788	B1	7/2001	Gray	7,161,175	B2	1/2007	Shau
6,287,949	B1	9/2001	Mori et al.	7,168,185	B2	1/2007	Ellis
6,308,439	B1	10/2001	Ellis	7,168,188	B2	1/2007	Auger et al.
6,308,813	B1	10/2001	Carlson	7,174,658	B2	2/2007	Ellis
D450,916	S	11/2001	Turner et al.	7,194,826	B2	3/2007	Ungari
6,314,662	B1	11/2001	Ellis	7,234,249	B2	6/2007	Ellis
6,326,245	B1	12/2001	Farnworth et al.	7,264,908	B2	9/2007	Kaneko et al.
6,351,853	B1	3/2002	Halstead et al.	7,287,341	B2	10/2007	Ellis
6,360,453	B1	3/2002	Ellis	7,334,349	B2	2/2008	Sokolowski et al.
6,360,457	B1	3/2002	Qui et al.	7,334,350	B2	2/2008	Ellis
6,366,472	B2	4/2002	Alina et al.	7,334,356	B2	2/2008	Ellis
6,391,405	B1	5/2002	Bonk et al.	7,412,588	B2	8/2008	Georgiou et al.
6,440,775	B2	8/2002	Khoury	7,467,406	B2	12/2008	Cox et al.
6,446,271	B1	9/2002	Ho	7,484,318	B2	2/2009	Finkelstein
6,457,261	B1	10/2002	Crary	7,506,020	B2	3/2009	Ellis
6,487,795	B1	12/2002	Ellis	7,546,699	B2	6/2009	Ellis
6,493,881	B1	12/2002	Picotte	7,562,211	B2	7/2009	Paya et al.
6,515,870	B1 *	2/2003	Skinner H01L 23/552 257/691	7,562,468	B2	7/2009	Ellis
6,519,780	B2	2/2003	Goodwin	7,591,084	B2	9/2009	Santa Ana
6,530,092	B2	3/2003	Pope	7,606,854	B2	10/2009	Ellis
6,537,639	B1	3/2003	Huang	7,634,529	B2	12/2009	Ellis
6,571,490	B2	6/2003	Scarfe et al.	7,647,710	B2	1/2010	Ellis
6,584,706	B1	7/2003	Ellis	7,707,742	B2	5/2010	Ellis
6,591,428	B2	7/2003	Halstead et al.	7,793,429	B2	9/2010	Ellis
6,591,429	B1	7/2003	Jaszai	7,793,430	B2	9/2010	Ellis
6,591,519	B1	7/2003	Ellis	7,805,756	B2	9/2010	Ellis
6,629,376	B1	10/2003	Ellis	7,814,233	B2	10/2010	Ellis
6,645,832	B2	11/2003	Kim et al.	7,840,997	B2	11/2010	Shevchenko
6,658,671	B1	12/2003	Von Holst et al.	7,908,650	B2	3/2011	Ellis
6,662,470	B2	12/2003	Ellis	7,926,097	B2	4/2011	Ellis
6,668,470	B2	12/2003	Ellis	7,984,301	B2	7/2011	Kaabouch et al.
6,675,498	B1	1/2004	Ellis	8,010,789	B2	8/2011	Witchey
6,675,499	B2	1/2004	Ellis	8,125,796	B2 *	2/2012	Ellis A43B 3/0005 257/316
6,681,408	B2	1/2004	Ku	8,141,276	B2	3/2012	Ellis
6,708,424	B1	3/2004	Ellis	8,164,170	B2 *	4/2012	Ellis G06F 1/16 257/682
6,722,058	B2	4/2004	Lucas et al.	8,255,986	B2	8/2012	Ellis
6,725,250	B1	4/2004	Ellis	8,378,474	B2	2/2013	Ellis
6,729,046	B2	5/2004	Ellis	8,561,164	B2 *	10/2013	Ellis, III A01N 25/30 709/250
6,732,141	B2	5/2004	Ellis	8,627,444	B2	1/2014	Ellis
6,748,674	B2	6/2004	Ellis	8,670,246	B2	3/2014	Ellis
6,763,616	B2	7/2004	Ellis	8,677,026	B2 *	3/2014	Ellis, III H04L 67/10 709/209
6,772,347	B1	8/2004	Xie et al.	8,726,303	B2 *	5/2014	Ellis, III G06F 9/5072 709/201
6,789,331	B1	9/2004	Ellis	8,739,195	B2 *	5/2014	Ellis, III G06F 9/5072 709/201
6,795,625	B1	9/2004	Ishii et al.	8,848,368	B2 *	9/2014	Ellis A43B 3/0005 312/223.2
6,797,545	B2	9/2004	Farnworth et al.	8,898,768	B2 *	11/2014	Ellis G06F 21/50 713/153
6,810,606	B1	11/2004	Ellis	9,003,510	B2 *	4/2015	Ellis G06F 21/50 713/153
D500,585	S	1/2005	Wahoske et al.				
6,837,951	B2	1/2005	Rapaport				
6,845,573	B2	1/2005	Litchfield et al.				
6,846,534	B2	1/2005	Bonk et al.				
6,848,200	B1	2/2005	Westin				
6,851,204	B2	2/2005	Aveni				

(56)	References Cited				DE	1918132	6/1965
U.S. PATENT DOCUMENTS					DE	1287477	1/1969
					DE	1290844	3/1969
					DE	2036062	7/1970
					DE	1948620	5/1971
9,009,809	B2 *	4/2015	Ellis	G06F 21/50	DE	1685293	7/1971
				713/153	DE	1685260	10/1971
9,183,410	B2 *	11/2015	Ellis	G06F 9/5072	DE	2045430	3/1972
2001/0032351	A1	10/2001	Nakayama et al.		DE	2522127	11/1976
2001/0046119	A1	11/2001	Hamano et al.		DE	2525613	12/1976
2001/0054159	A1	12/2001	Hoshino		DE	2602310	7/1977
2002/0002730	A1	1/2002	Dennis et al.		DE	2613312	10/1977
2002/0023291	A1	2/2002	Mendoza		DE	2706645	8/1978
2002/0152541	A1	10/2002	Halstead et al.		DE	2654116	1/1979
2002/0152542	A1	10/2002	Dennis et al.		DE	2737765	3/1979
2002/0166157	A1	11/2002	Pope		DE	2805426	8/1979
2003/0046830	A1	3/2003	Ellis, III		DE	3021936	4/1981
2003/0079276	A1	5/2003	Forsyth et al.		DE	3024587	1/1982
2003/0096899	A1	5/2003	Pearce		DE	8219616.8	9/1982
2003/0200598	A1	10/2003	Jessie		DE	3113295	10/1982
2004/0073603	A1	4/2004	Ellis		DE	3245182	5/1983
2004/0098621	A1	5/2004	Raymond		DE	3317462	10/1983
2004/0158744	A1	8/2004	Deng et al.		DE	8431831.7	12/1984
2004/0162992	A1	8/2004	Sami et al.		DE	3347343	7/1985
2004/0215931	A1	10/2004	Ellis		DE	8530136.1	2/1988
2004/0226077	A1	11/2004	Toth		DE	3629245	3/1988
2005/0039346	A1	2/2005	Thomas et al.		DE	40 08 335	9/1991
2005/0076536	A1	4/2005	Hatfield et al.		EP	0048965	4/1982
2005/0138169	A1	6/2005	Bahr		EP	0083449	A1 12/1982
2005/0144808	A1	7/2005	Vito et al.		EP	0130816	1/1985
2005/0180095	A1	8/2005	Ellis		EP	0185781	7/1986
2005/0246824	A1	11/2005	Berger et al.		EP	0207063	10/1986
2005/0262739	A1	12/2005	McDonald et al.		EP	0206511	12/1986
2006/0010717	A1	1/2006	Finkelstein		EP	0213257	3/1987
2006/0075001	A1	4/2006	Canning et al.		EP	0215974	4/1987
2006/0095497	A1	5/2006	Ellis		EP	0238995	9/1987
2006/0137221	A1	6/2006	Dojan et al.		EP	0260777	3/1988
2006/0177226	A1	8/2006	Ellis		EP	0301331	2/1989
2006/0179683	A1	8/2006	Weiss et al.		EP	0329391	8/1989
2006/0190565	A1	8/2006	Ellis		EP	0410087	1/1991
2006/0230636	A1	10/2006	Kokstis et al.		EP	0 647 052	4/1995
2006/0248749	A1	11/2006	Ellis		EP	0697825	2/1996
2007/0084081	A1	4/2007	Fallon		EP	0 840216	5/1998
2007/0094896	A1	5/2007	Hatfield et al.		EP	0 853 279	7/1998
2007/0162974	A1	7/2007	Speidel		EP	0910964	4/1999
2007/0196948	A1	8/2007	Trezza		EP	1068460	1/2001
2007/0300305	A1	12/2007	Gonsalves et al.		EP	1374808	1/2004
2008/0083140	A1	4/2008	Ellis		EP	1414322	5/2004
2008/0083976	A1	4/2008	Haba et al.		EP	1480534	12/2004
2008/0134290	A1	6/2008	Olsson		EP	1529457	5/2005
2008/0229614	A1	9/2008	Santa Ana		EP	1 164 766	12/2011
2009/0026524	A1	1/2009	Kreupl et al.		FR	602501	3/1926
2009/0031412	A1	1/2009	Ellis		FR	925961	9/1947
2009/0200661	A1	8/2009	Ellis		FR	1004472	3/1952
2009/0254986	A1	10/2009	Harris et al.		FR	1245672	10/1960
2009/0282092	A1	11/2009	Ellis		FR	1323455	2/1963
2010/0011083	A1	1/2010	Ellis		FR	2006270	11/1972
2011/0004930	A1	1/2011	Ellis		FR	2261721	9/1975
2011/0004931	A1	1/2011	Ellis		FR	2511850	3/1983
2012/0096537	A1	4/2012	Ellis		FR	2622411	5/1989
2012/0155002	A1 *	6/2012	Ellis	A43B 3/0005	GB	9591	0/1913
				361/679.02	GB	16143	0/1892
2012/0175752	A1	7/2012	Ellis		GB	764956	1/1957
2012/0224819	A1 *	9/2012	Ellis	A43B 7/1425	GB	792034	3/1958
				385/100	GB	807305	1/1959
2013/0201623	A1	8/2013	Ellis		GB	1504615	3/1978
2014/0013411	A1 *	1/2014	Ellis	A43B 3/0005	GB	2023405	1/1980
				726/11	GB	2039717	8/1980
2014/0244085	A1 *	8/2014	Ellis	B60K 6/365	GB	2076633	12/1981
				701/22	GB	2133668	8/1984
2015/0137307	A1 *	5/2015	Stuber	H01L 23/60	GB	2136670	9/1984
				257/503	JP	39-15597	8/1964
FOREIGN PATENT DOCUMENTS					JP	45-5154	3/1970
					JP	50-71132	11/1975
					JP	57-139333	8/1982
					JP	59-23525	7/1984
CA	1176458	10/1984			JP	61-55810	4/1986
DE	23257	5/1956			JP	1129505	6/1986
DE	1888119	12/1963			JP	61-167810	10/1980
DE	1918131	6/1965					

(56)

References Cited

FOREIGN PATENT DOCUMENTS

JP	1-195803	8/1989
JP	2136505	5/1990
JP	2279103	11/1990
JP	3-85102	4/1991
JP	3086101	4/1991
JP	4-279102	10/1992
JP	5-123204	5/1993
NZ	189890	9/1981
WO	WO 87/07480	12/1987
WO	WO 87/07481	12/1987
WO	WO 88/08263	11/1988
WO	WO 89/06500	7/1989
WO	WO 90/00358	1/1990
WO	WO 91/00698	1/1991
WO	WO 91/03180	3/1991
WO	WO 91/04683	4/1991
WO	WO 91/05491	5/1991
WO	WO 91/10377	7/1991
WO	WO 91/11124	8/1991
WO	WO 91/11924	8/1991
WO	WO 91/19429	12/1991
WO	WO 92/07483	5/1992
WO	WO 92/18024	10/1992
WO	WO 93/13928	7/1993
WO	WO 94/01964	1/1994
WO	WO 94/03080	2/1994
WO	WO 95/01060	1/1995
WO	WO 96/14768	5/1996
WO	WO 97/00029	1/1997
WO	WO 98/26366	6/1998
WO	WO 99/04561	1/1999
WO	WO 99/32972	7/1999
WO	WO 00/054616	9/2000
WO	WO 00/064293	11/2000
WO	WO 00/070981	11/2000
WO	WO 01/080678	11/2001
WO	WO 02/09547	2/2002
WO	WO 2005/115190	12/2005
WO	WO 2005/117628	12/2005
WO	WO 2005/117629	12/2005
WO	WO 2006/113212	10/2006
WO	WO 2006/124116	11/2006
WO	WO 2007/024523	3/2007
WO	WO 2011/094616	8/2011
WO	WO 2011/103299	8/2011

OTHER PUBLICATIONS

Partial file history for U.S. Appl. No. 13/769,292.
 European Search Report EP-05825112.5-2318 (Jul. 14, 2010).
 Examiner's Comments, European Search Report EP-05825112.5-2318 (Jul. 14, 2010).
 Partial file history for U.S. Appl. No. 12/292,769.
 Partial file history for U.S. Appl. No. 12/292,553.
 Partial file history for U.S. Appl. No. 13/404,888.
 Partial file history for U.S. Appl. No. 13/426,133.
 Fengjing Shao et al., "A new secure architecture of network computer based on single CPU and Dual Bus" Fifth IEEE International Symposium on Embedded Computing, pp. 309-314 (2008).
 Tiedong Wang et al., "A Hardware implement of Bus Bridge Based on Single CPU and Dual Bus", 2008 International Symposium on Computer Science and Computational Technology, pp. 17-20 (2008).
 Famatech "Radmin V3.0, User Manuel", Jun. 3, 2007 (203 pages).
 Connect One, iChip CO2064/CO2128/CO2144, 2011 (64 pages).
 O. M. Woodward et al., "The Omniguide Antenna: An Omnidirectional Waveguide Array for UHF-TV Broadcasts", IRE international Convention Records, pp. 37-39, Mar. 1955.
 Litzkow, et al., "Condor—A Hunter of Idle Workstations", 1988 IEEE, pp. 104-111.
 Theimer, et al., "Finding Idle Machines in a Workstation-Based Distributed System", IEEE Transactions on Software Engineering, Nov. 1989, vol. 15, No. 11, pp. 1444-1458.

Brown et al., Special Edition Using Netscape TM 2 Second Edition, Que Corporation, 1995, Ch. 1-2.
 Gilder, "Angst and Awe on the Internet by George Gilder", Forbes ASAP, Dec. 4, 1995.
 Tanduary, et al., "Batrun: Utilizing Idle Workstations for Large-scale Computing", Summer 1996, pp. 41-48.
 Brisbin, "Shopping for Internet Access", MacUser, Dec. 1994, v. 10, p. 133(2).
 Gilder, "The Bandwidth Tidal Wave", Forbes ASAP, Dec. 5, 1994.
 N/A, "Special Report—Wonder Chips", Business Week, Jul. 4, 1994.
 N/A, "Supercomputers Divide and Conquer", The Economist, Dec. 14, 1996.
 N/A, "Cyber View World Widgets", Scientific American, May 1997, p. 48.
 Gibbs, "Bandwidth, Unlimited", Scientific American, Jan. 1997, p. 41.
 Markoff, "A New Standard to Govern PC's with Multiple Chips, Work Stations Mimicking Supercomputers", The New York Times, Oct. 28, 1997.
 N/A, "Aliens on your desktop", The Economist, Apr. 18, 1998, p. 78.
 Hare et al., "Master the Complexities of Network Security", Internet Firewalls and Network Security, Second Edition, pp. 325-350 and 516, 1996.
 Fox et al., Petcops and Exaops: Supercomputing on the Web, "IEEE Internet Computing", vol. 1 No. 2 Mar.-Apr. 1997, pp. 38-46.
 Dincer et al., Building a World-Wide Virtual Machine Based on Web and HPC Technologies, "Student Technical Papers", <http://www.supercomp.org/sc96/proceedings/SC96PROC/DINCER/IN-DEX.HTM>, pp. 1-18 (1996).
 Hobbs et al., A Remote Process Creation and Execution Facility Supporting Parallel Execution on Distributed Systems, "IEEE 1996", pp. 92-99.
 Boku et al., The Architecture of Massively Parallel Processor CP-PACS, "IEEE 1997", pp. 31-40.
 Choi et al., A Diagnostic Network for Massively Parallel Processing Systems, "IEEE 1994", pp. 348-353.
 Bellovin et al., Network. Firewallis, "IEEE Communications Magazine 1994", pp. 50-57.
 Weiyi et al., "Java-to-Go—Iterative Computing Using Java", Sep. 10, 1996 <http://ptolemy.eecs.berkeley.edu/dgm/javatools/java-to-go/>.
 Sullivan et al., "A New Major SETI Project Based on Project Serendip Data and 100,000 Personal Computers". <http://setiathome.ss.berkeley.edu/woody-paper.htm/>; (1997).
 "Ein-Chip-Firewall", Elektroniknet Top News, XP-002164257; Mar. 31, 1999.
 "Means for Implementing Optical Interconnections for Parallel Processors", IBM Technical Disclosure Bulletin, Apr. 1991, vol. 33, No. 11, pp. 56-58, XP 000110310.
 Alexandrov et al., "SuperWeb: Research Issues in Java-Based Global Computing", Concurrency, vol. 9, No. 6, Jun. 1997, pp. 535-553.
 Baratloo et al., "Charlotte: Metacomputing on the Web", 9th International Conference on Parallel and Distributed Computing Systems (PDCS), 1996, pp. 1-8.
 Bevinakoppa et al., "Digital Image Compression on a Network of Transputers", Proc. of 5th Australian Transputer & OCCAM User Group Conference, IOS Press, Amsterdam, pp. 25-32; Nov. 4, 1992.
 Blumofe; R. et al., "Scheduling Large-Scale Parallel Computations on Networks of Workstations", Proc. Of the 3rd IEEE Int'l Sump on High Performance Distributed Computing, pp. 96-105; Aug. 1994.
 Fields, Scott, "Hunting for Wasted Computing Power—New Software for Computing Networks Plus Idle PC's to Work", 1993 University of Wisconsin—Madison. Internet: <http://www.cs.wise.edu/condor/doc/Wiseidea.html>.
 Brecht et al., "ParaWebs Towards World-Wide Supercomputing", Proceedings of the Seventh AcM SIGOPS European Workshop, Sep. 1996, 8 pgs.
 Capello et al., "Market-Based Massively Parallel Internet Computing", Proceedings Third Working Conference on Massively Parallel Programming Models, 1998; pp. 118-129.

(56)

References Cited

OTHER PUBLICATIONS

- Celenk, M. et al., "Parallel Task Execution in LANs and Performance Analysis"; Proc. Of the 1995 IEEE 14th Annual Int'l Phoenix Conf. on Computers and Communications, pp. 423-429, Mar. 1995.
- Chen, C. et al., "The DSC: Processing Scientific Data Over the Internet"; Proc. Of the 16th Int'l Conf. on Distributed Computing Systems, pp. 673-679, May 1996.
- Clark, H. et al., "DAWGS—A Distributed Computer Server Utilizing Idle Workstations"; Proc. Of the 5th Distributed Memory Computing Conf., IEEE, pp. 732-741, Apr. 1990.
- Fogg, C., "Survey of Software and Hardware VLC Architectures", SPIE, vol. 2186, Image and Video Compression (1994), pp. 29-37.
- Fox et al., "Towards Web/Java based High Performance Distributed Computing—an Evolving Virtual Machine", as presented at 5th IEEE Int'l Symposium on High Performance Distributed Computing, Aug. 6-9, 1996; 86 pages.
- Fox, E., "Advances in Interactive Digital Multimedia Systems", Computer, Oct. 1991, pp. 9-21.
- Gemmell, et al., "Multimedia Storage Servers: a Tutorial", Computer, May 1995, pp. 40-49.
- Hayes, "Computing Science: Collective, Wisdom", American Scientist, Mar.-Apr. 1998, pp. 1-8.
- Kim, B., "ATM Network: Goals and Challenges", Communications of the ACM, Feb. 1995, Vol. 38, No. 2, pp. 39-44, 109.
- Kremien, O., "Buying and Selling Computational Power Over the Network", Proc. Of the 4th Int'l Conf. on Computer Communications and Networks, IEEE, pp. 616-619, Sep. 1995.
- Lindley, C.; "JPEG-Like Image Compression, Part 2", Dr. Dobbs Journal, Aug. 1995, pp. 62-66, 102-105.
- Lindley, C., "JPEG-Like Image Compression, Part 1", Dr. Dobbs Journal, Jul. 1995, pp. 50-58, 101-102.
- Morris, J. et al., "Fault Tolerant Networks of Workstations", Proc. Of the 3rd Int'l. Conf. on High Performance Computing, IEEE, pp. 271-276, Dec. 1996.
- Nass, R., "Hardware-software combo could simplify MPEG real-time video compression", Electronic Design, May 3, 1993, p. 36.
- Nowatzky et al., "Are Crossbars Really Dead? The Case for Optical Multiprocessor Interconnect Systems", Proceedings of the Annual Symposium on Computer Architecture, ACM, vol. 22, Jun. 1995, pp. 106-115, XP 000687800.
- Ozer, "Digital Video: Shot by Shot", PC Magazine, Apr. 11, 1995, pp. 104-107, 110.
- Ozer, J., "Why MPEG is Hot", PC Magazine, Apr. 11, 1995, pp. 130-131.
- Plotkin, "The Digital Compression Facility—A Solution to Today's Compression Needs", 1994 IEEE, pp. 445-449.
- Qiao et al., "Time-Division Optical Communications in Multiprocessor Arrays", ACM, 1991, pp. 644-653, XP 000337522.
- Rincon et al., "The Changing Landscape of System-on-a-chip Design", MicroNews, Third Quarter 1999. www.chips.ibm.com/micronews/vol5_no3/rincon.html, pp. 1-10, vol. 5, No. 3.
- Sachs, M. et al., "LAN and I/O Convergence: A Survey of the Issues". Computer, Dec. 1994, pp. 24-32.
- Sakano, et al., "A Three-Dimensional Mesh Multiprocessor System Using Board-to-Board Free-Space Optical Interconnects: COSINE-III", IEEE, 1993, pp. 278-283, XP 000463415.
- Schroeder, E., "New Offerings Buoy MPEG as Video Standard", Desktop Computing, PC Week, May 8, 1995, pp. 1 and 29.
- Shen et al., "A Parallel Implementation of an MPEG1 Encoder: Faster Than Real-Time!", Proc. Of SPIE Conf. on Digital Video Compression, San Jose, CA, Feb. 1995.
- McHenry et al., "An FPGA-Based Coprocessor for ATM Firewalls", Field-Programmable Custom Computing Machines 1997, Apr. 16-18, 1997, pp. 30-39, XP-002157218.
- Shivan, J. et al., "Client-Server Based Ray-Tracer Using ASTRA: An Asynchronous RPC Mechanism", Computer Communications, vol. 19, No. 5, pp. 445-455, May 1996.
- Szabo, B. et al., "Design Considerations for JPEG Video and Synchronized Audio in a Unix Workstation Environment", USENIX—Summer '91, pp. 353-368.
- Vetter, R. "ATM Concepts, Architectures, and Protocols", Communications of the ACM; Feb. 1995; vol. 38, No. 2. pp. 30-38, 109.
- Vetter, R. et al., "Issues and Challenges in ATM Networks", Communications of the ACM, Feb. 1995; vol. 38, No. 2, pp. 28-29.
- Waldspurger et al., "Spawn: A Distributed Computational Economy", IEEE Transactions on Software Engineering, vol. 18, No. 2, Feb. 1992, pp. 103-117 XP002124500.
- Yoshida, J., "MPEG-2 Encoders Make Show of Force at NAB", Electronic Engineering Times, Apr. 10, 1995.
- Yu, Y. et al., "Software Implementation of MPEG-II Video Encoding Using Socket Programming in LAN", SPIE vol. 2137, pp. 229-240, 1994.
- Waltz et al. "Make 'em pay: Billing Net Usage", MacWeek, vol. 6 (No. 27), p. 24 (2) (Dialog full text), (Jul. 27, 1992).
- "The Economics of Network Billing: Network Billing and Monitoring Systems can Improve Efficiency and Cut Costs", IBM System User, vol. 14 (No. 11), p. 53 (1) (Dialog Fulltext) (Nov. 1993).
- "Let Your Computer Make Money While You Sleep", Newsbytes, p. 1 (Dialog Fulltext), (Aug. 16, 1996).
- Regenold et al., "A Single-Chip Multiprocessor DSP Solution for Communication Applications", ASIC Conference and Exhibit 1994, pp. 437-440.
- Geppert, L. Solid State [Trend/Development], IEEE Spectrum. v. 33. iss. 1, 1996, pp. 51-55.
- Li, Yao, "Free-space Optical Bus-based WDMA Interconnects for Parallel Computation", LEOS '92 Conference Proceedings, Lasers and Bet-Aron-Optics Society Annual Meeting, p. 588-589, No. 16-19, 1992.
- Dickinson et al., "An Integrated Free Space Optical Bus", 1989 IEEE International Conference on Computer Design, VLSI in Computers and Processors, p. 62-65, Oct. 2-4, 1989.
- Natarajan et al., "Bi-Directional Optical Backplane Bus for General Purpose Multi-Processor", Journal of Lightwave Technology, vol. 14, No. 6, p. 1031-1040, Jun. 6, 1995.
- Zhao et al., "General Purpose Bidirectional Optical Backplane: High Performance Bus for Multiprocessor Systems", Massively Parallel Processing Using Optical Interconnections, 2nd International Conference, p. 188-195, Oct. 23-24, 1995.
- Wu et al., "Microprocessor Control Signal Transmission Through Optical Fiber", Conference Record of 1992, IEEE Industry Applications Society Annual Meeting. p. 1747-1750, Oct. 4-9, 1992.
- Fox et al., "Towards Web/Java based High Performance Distributed Computing—an Evolving Virtual Machine", www.npac.syr.edu/projects/webpace/doc/hpdc5/paper, Jun. 10, 1996, 11 pages.
- None, "PC Vision: Intel unveils plans to bring PCs to Vehicles"; EDGE: Work-Group Computing Report, EDGE Publishing, p. 1-2 (Oct. 28, 1996).
- The American Heritage College Dictionary 4th Ed.—definition of "firewall" (2007).
- White, "Covert Distributed Processing with Computer Viruses", Advances in Cryptology, Crypto 89, Springer LNCS, v. 435, pp. 616-619.
- Foster et al., "The Grid: Blueprint for a New Computing infrastructure", Morgan Kaufman Publishers, Inc., 1998.
- Hwang et al., "Scalable Parallel Computing", WCB McGraw-Hill, 1998.
- Wilkinson, et al., "Parallel Programming", Prentice Hall, 1998.
- Patterson et al., "Computer Architecture: A Quantitative Approach" (2nd Edition), Morgan Kaufmann Publishers, Inc., 1996.
- Culler et al., "Parallel Computer Architecture", Morgan Kaufmann Publishers, Inc., 1998.
- Hennessy et al., "Computer Organization and Design", Morgan Kaufmann Publishers, Inc., 1998.
- Slater, "The Microprocessor Today", IEEE Micro 1996, pp. 32-44.
- Steinert-Threlkeid; "New Breed of Chip TI develops a Super circuit"; The Sun Baltimore; May 4, 1992.
- Dallas Morning News; "LSI holds big plans for tiny chips Versatility of 'system on a chip' creates niche in microelectronics," Mar. 4, 1996.
- Mokhoff; "System-on-a-chip comes to wireless arena," Electronic Engineering Times; Feb. 12, 1996.
- Cindi; "System on a Chip stars at ISSCC," Electronic News; Feb. 19, 1996.

(56)

References Cited

OTHER PUBLICATIONS

- Ang., "System-on-a-chip to define next-generation set-top box"; Electronic Engineering Times; Dec. 15, 1995.
- Marc; "New family of microprocessor cores from LSI Logic extends customers' system-on-a-chip design capability" Nov. 7, 1994.
- Wall Street Journal; "Technology Brief—Advance Micro Devices Inc.: Company unveils Microchip for Hand-Held Computers"; Oct. 18, 1993.
- Gelsinoer, Patrick of al. "Microprocessors circa 2000," IEEE Spectrum, Oct. 1989 pp. 43-47.
- Yu, Abet "The Future of Microprocessors," IEEE Micro, Dec. 1996, pp. 46-53.
- McWilliams. "Dell to Phase Out Computers Using Intel's Itanium," The Wall Street Journal, Online, Sep. 15, 2005.
- David Pescovitz, "Power of the PC", Scientific American, pp. 27-28 (Apr. 2000).
- Stephen H. Wildstrom, "The Problem with Firewalls", Business Week, pp. 25 (Mar. 20, 2000).
- J. McH., "Build Your Own Supercomputer", Forbes, pp. 228 (Nov. 15, 1999).
- Wilkinson, Barry et al., "Parallel Programming: Techniques and Applications Using Networked Workstations and Parallel Computers", Chapter 4, Prentice-Hall, Inc., 1999.
- Baker, Lou et. al., "Parallel Programming", Chapter 4, McGraw Hill Companies, Inc., 1996.
- Kayssi, A.; Harik, L.; Ferzli, R.; Fawaz, M. "FPGA-based internet Protocol Firewall Chip": Electronics, Circuits and System, 2000. ICECS 2000. The 7th International Conference on vol. 1, Dec. 17-20, 2000 pp. 316-319 vol. 1.[retrieved from IEEE database Jun. 9, 2008].
- English language abstract of EP 0 647 052, published Apr. 5, 1995.
- Newton's Telecom Dictionary, "Mobile IP"; p. 459. Mar. 1998.
- Holographic Quantum Computer, <http://www.unite/nw.com/holo1/index> (May 1999).
- Jonathan Fahey, "Screen Grab", Forbes; pp. 52-53, Mar. 5, 2001.
- Ronald Grover et al., "TV Guy", Business Week, pp. 66-76, Mar. 12, 2001.
- "Distributed Computing", Red Herring, No. 87, pp. 166-202; Dec. 18, 2000.
- Om Malik, "Distributed Computing Redefines Computer Networks, Underpinning Innovation, Company Formation; and Investments", Red Herring, No. 88, pp. 95-96 and 105, Dec. 4, 2000.
- Alan Zeichick, "PCP Network Expalined", Red Herring, No. 86, pp. 204 and 206, Dec. 4, 2000.
- "Napster is Clouding Groves Crystal Ball"; Fortune, pp. 271-272, May 29, 2000.
- Gordon Force, Sr., "Portable Data Enoryctlon Approaches", WESTCON/95, Conference Record: Microelectronics Communications Technology Producing Quality Production Mobile and Portable Power emerging Technologies, Nov. 7-9, 1995, pp. 413-419.
- A.D. Romig, Jr., "High Performance Microsystems Packaging: A Perspective", Microelectron Reliab., vol. 37, No. 10/11, pp. 1771-1781 (1997).
- Adidas Catalog 1986.
- Adidas Catalog 1988.
- Adidas Catalog 1989.
- Adidas Catalog 1990.
- Adidas Catalog 1991.
- Adidas Catalog, Spring 1987.
- Adidas' Second Supplemental Responses to Interrogatory No. 1.
- Adidas shoe, Model "Buffalo" 1985.
- Adidas shoe, Model "London", 1986.
- Adidas shoe, Model "Marathon 86" 1985.
- Adidas shoe, Model "Skin Racer", 1988.
- Adidas shoe, Model "Tokio H" 1985.
- Adidas shoe, Model "Torsion Grand Slam Indoor," 1989.
- Adidas shoe, Model <<Boston Super>> 1985.
- Adidas shoe, Model <<Fire>> 1985.
- Adidas shoe, Model <<Indoor Pro>> 1987.
- Adidas shoe, Model <<Kingscup Indoor>>, 1986.
- Adidas shoe, Model <<Marathon>> 1986.
- Adidas shoe, Model <<Questar>>, 1986.
- Adidas shoe, Model <<Tauern>> 1986.
- Adidas shoe, Model <<Tennis Comfort>>, 1988.
- Adidas shoe, Model <<Torsion Special HI>> 1989.
- Adidas shoe, Model <<Torsion ZX9020S>> 1989.
- Adidas shoe, Model <<Water Competition>> 1980.
- Adidas shoe, Model, <<Indoor Pro>> 1987.
- Adidas Spring Catalog 1989.
- Areblad et al. <<Three-Dimensional Measurement of Rearfoot Motion During Running>> *Journal of Biomechanics*, vol., 23, vol. 9, pp. 933-940 (1990).
- Avia Catalog 1986.
- Avia Fall Catalog, 1988.
- Brooks Catalog 1986.
- Cavanagh et al., "Biomechanics of Distance Running," Human Kinetics Books, pp. 155-164, 1990.
- Clark Shoe Designed by Sven Coomer 1991.
- Fineagan, "Comparison of the Effects of a Running Shoe and A Racing Flat on the Lower Extremity Biochemical Alignment of Runners," *Journal of the American Physical Therapy Association*, vol. 68, No. 5, p. 806 (1988).
- Fixx, *The Complete Book of Running*, pp. 134-137, 1977.
- Footwear Journal, Nike Advertisement, Aug. 1987.
- Footwear News, vol. 44, No. 37, Nike Advertisement (1988).
- Footwear News, Special Supplement, Feb. 8, 1988.
- Footwear News, vol. 45, No. 5, Nike Advertisement 1989.
- Frederick, *Sports Shoes and Playing Surfaces. Biomechanical Properties*, Entire Book, 1984.
- Johnson et al., <<A Biomechanical Approach to the Design of Football Boots>>, *Journal of Biomechanics*, vol. 9, pp. 581-585 (1976).
- Komi et al., "Interaction Between Man and Shoe in Running: Considerations for More Comprehensive Measurement Approach," *International Journal of Sports Medicine*, vol.8, pp. 196-202 1987.
- Kronos Catalog, 1988.
- K-Swiss Catalog, Fall 1991.
- Leuthi et al., <<Influence of Shoe Construction on Lower Extremity Kinematics and Load During Lateral Movements in Tennis>>, *International Journal of Sport Biomechanics*, vol. 2, pp. 166-174, 1986.
- Nawoczinski et al., <<Effect of Rocker Sole Design on Plantar Forefoot Pressures>> *Journal of the American Podiatric Medical Association*, vol. 79, No. 9, pp. 455-460, 1988.
- Nigg et al., "Biomechanical Aspects of Sport Shoes and Playing Surfaces," Proceedings of the International Symposium on Biomechanical Aspects of Sport Shoes and Playing Surfaces, 1983.
- Nigg et al., "Influence of Heel Flare and Midsole Constructions on Pronation" *International Journal of Sport Biomechanics*, vol. 4, No. 3, pp. 205-219 (1987).
- Nigg et al., <<The Influence of Lateral Heel Flare of Running Shoes on Pronation and Impact Forces>>, *Medicine and Science in Sports and Exercise*, vol. 19, No. 3, pp. 294-302, 1987.
- Nigg et al., *Biomechanics of Running Shoes*, entire book, 1986.
- Nigg, <<Biomechanical Analysis of Ankle and foot Movement Medicine and Sport Science>>, vol. 23, pp. 22-29 1987.
- Nike Catalog, Footwear Fall, 1988.
- Nike Fall Catalog 1987, pp. 50-51.
- Nike Shoe, men's cross-training Model "Air Trainer SC" 1989.
- Nike shoe, men's cross-training Model <<Air Trainer>> TW 1989.
- Nike shoe, Model "Air Force" #1978, 1988.
- Nike shoe, Model "Air" #1553, 1988.
- Nike shoe, Model <<Air Flow>> #718, 1988.
- Nike shoe, Model <<Air Revolution>> #15075, 1988.
- Nike shoe, Model <<Air>>, #13213, 1988.
- Nike shoe, Model <<Air>>, #4183, 1988.
- Nike shoe, Model <<High Jump>> 88, 1988.
- Nike shoe, Model <<Leather Cortex® >>, 1988.
- Nike shoe, Model <<Zoom Street Leather>> 1988.
- Nike Spring Catalog 1989 pp. 62-63.

(56)

References Cited

OTHER PUBLICATIONS

Palamarchuk et al., "In shoe Casting Technique for Specialized Sport Shoes," *Journal of the American Podiatric Medical Association*, vol. 79, No. 9, pp. 462-465, 1989.

Prince Cross-Sport 1989.

Puma basketball shoe, *The Complete Handbook of Athletic Footwear*, pp. 315, 1987.

Romika Catalog, Summer 1978.

Runner's World, "Shoe Review" Nov. 1988 pp. 46-74.

Runner's World, "Spring Shoe Survey," pp. 45-74, 1989.

Runner's World, Apr. 1988.

Runner's World, Oct. 1986.

Saucony Spot-bilt Catalog 1988.

Saucony Spot-bilt Catalog Supplement, Spring 1985.

Saucony Spot-bilt shoe, *The Complete Handbook of Athletic Footwear*, pp. 332, 1987.

Segesser et al., "Surfing Shoe," *The Shoe in Sport*, 1989, (Translation of a book published in Germany in 1987), pp. 106-110.

Sporting Goods Business, Aug. 1987.

Sports Illustrated, Nike Advertisement, Aug. 8, 1988.

Sports Illustrated, Special Preview Issue, The Summer Olympics <<Seoul '88 >> Reebok Advertisement.

The Complete Handbook of Athletic Footwear, Entire B000k, 1987.

Vagenas et al., "Evaluation of Rearfoot Asymmetries in Running with Worn and New Running Shoes," *International Journal of Sport Biomechanics*, vol. 4, No. 3, pp. 220-230.

Valiant et al., <<A Study of Landing from a Jump: Implications for the Design of a Basketball Shoe>>, *Scientific Program of IX International Congress of Biomechanics*, 1983.

Williams et al., <<The Mechanics of Foot Action During the Golf Swing and Implications for Shoe Design>>, *Medicine and Science in Sports and Exercise*, vol. 15, No. 3, pp. 247-255, 1983.

World Professional Squash Association Pro Tour Program, 1982-1983.

Williams, Walking on Air, *Case Alumnus*, vol. LXVII, No. 6, Fall 1989, pp. 4-8.

Brooks advertisement in *Runner's World* etc., Jun. 1989, pp. 56+.

Cavanaugh et al., Biological Aspects of Modeling Shoe/Foot Interaction During Running, *Sports Shoes and Playing Surfaces*, 1984, pp. 24-25, 32-35, 46.

Blechschiidt, The Structure of the Calcaneal Padding, *Foot & Ankle* vol. 2, No. 5, Mar. 1982, pp. 260-283.

Cavanaugh "The Running Shoe Book," © 1980, pp. 176-180, Anderson World, Inc., Mountain View, CA.

Executive Summary with Seven Figures, 1993.

German Description of adidas badminton shoe, pre-1989(?), 1 page.

The Reebok Lineup, Fall 1987, 2 pages.

* cited by examiner

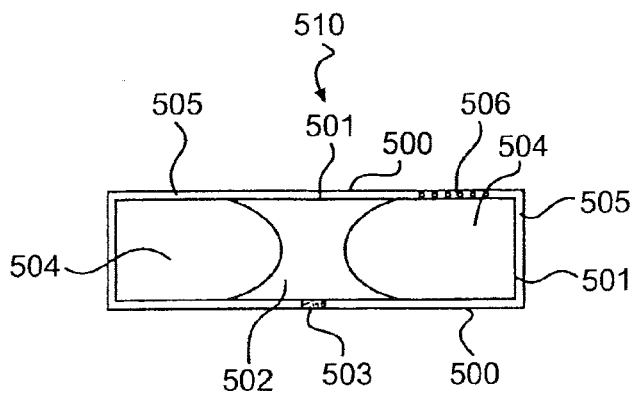


FIG. 1A

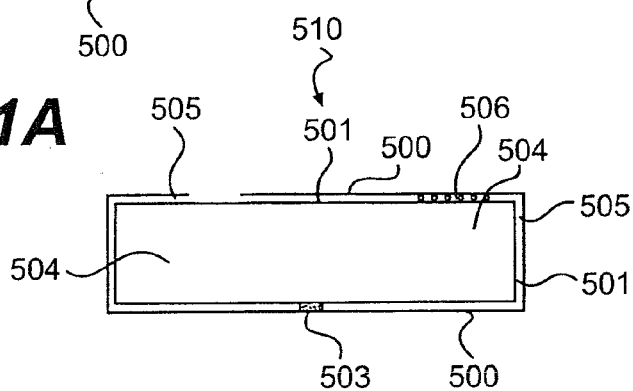


FIG. 2A

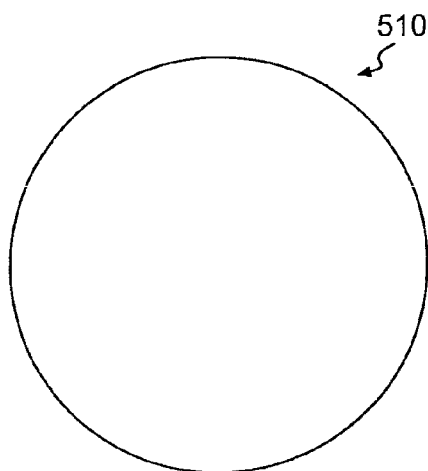


FIG. 1B

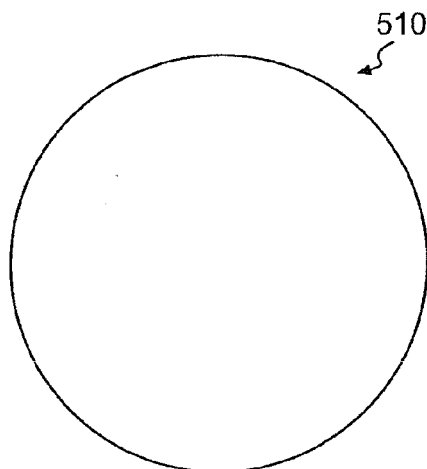


FIG. 2B

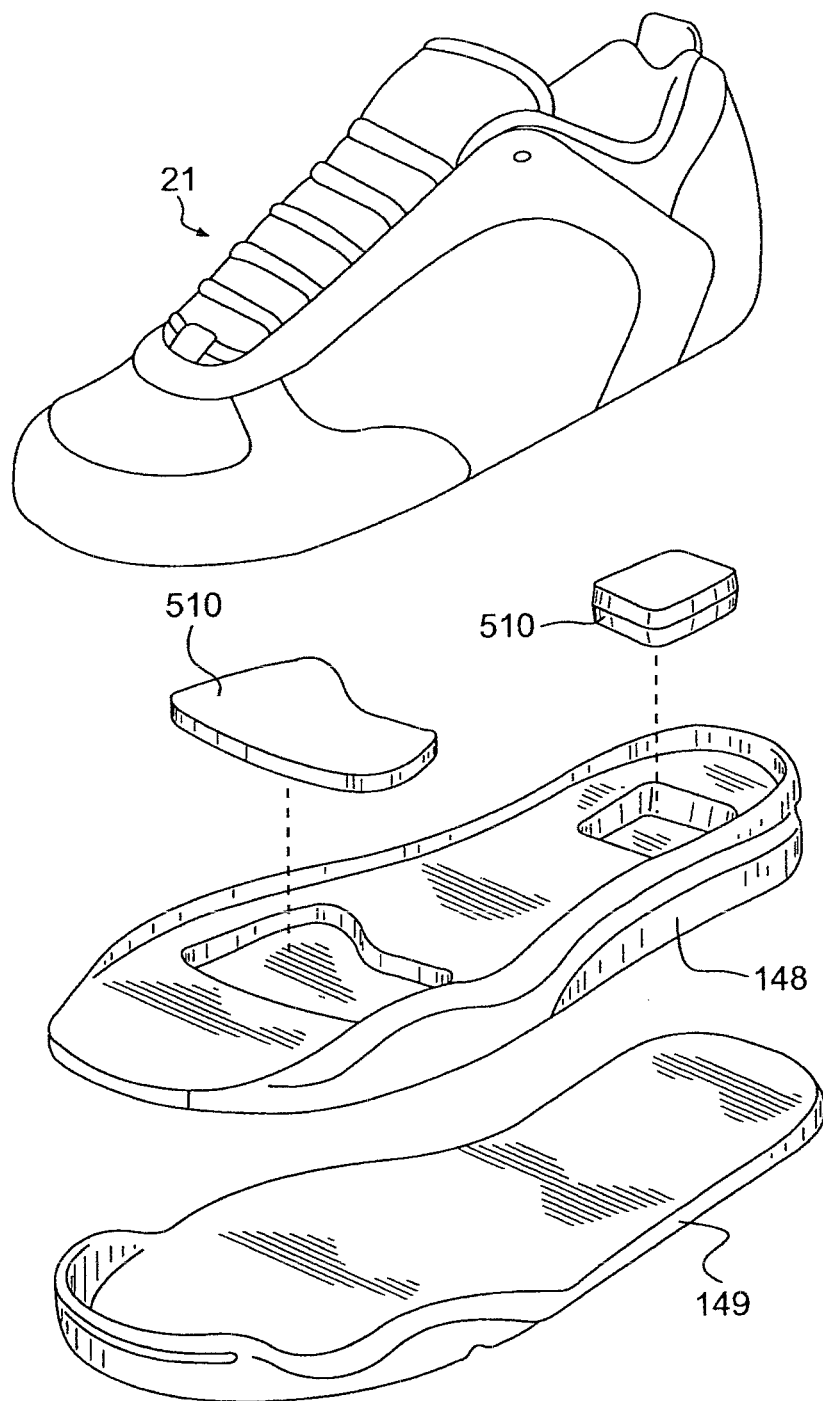


FIG. 1C

FIG. 1D

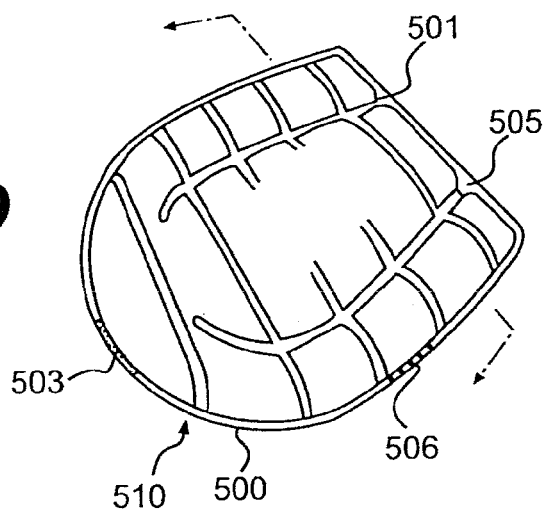


FIG. 1E

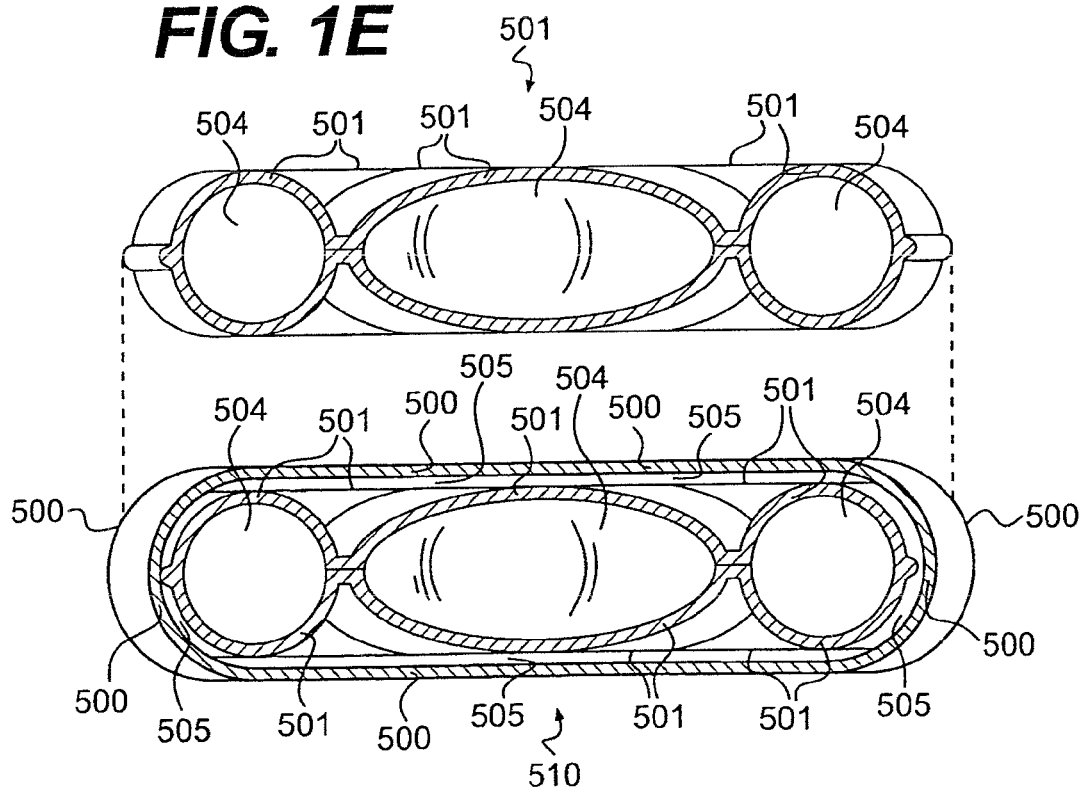


FIG. 1F

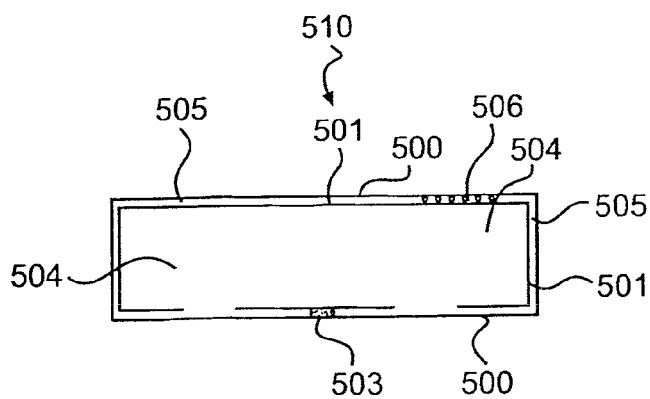


FIG. 3A

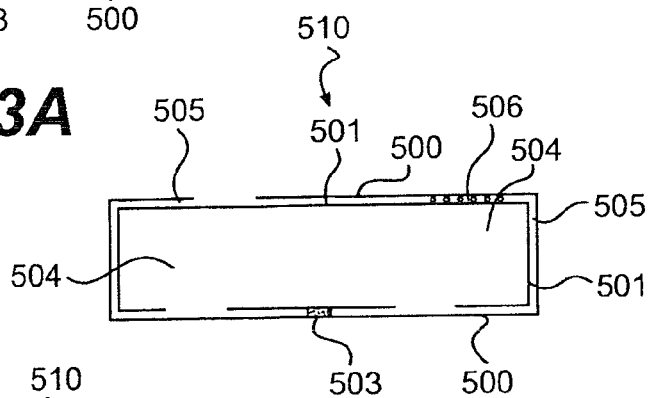


FIG. 4A

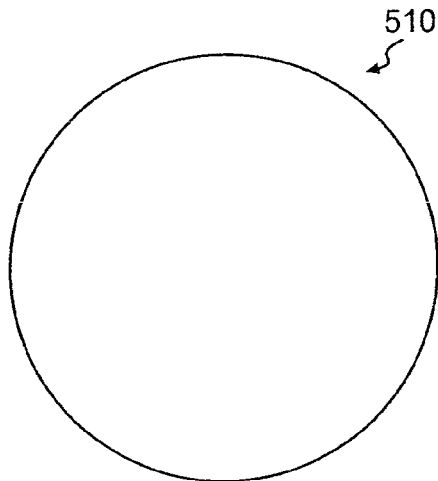


FIG. 3B

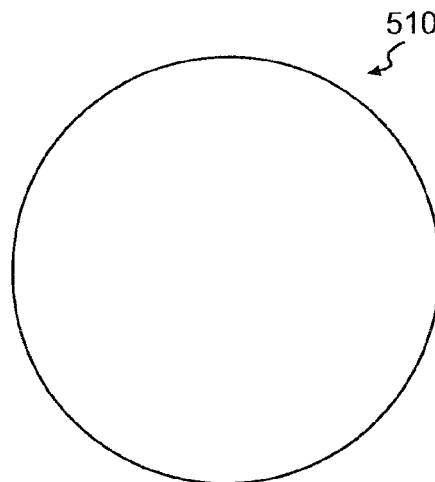


FIG. 4B

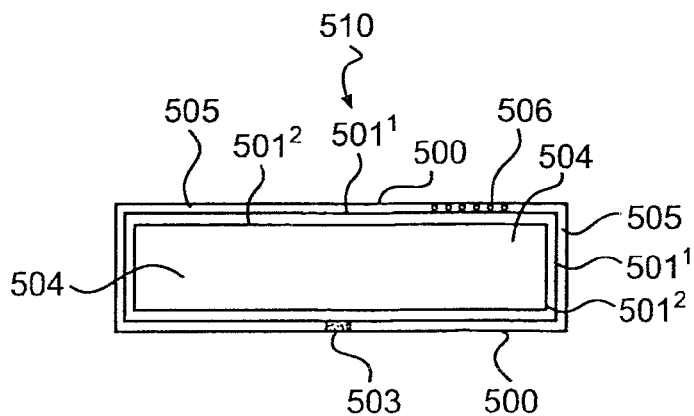


FIG. 5A

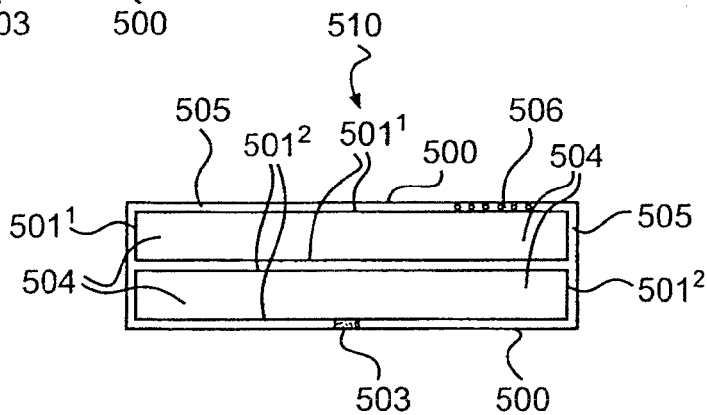


FIG. 6A

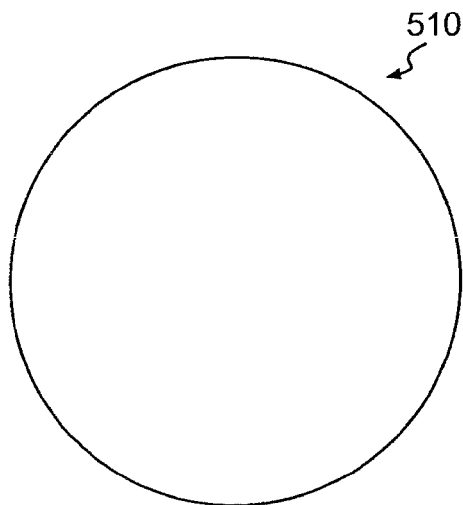


FIG. 5B

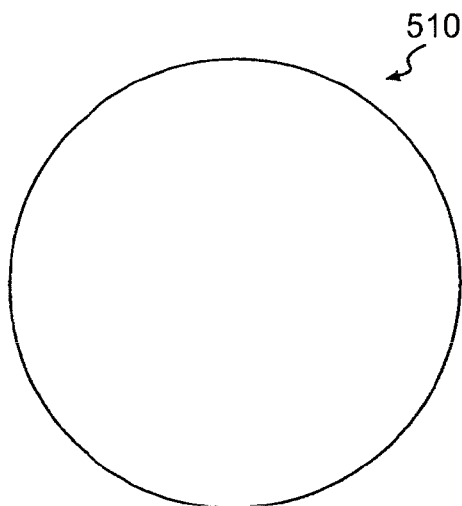


FIG. 6B

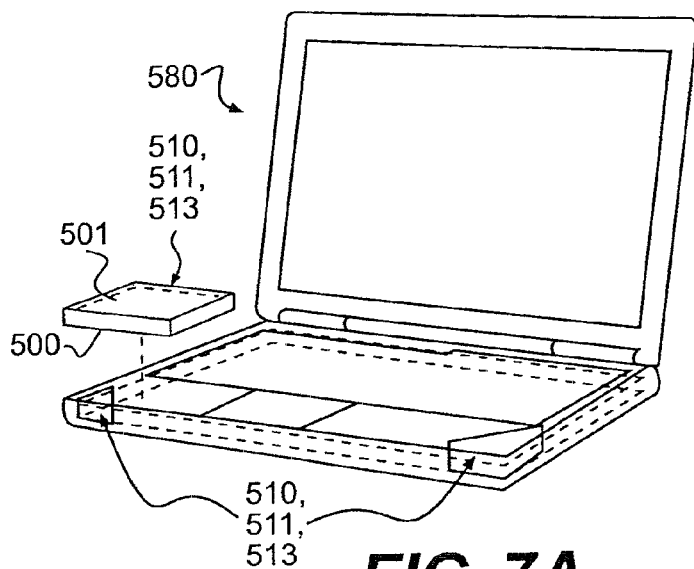


FIG. 7A

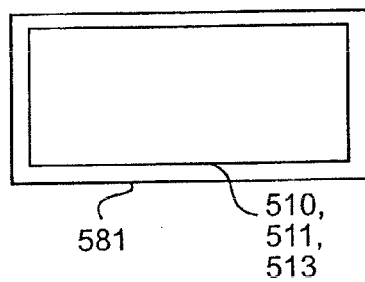


FIG. 7B

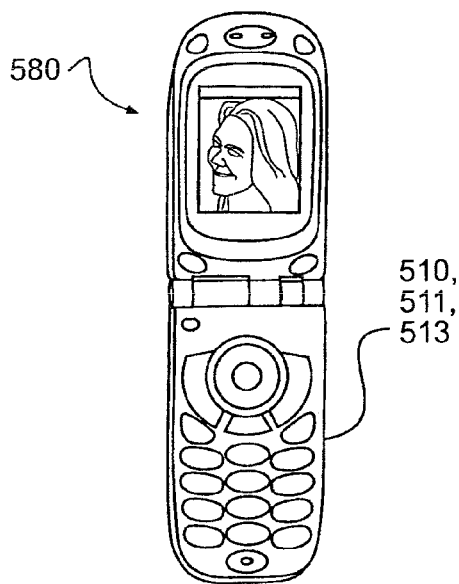


FIG. 7D

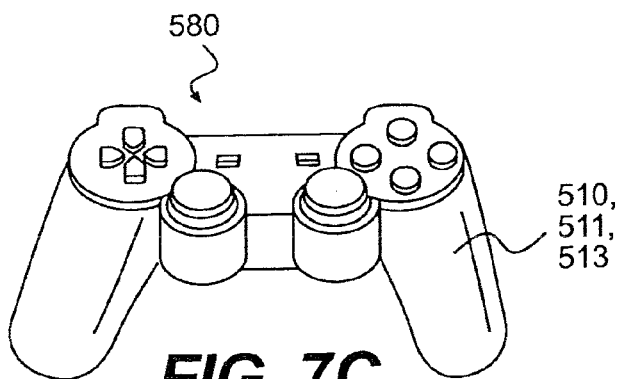


FIG. 7C

APPROXIMATE COMPARISON OF HARDNESS SCALES

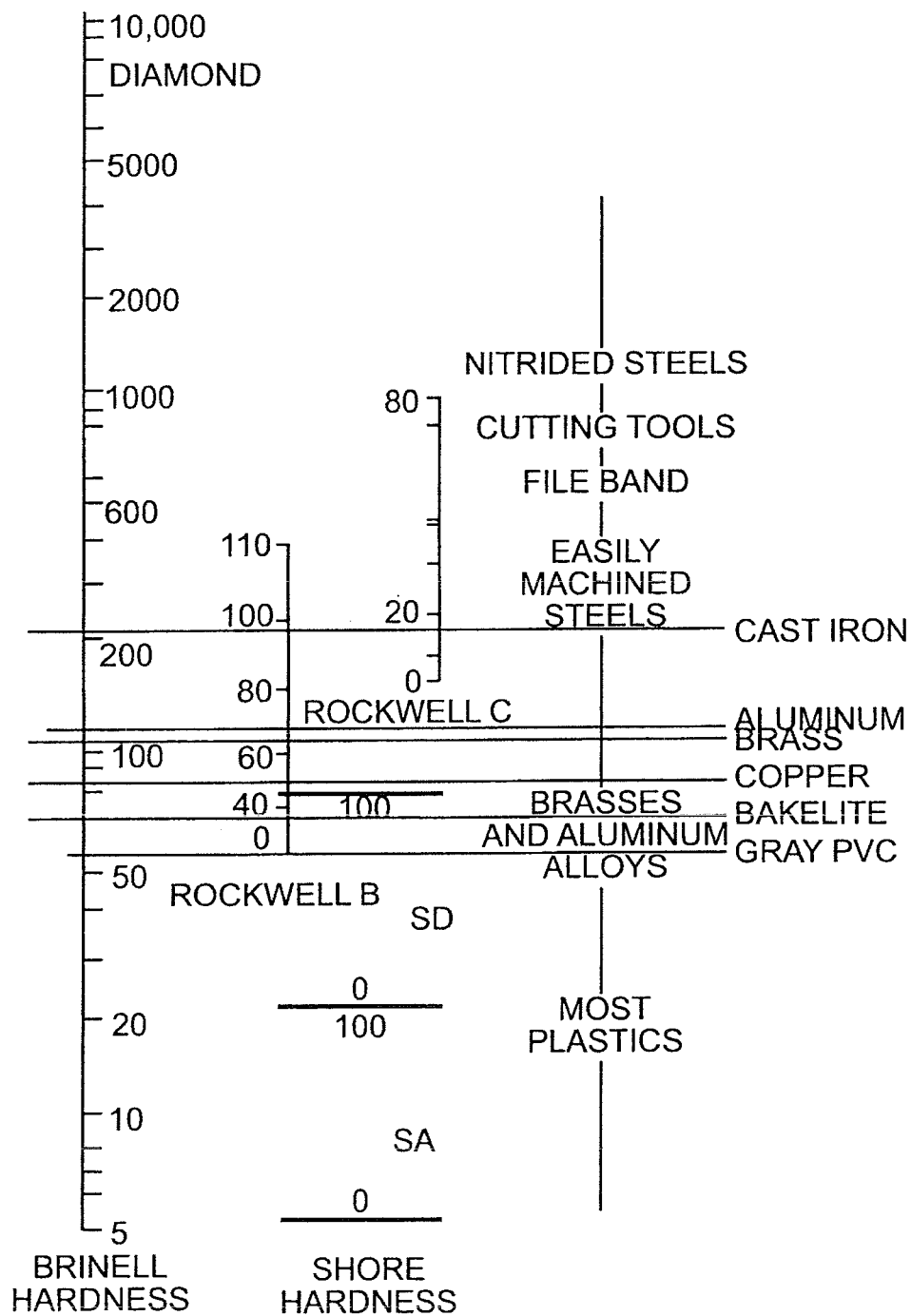
**FIG. 8**



FIG. 9A



FIG. 9B

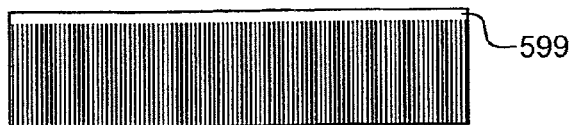
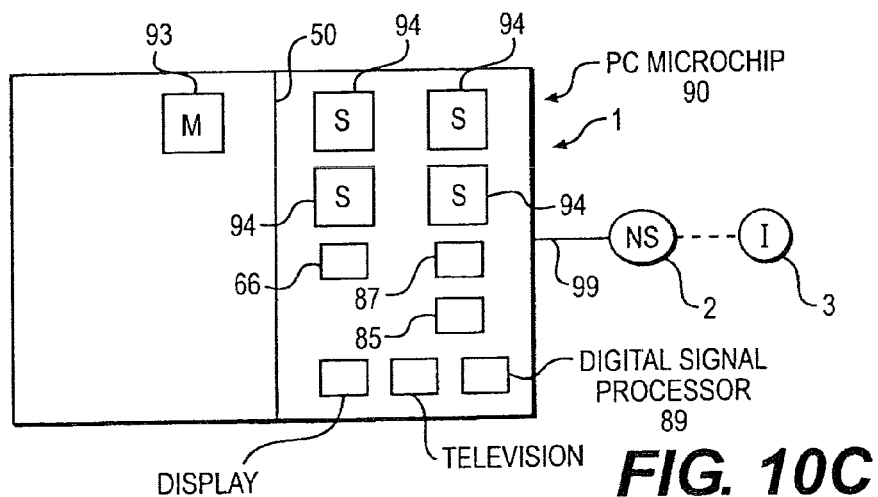
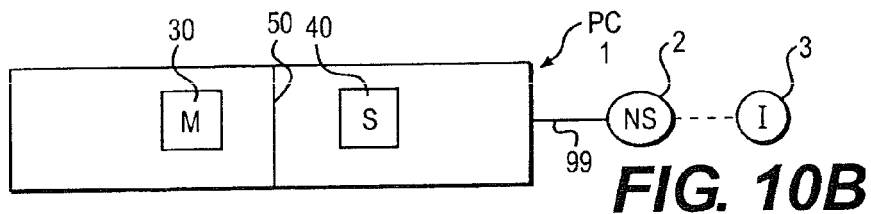
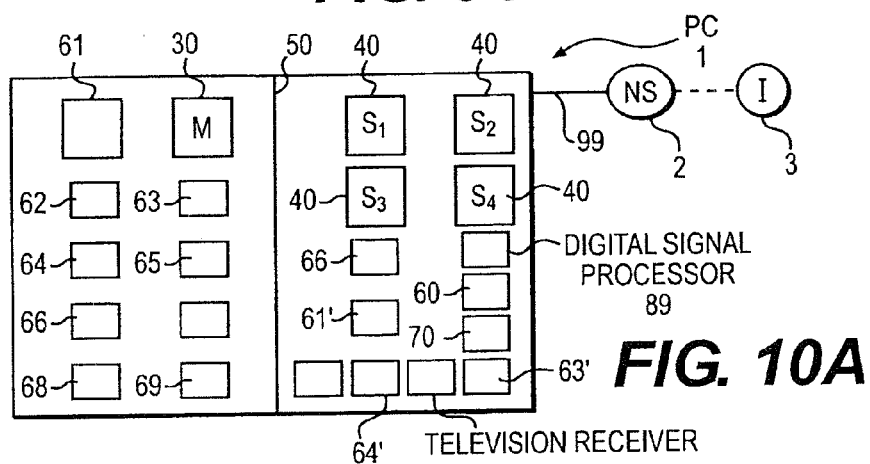


FIG. 9C



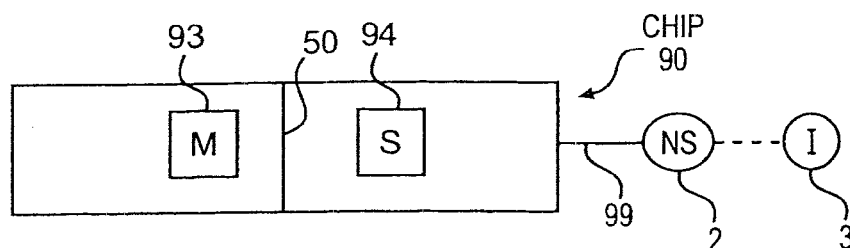


FIG. 10D

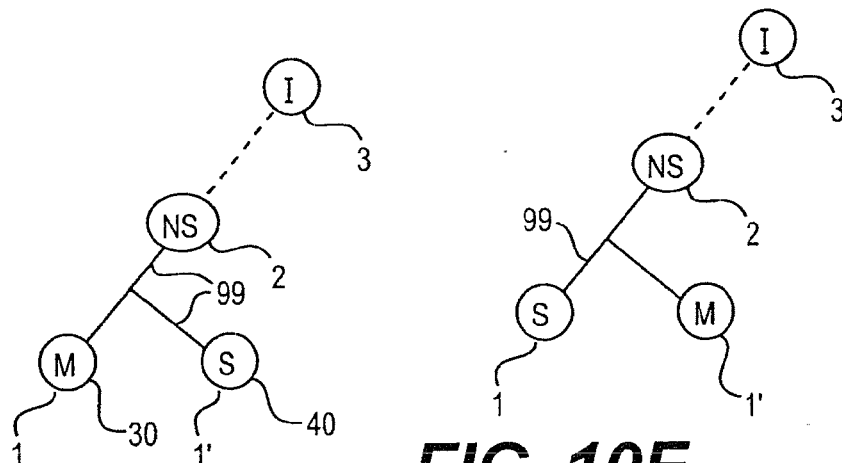


FIG. 10E

FIG. 10F

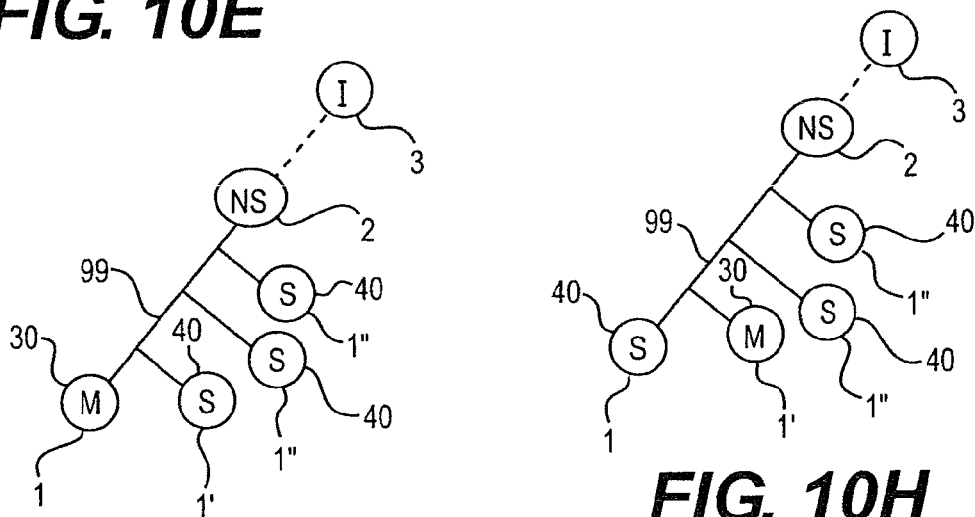


FIG. 10G

FIG. 10H

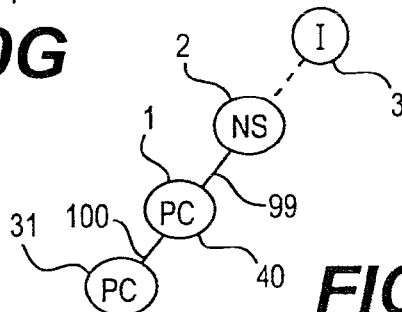
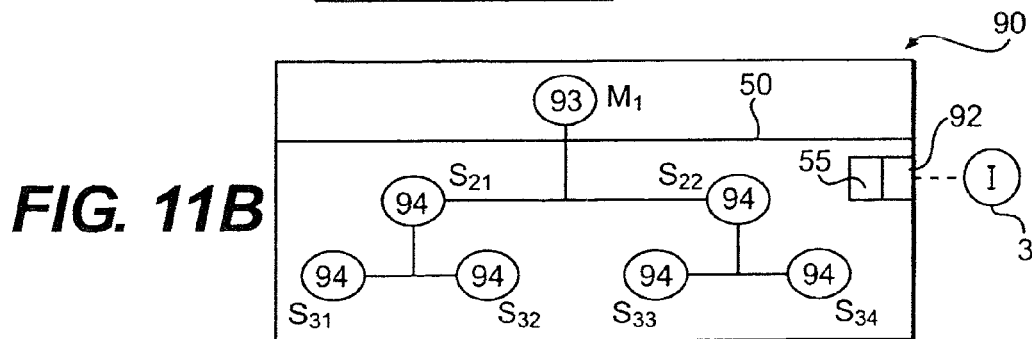
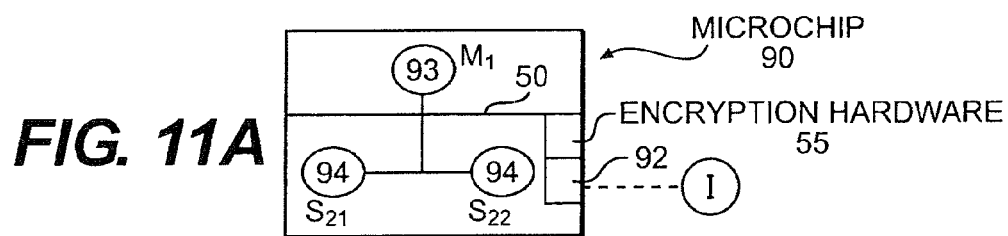
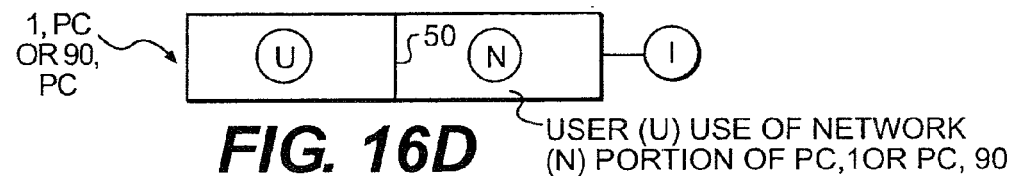
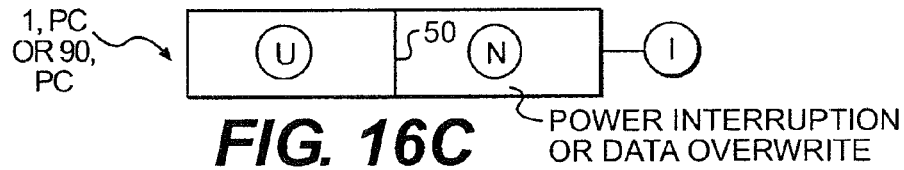
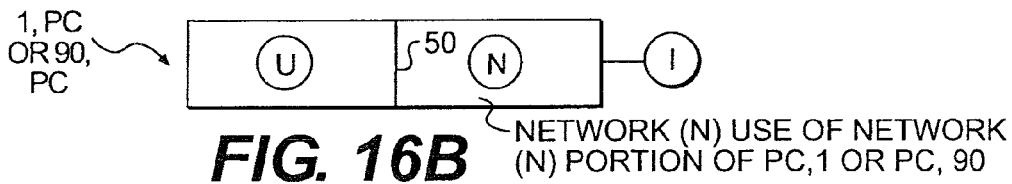
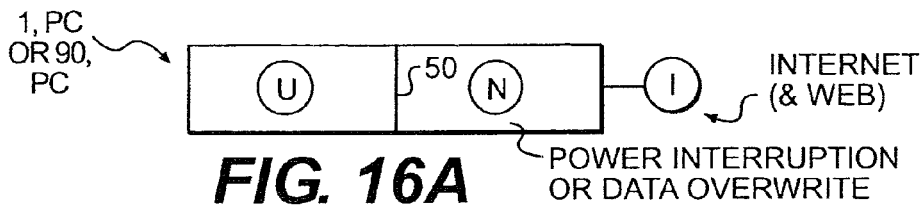


FIG. 10I



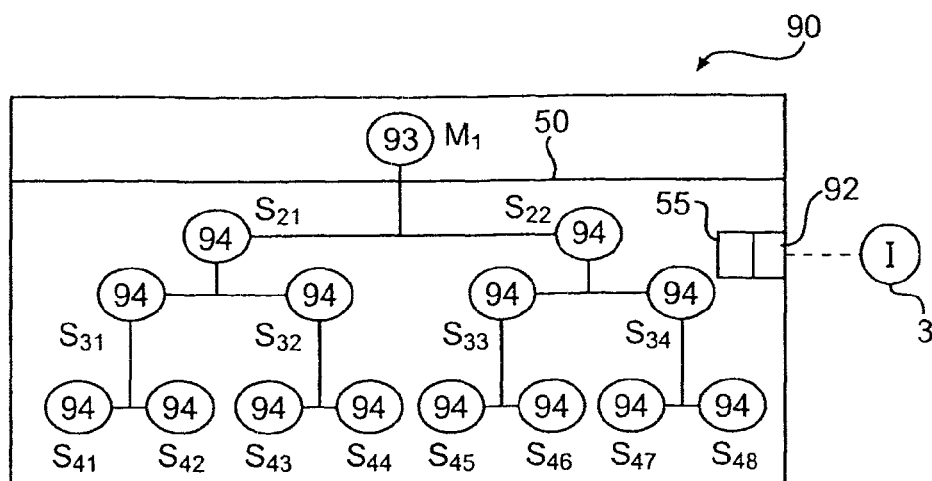


FIG. 11C

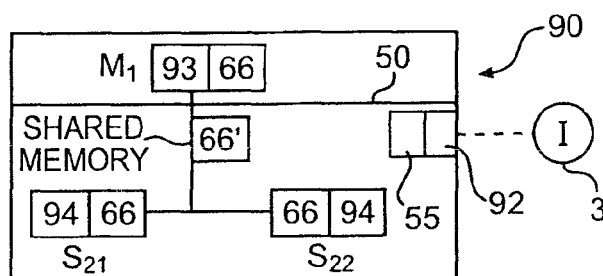


FIG. 11D

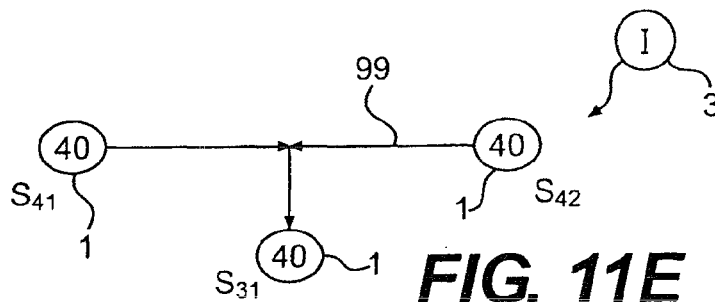


FIG. 11E

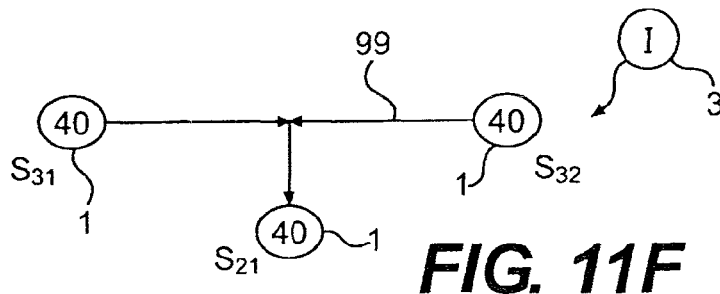
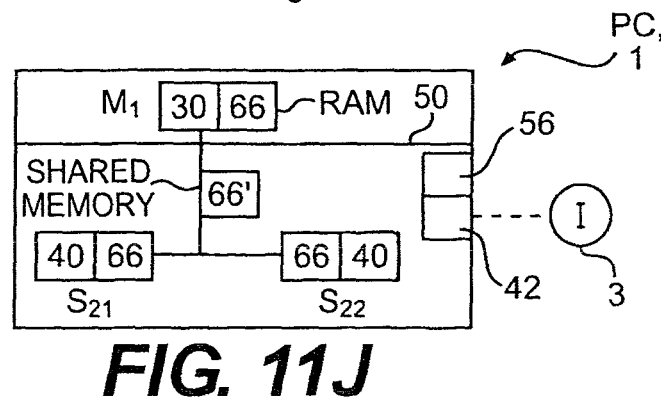
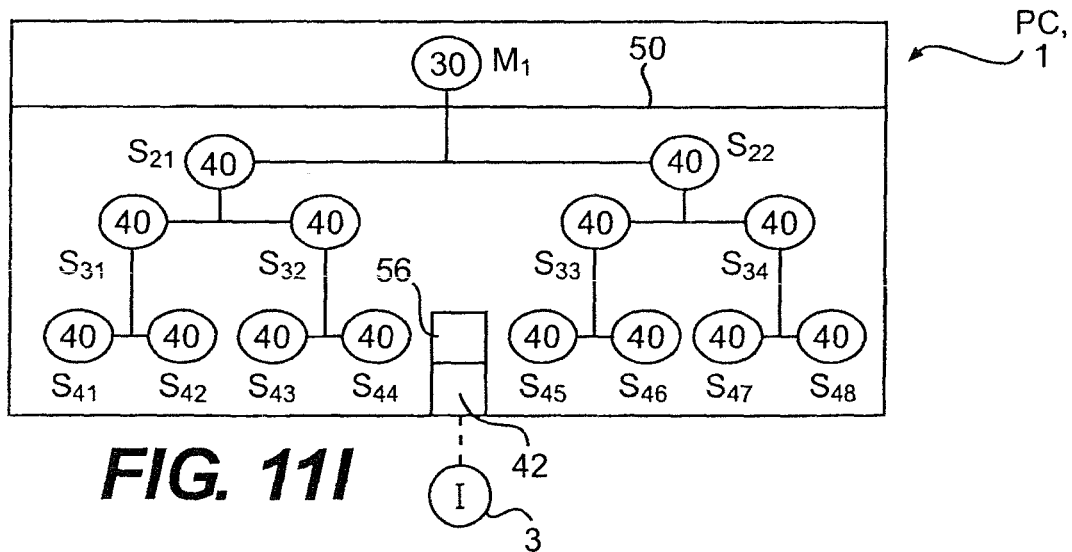
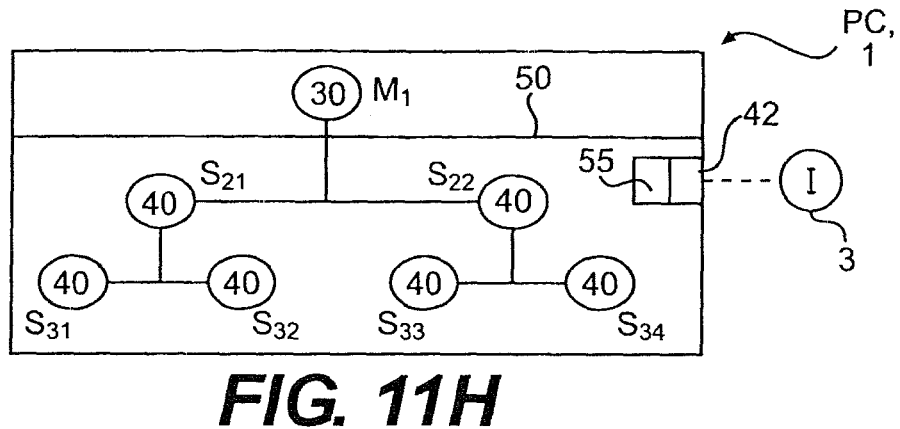
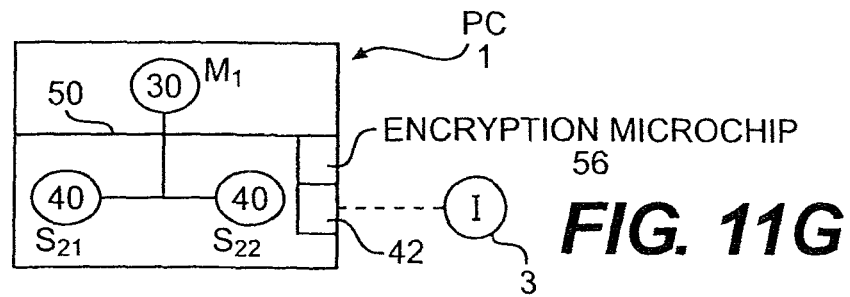


FIG. 11F



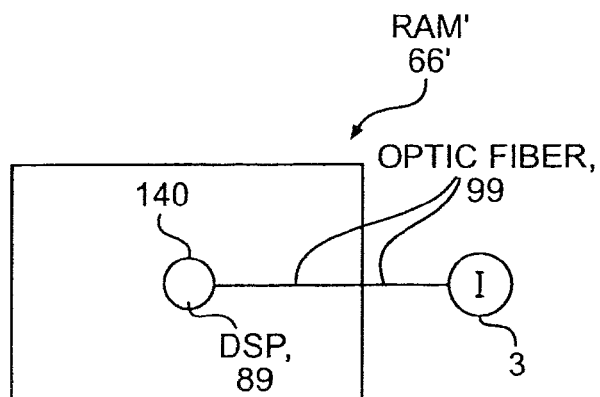


FIG. 11K

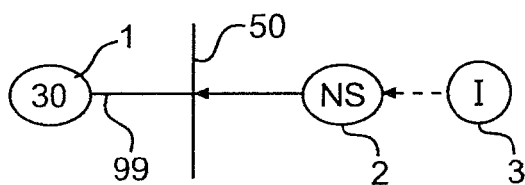


FIG. 12A

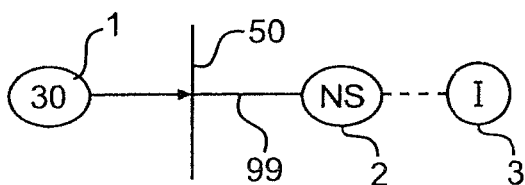


FIG. 12B

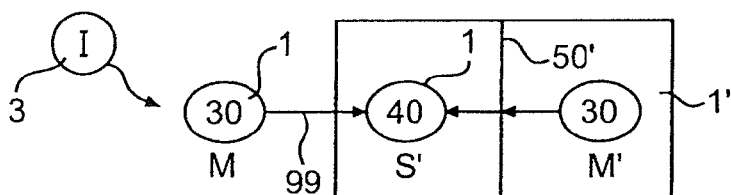


FIG. 12C

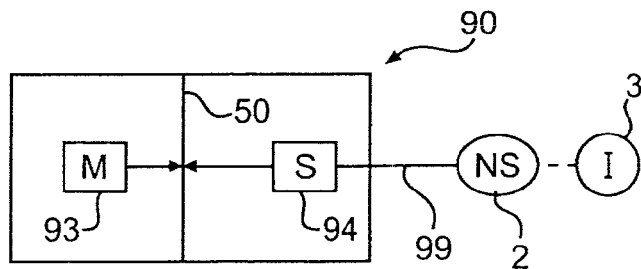


FIG. 12D

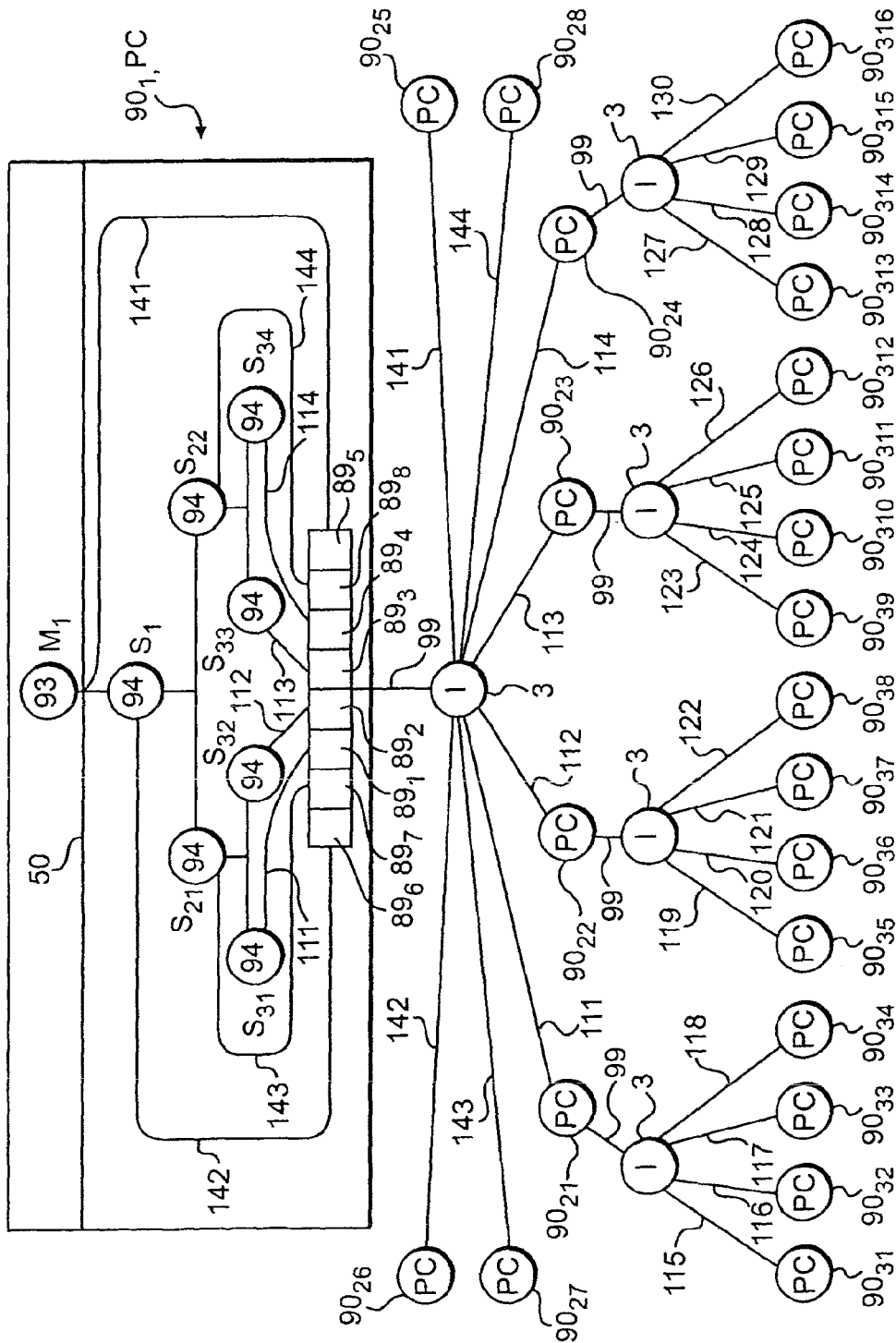


FIG. 13

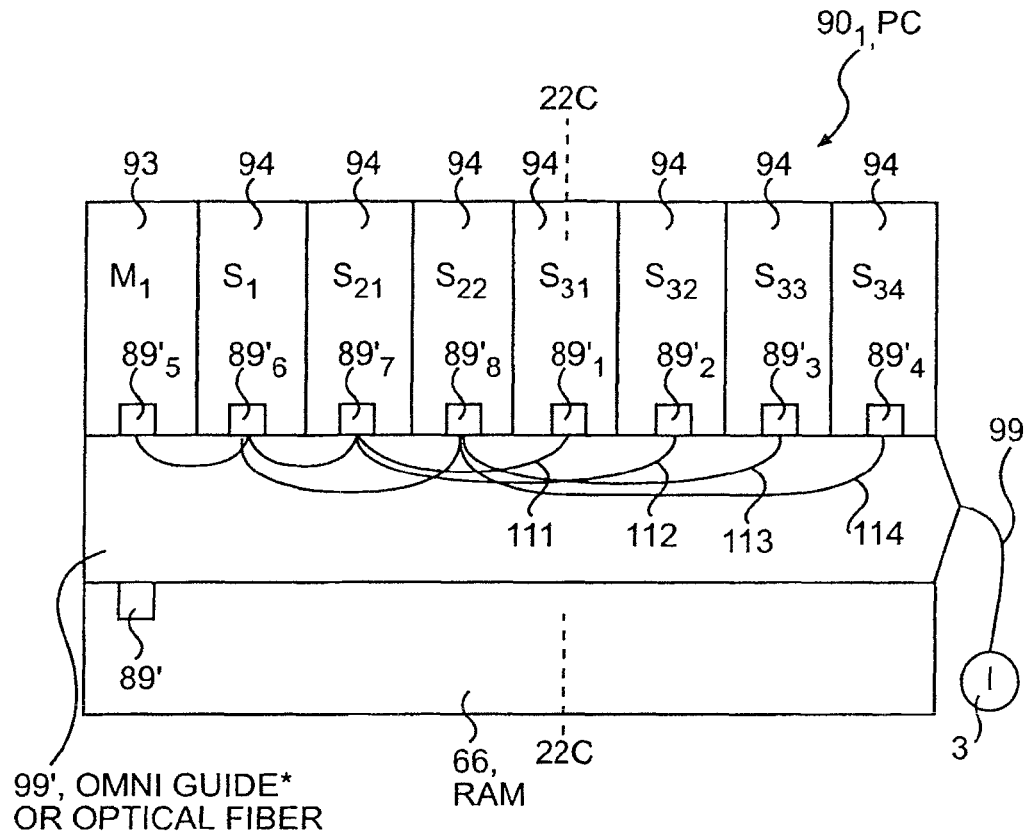


FIG. 14A

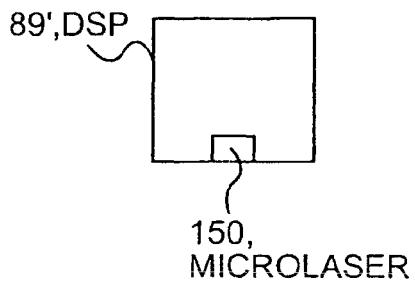


FIG. 14B

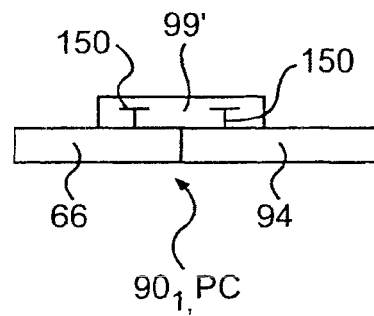


FIG. 14C

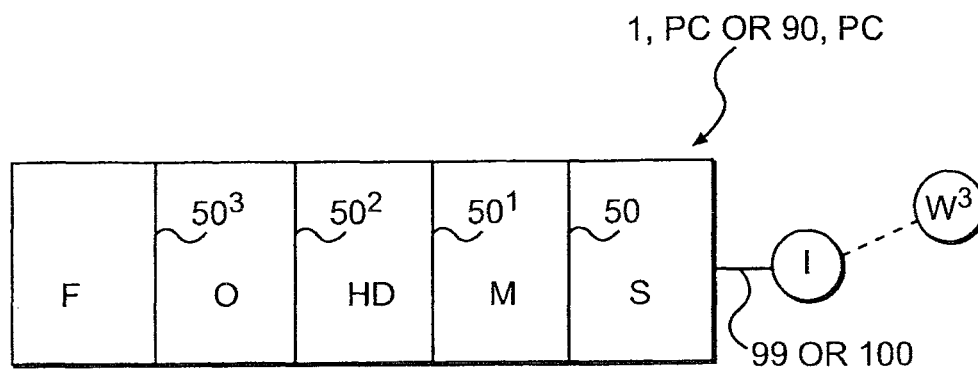


FIG. 15A

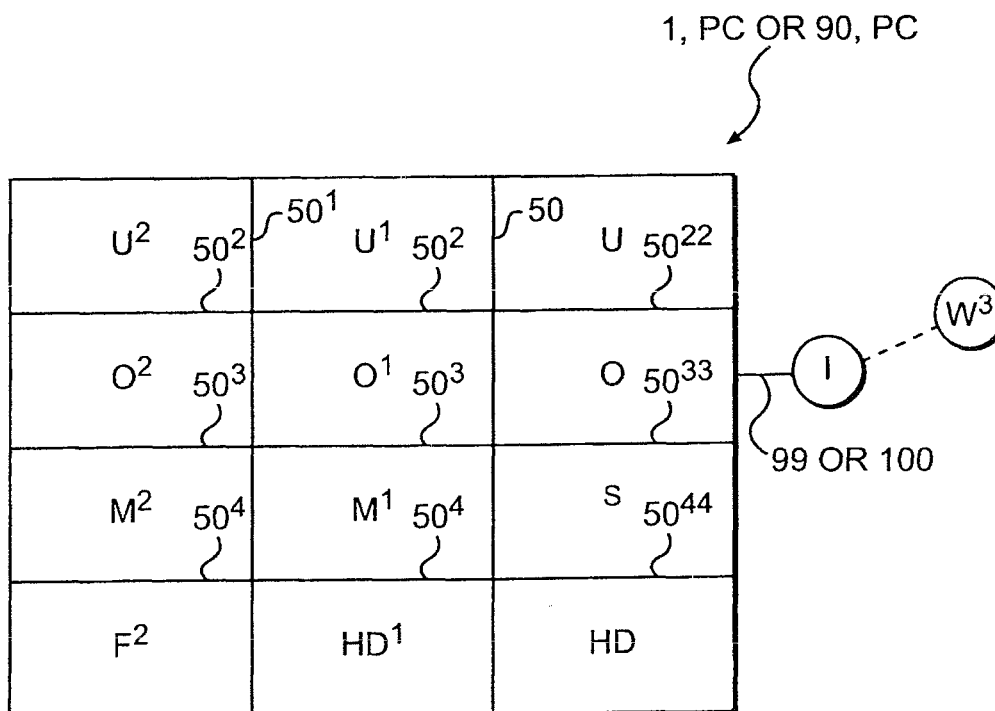


FIG. 15B

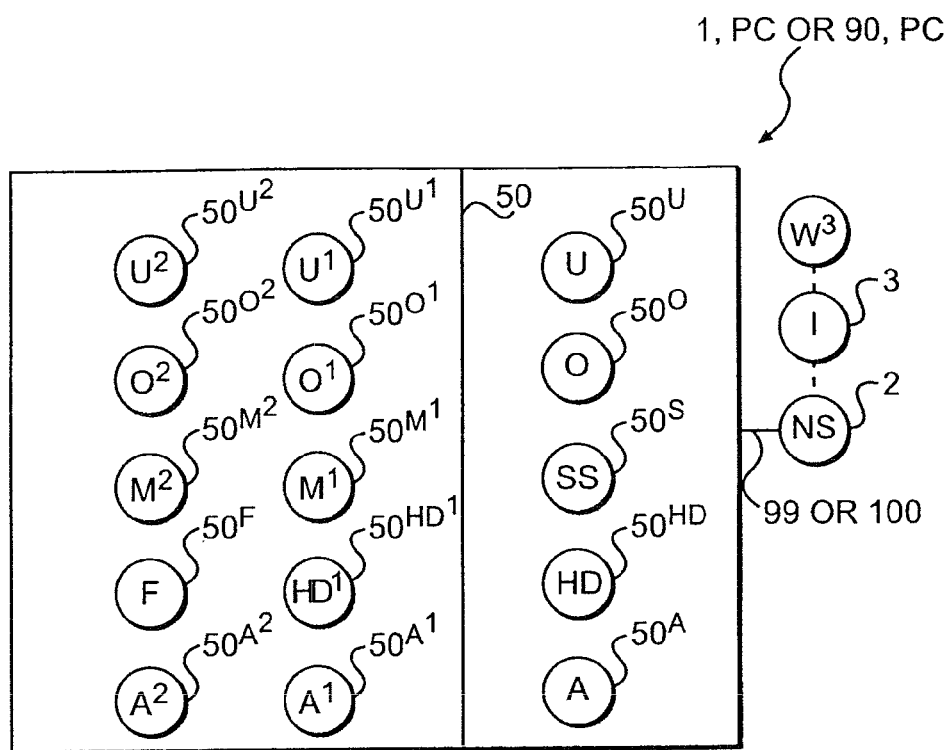


FIG. 15C

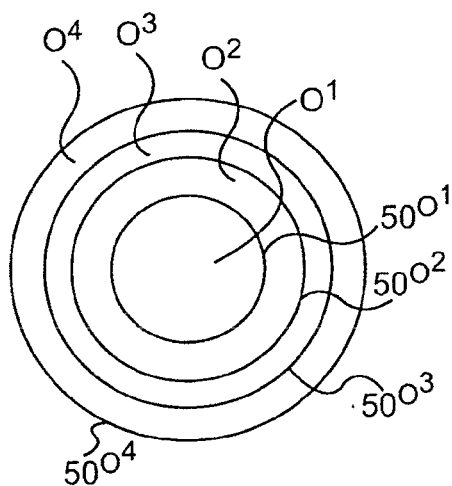


FIG. 15D

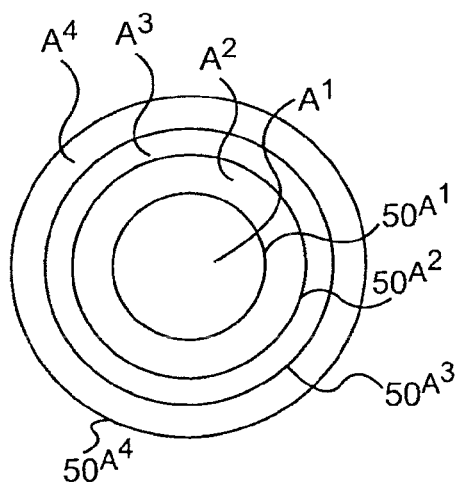
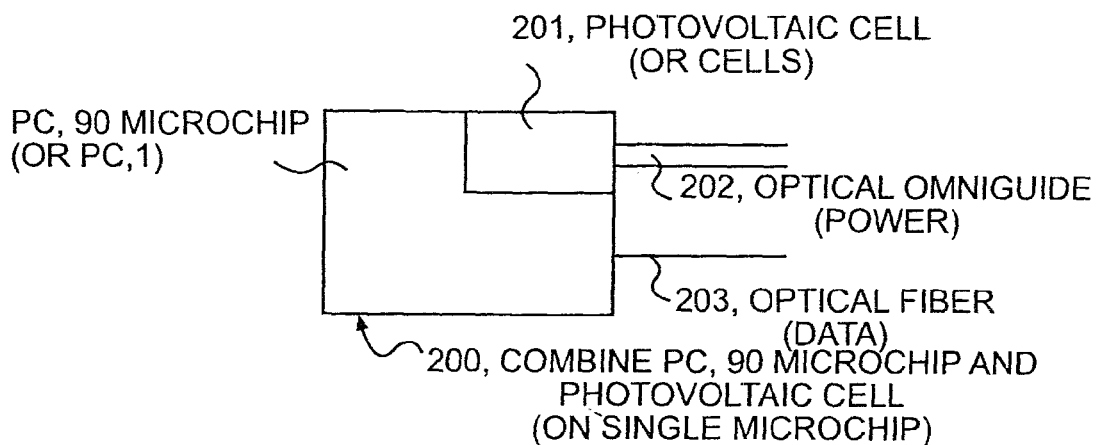
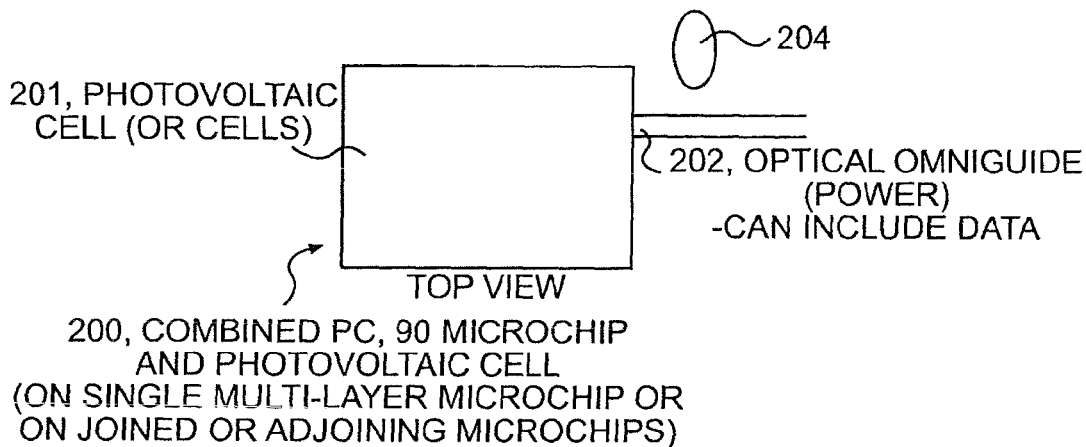
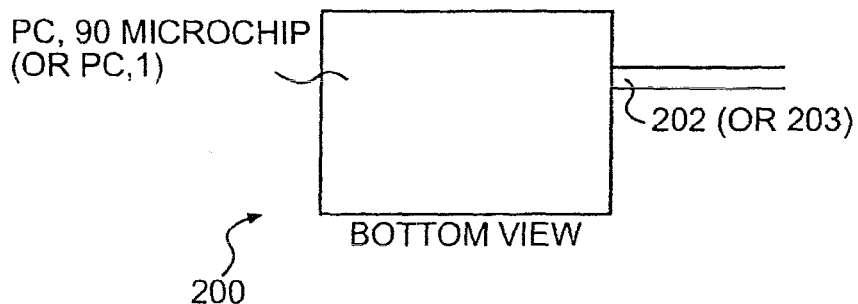


FIG. 15E

**FIG. 17A****FIG. 17B****FIG. 17C**

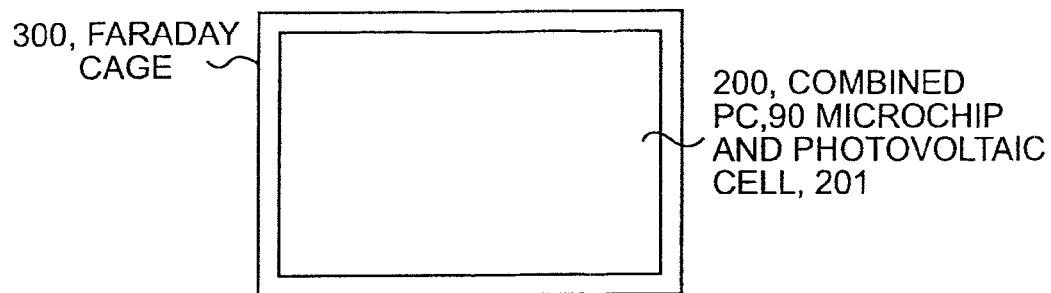


FIG. 18A

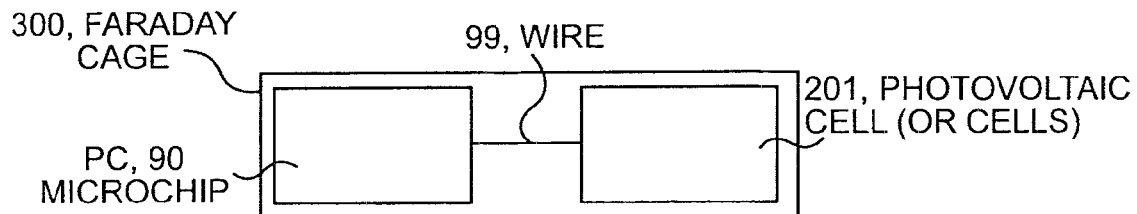


FIG. 18B

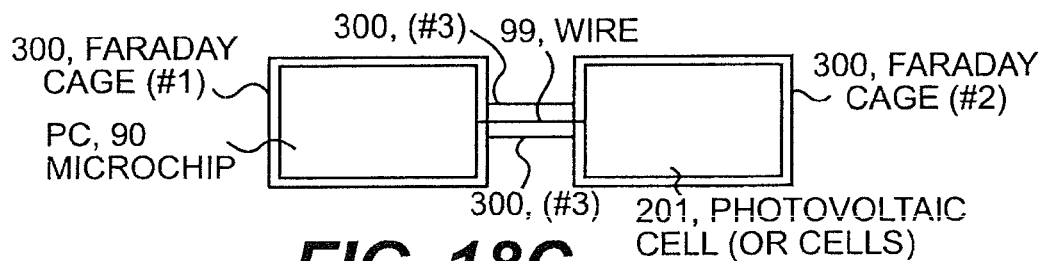


FIG. 18C

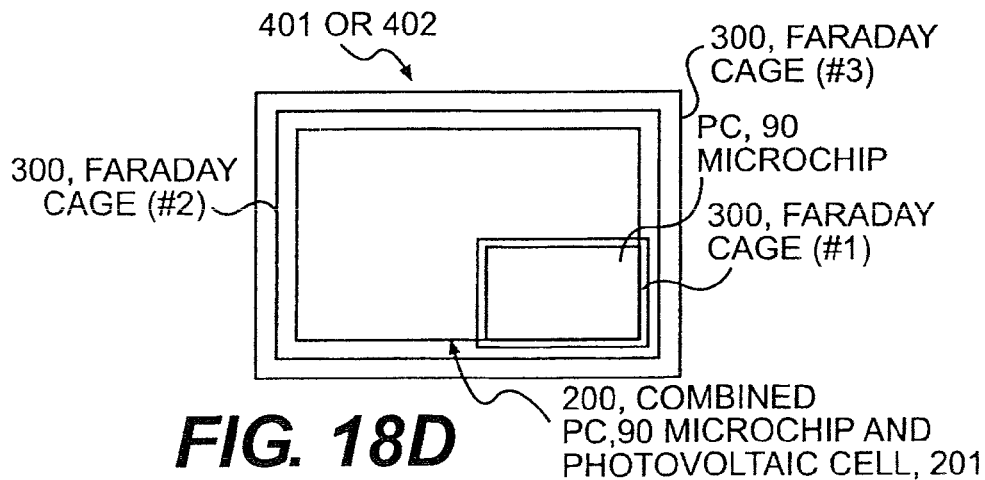


FIG. 18D

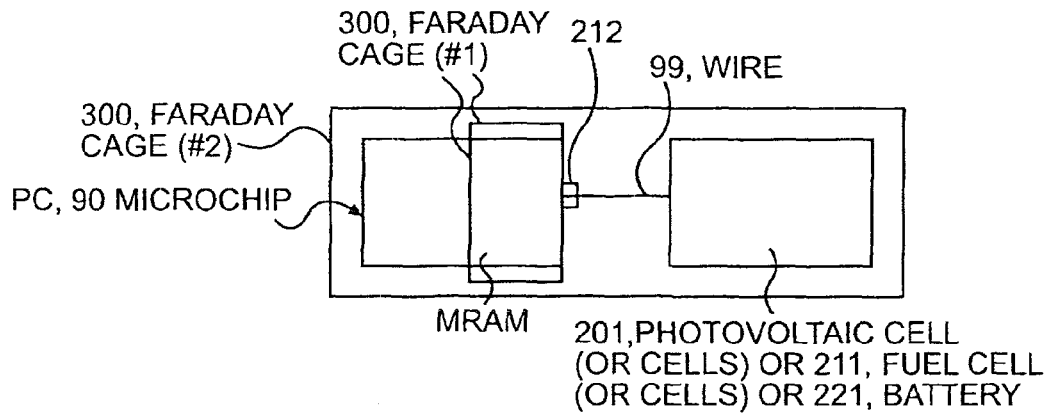


FIG. 18E

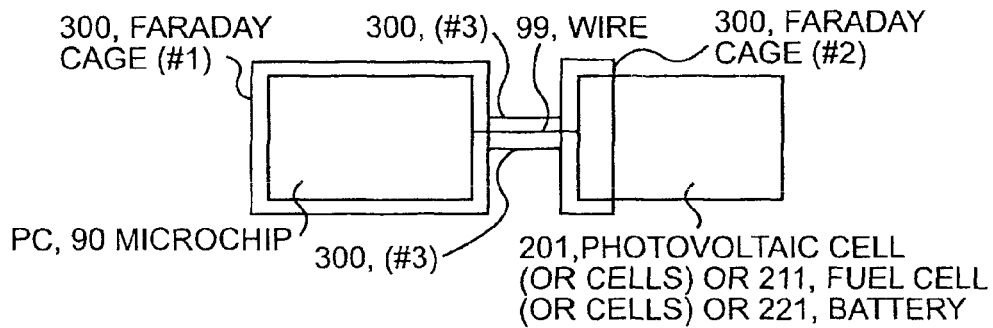


FIG. 18F

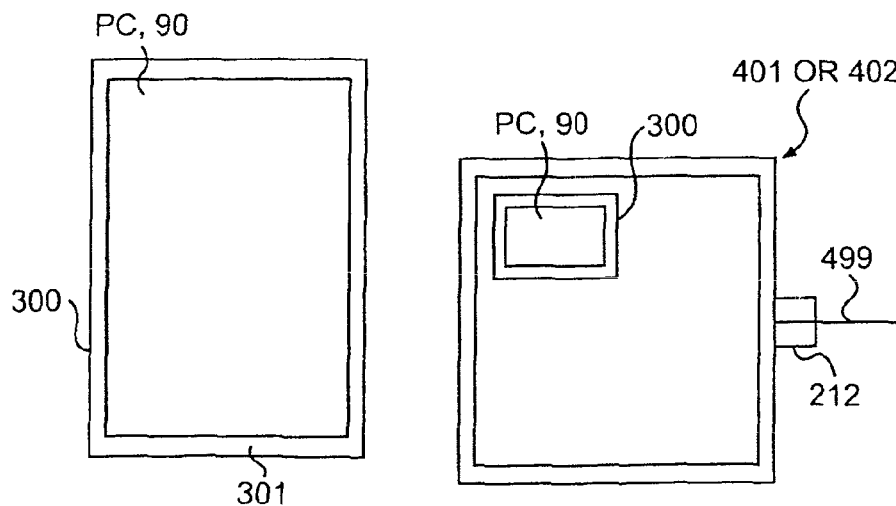
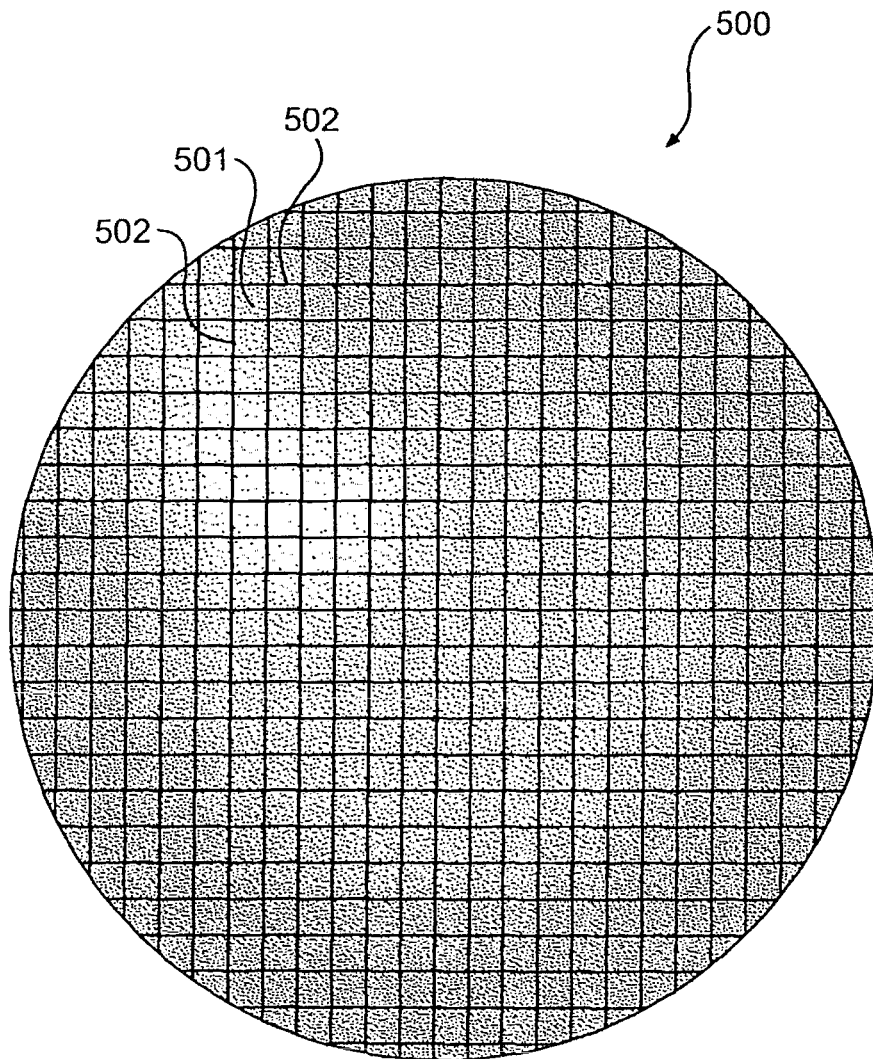


FIG. 18G

FIG. 18H

**FIG. 19**

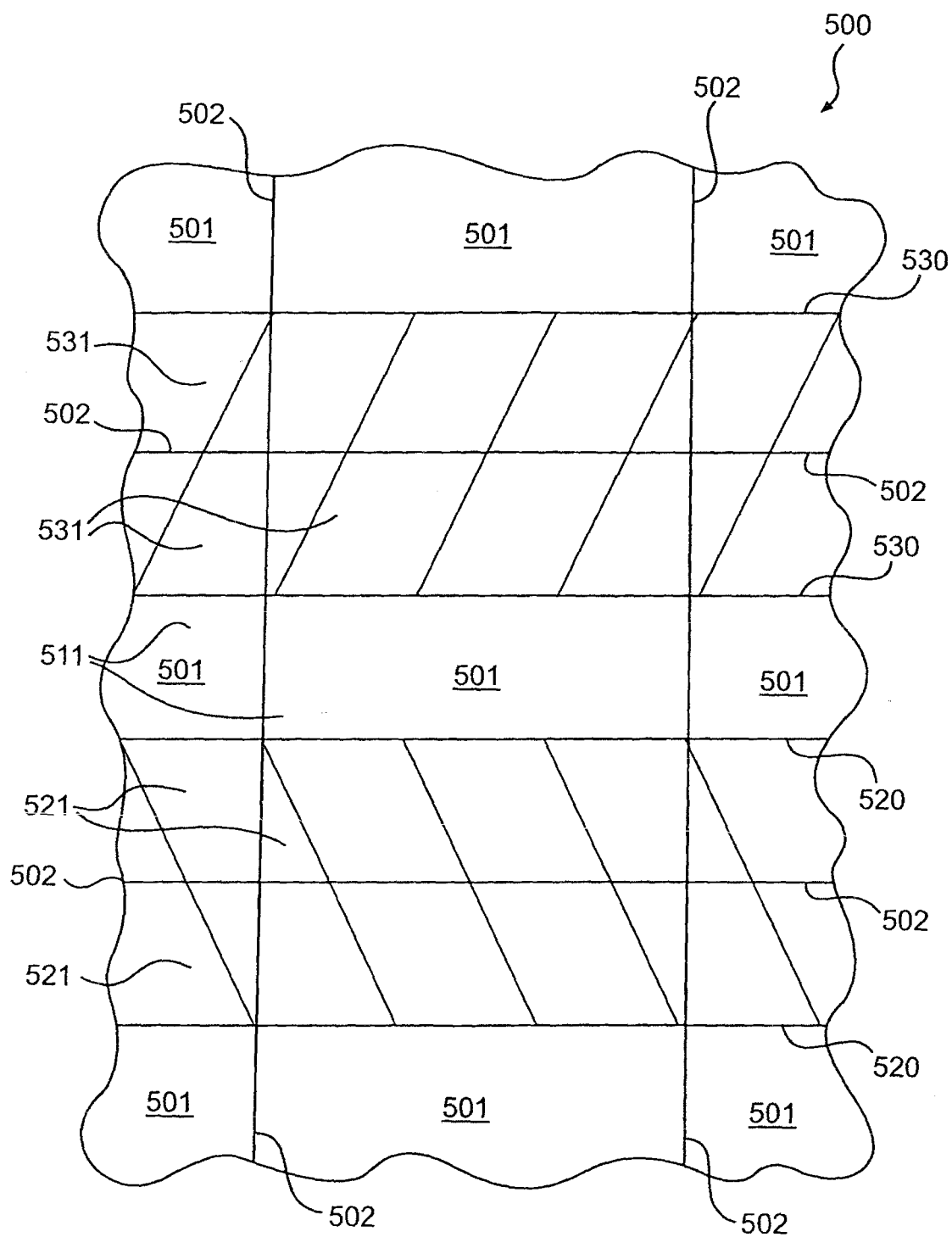


FIG. 20A

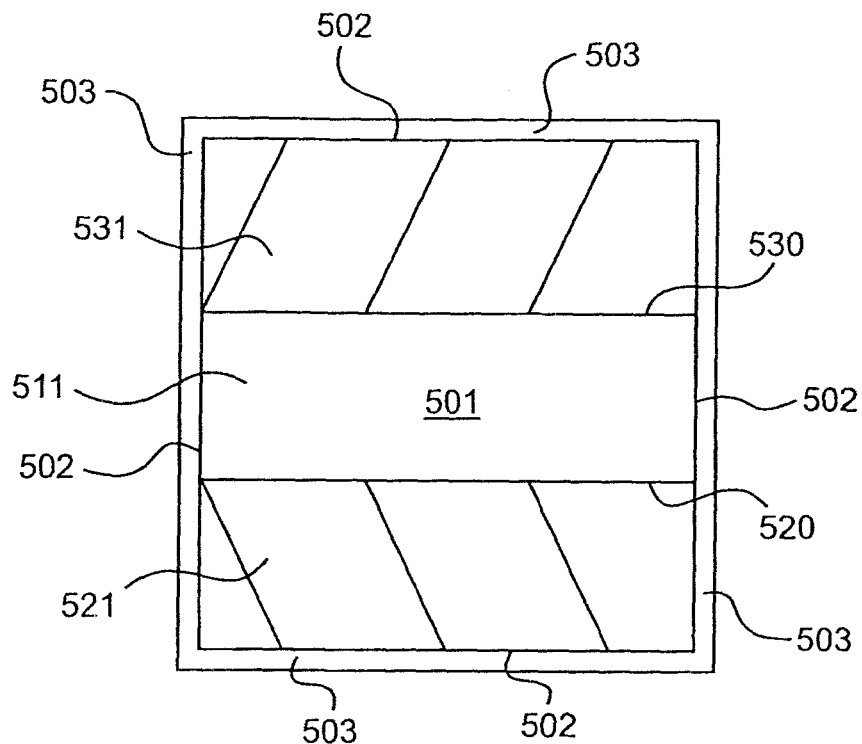


FIG. 20B

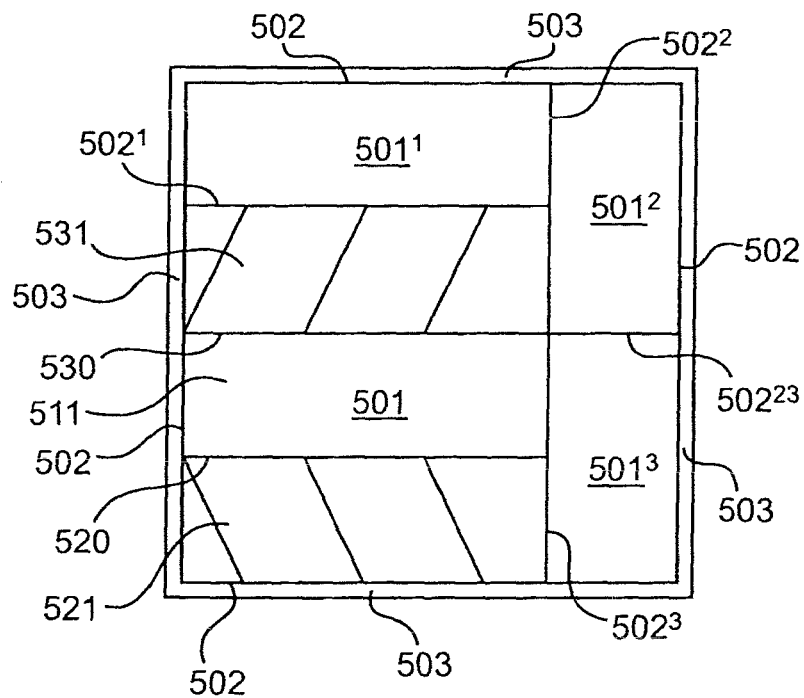


FIG. 22

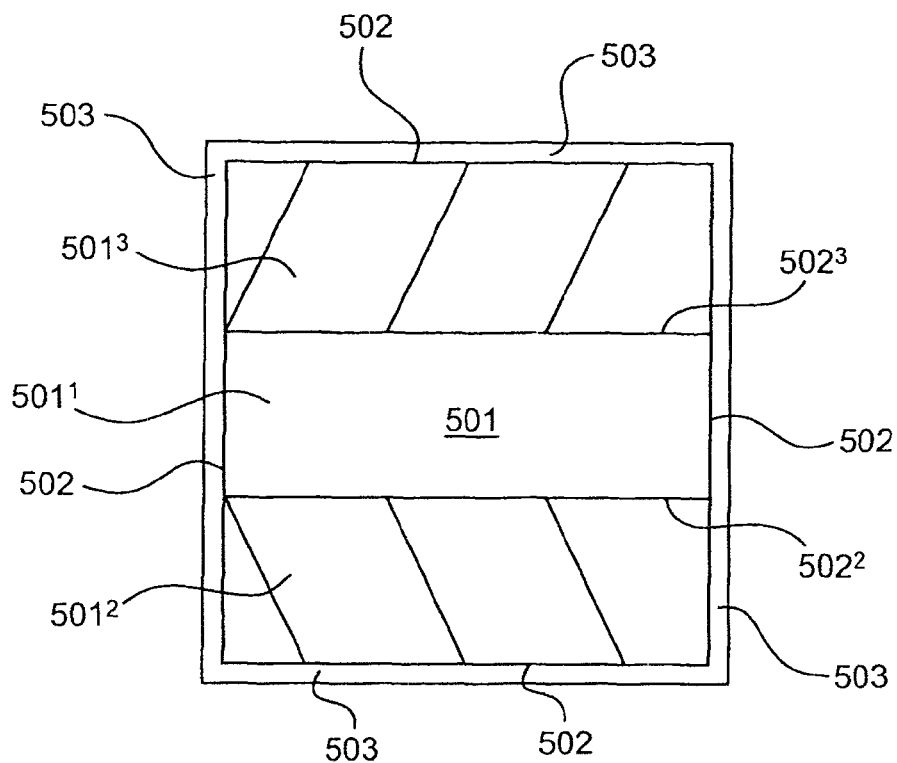


FIG. 21A

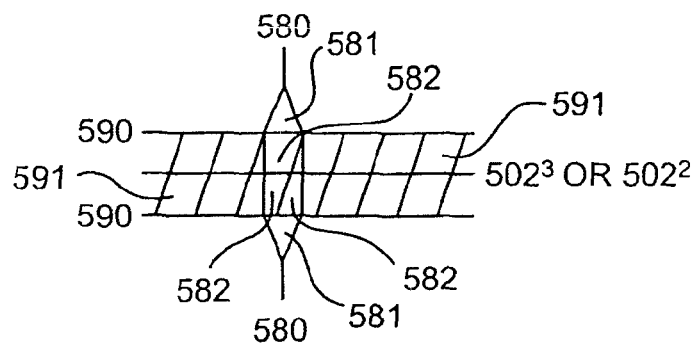


FIG. 21B



FIG. 21C

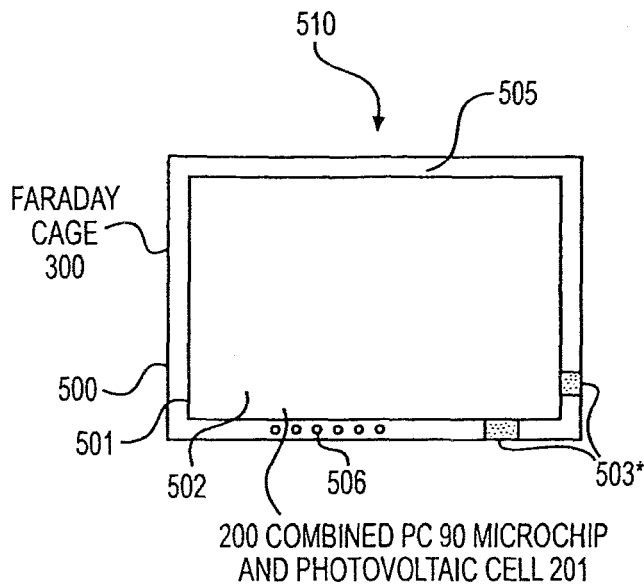


FIG. 23A

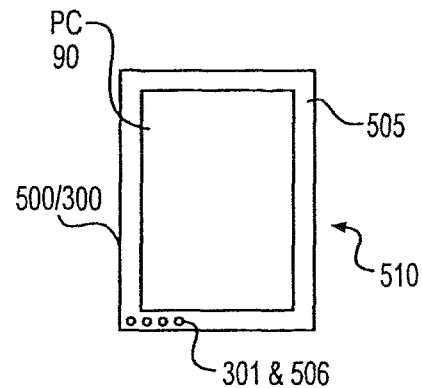


FIG. 23G

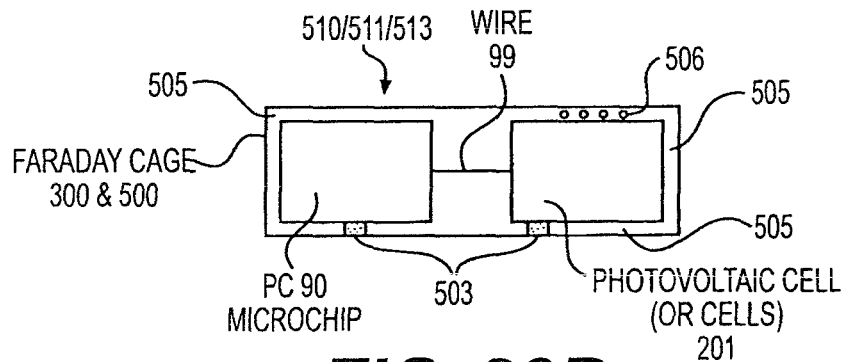


FIG. 23B

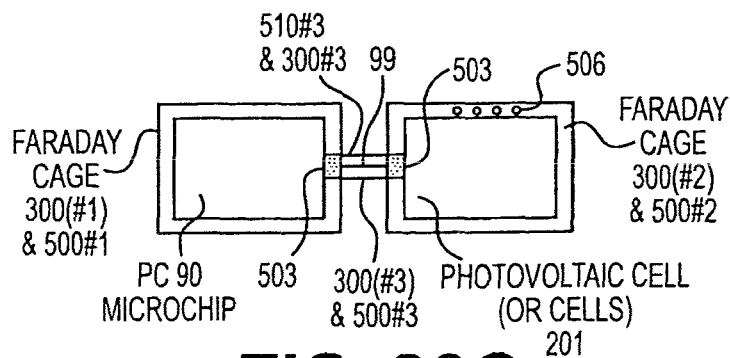
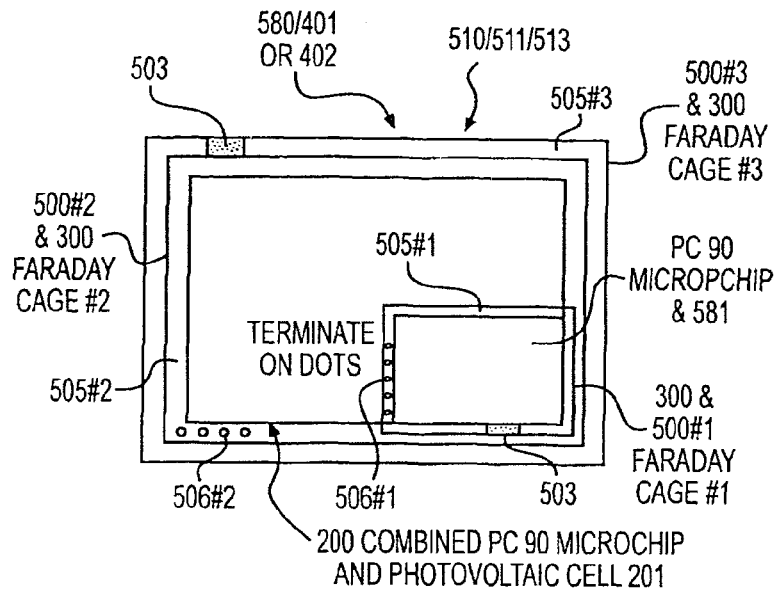
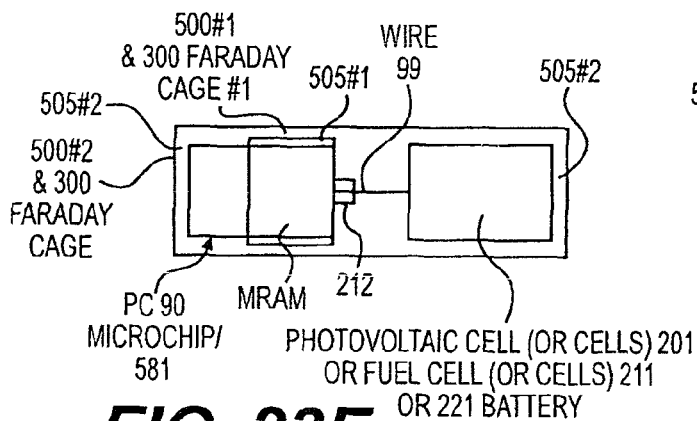
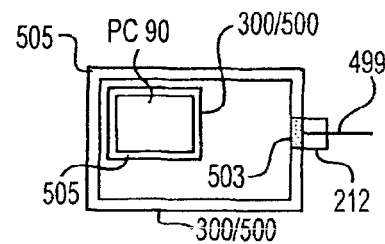
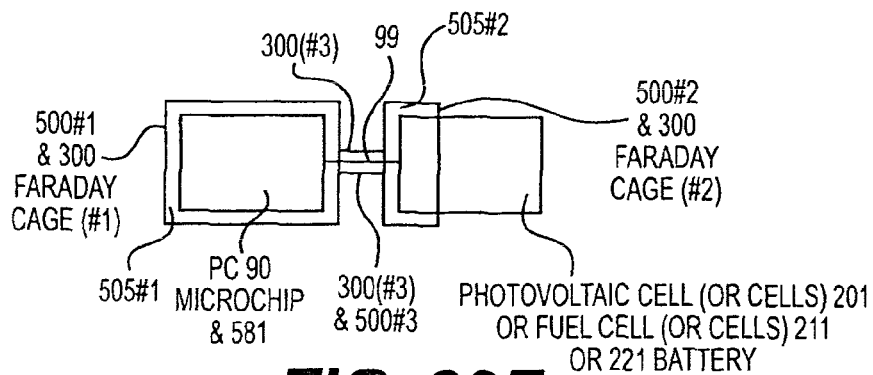


FIG. 23C

*CAN INCLUDE EXTERNAL CONNECTION
FROM MICROCHIP/CELL 200

**FIG. 23D****FIG. 23E****FIG. 23H****FIG. 23F**

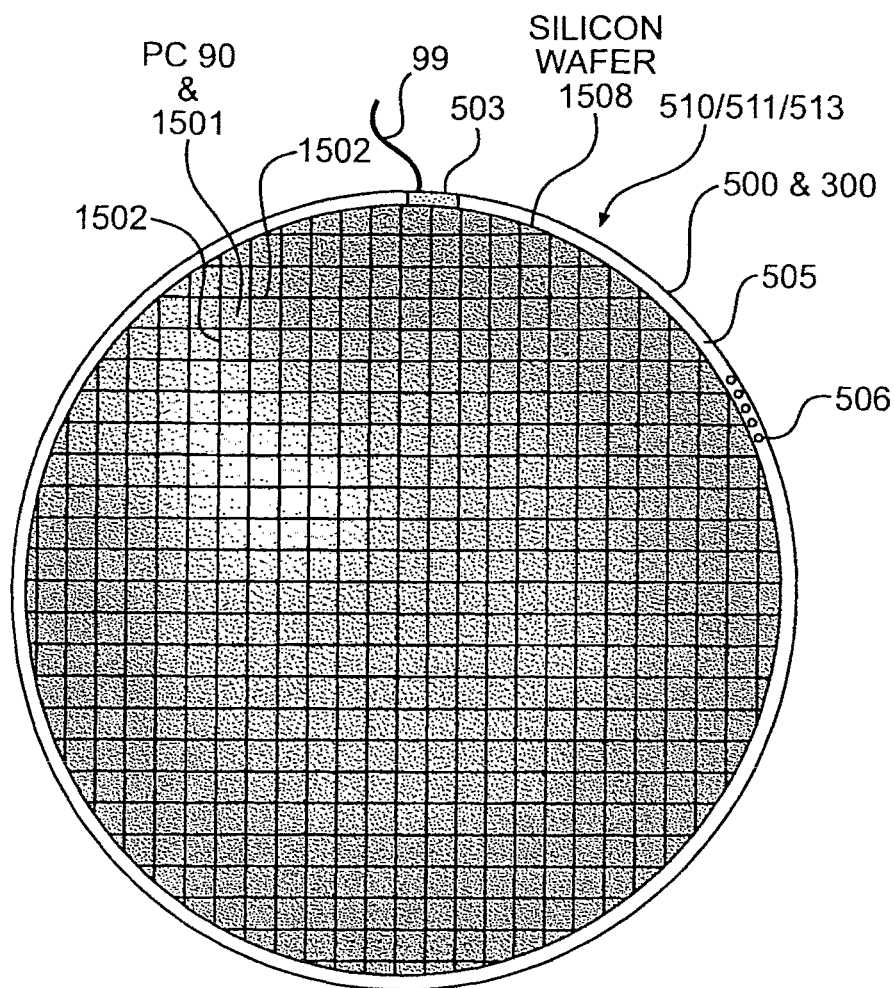


FIG. 24A

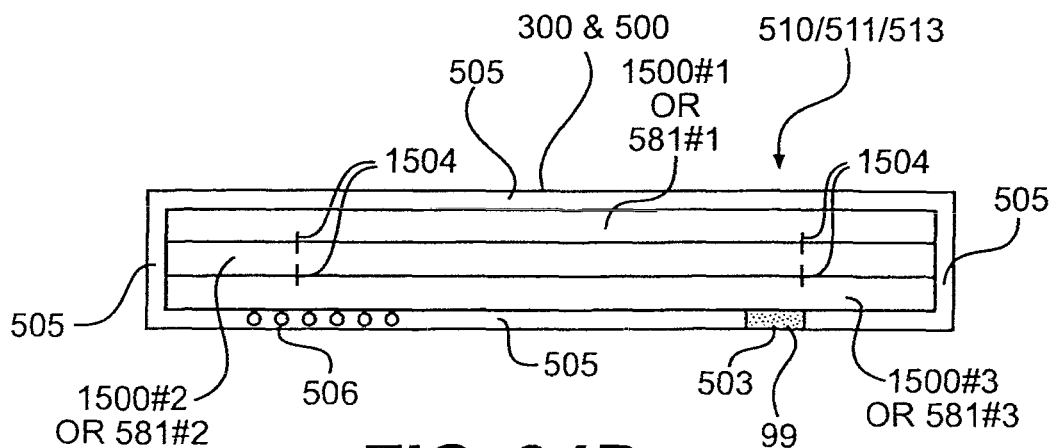


FIG. 24B

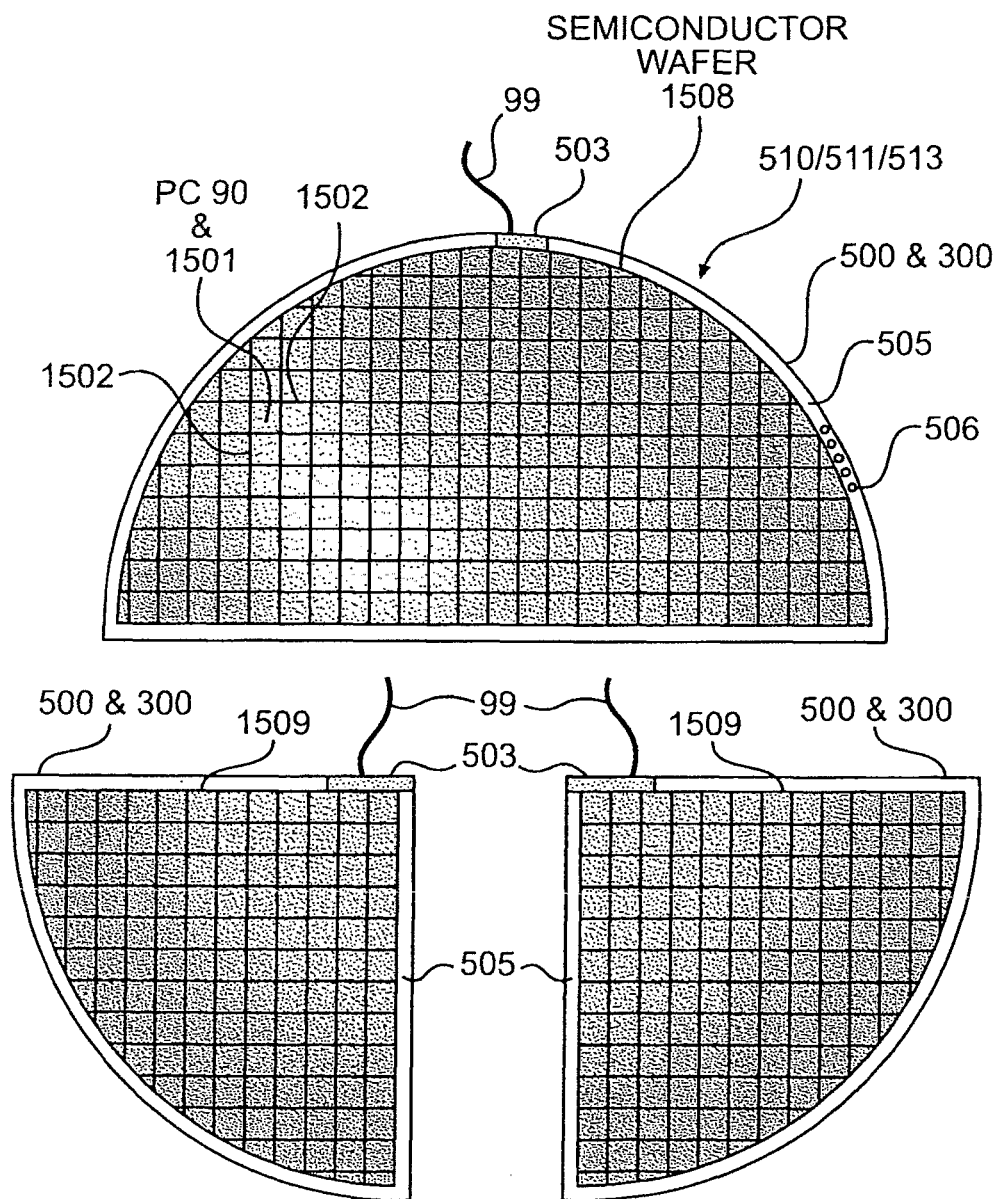


FIG. 24C

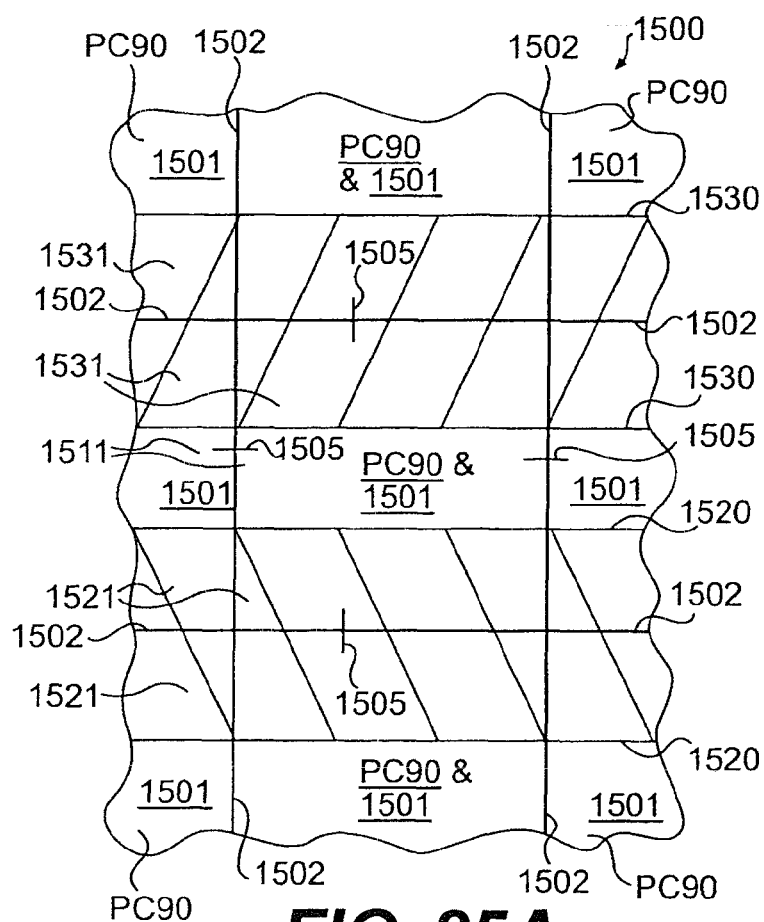


FIG. 25A

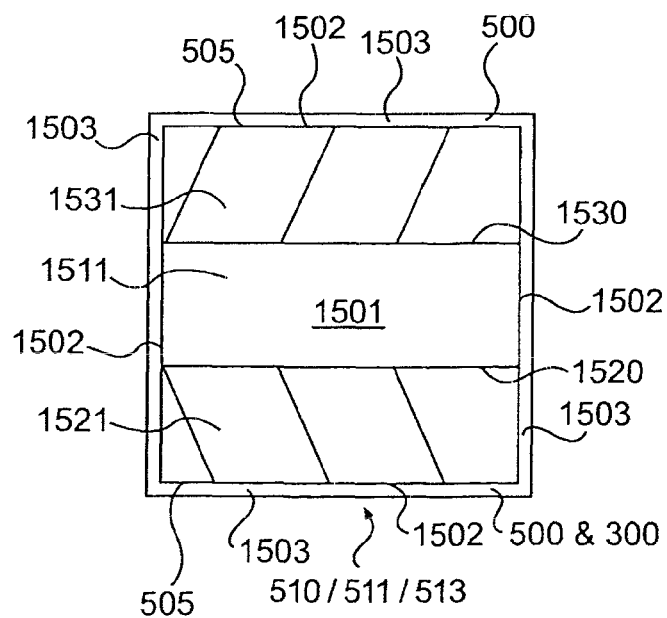


FIG. 25B

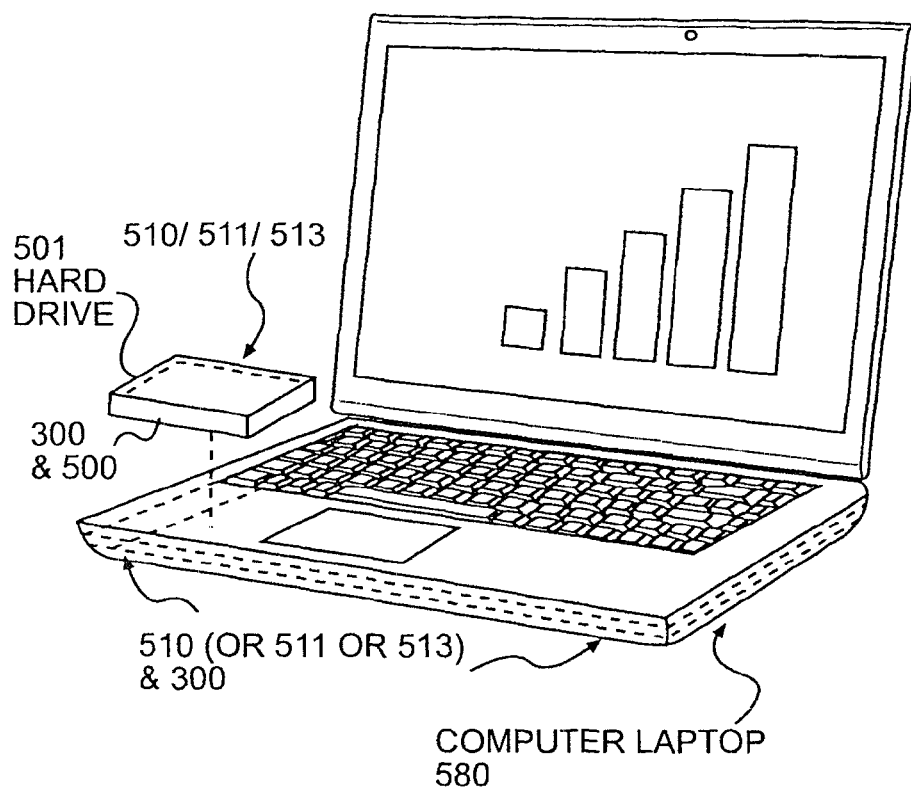


FIG. 26A

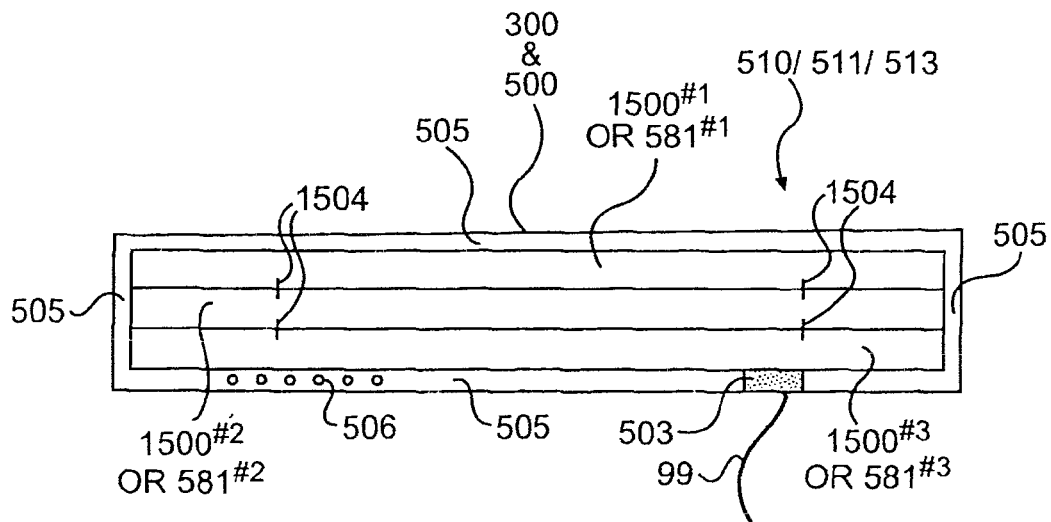


FIG. 26B

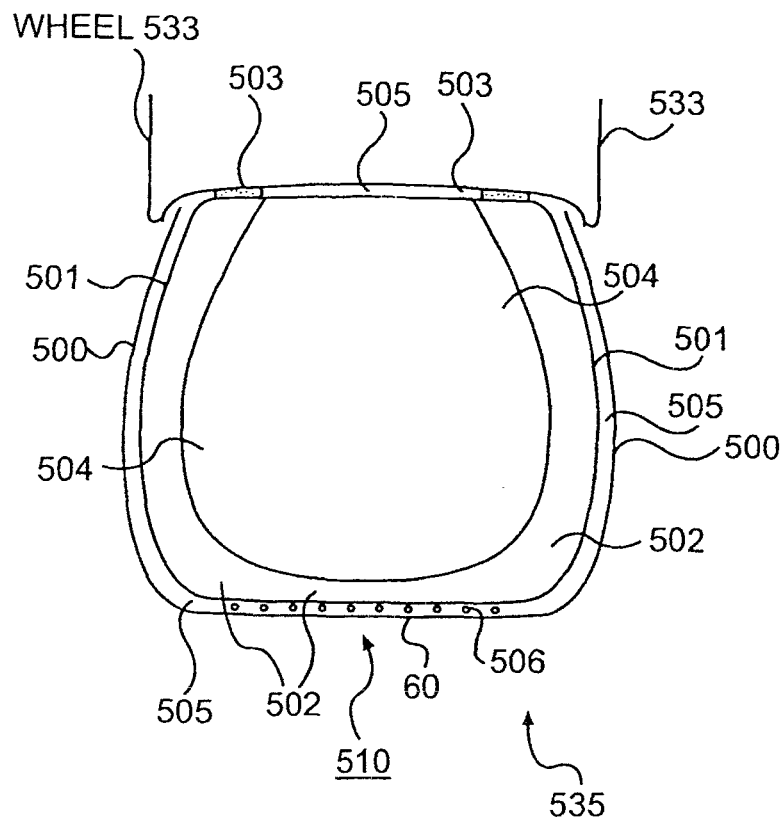


FIG. 27A

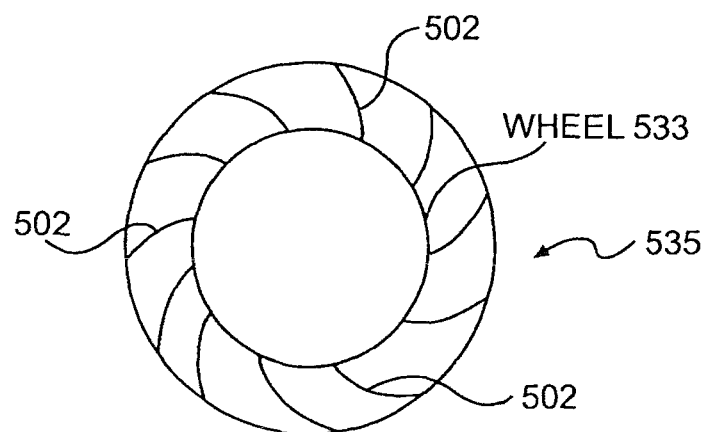


FIG. 27B

MICROCHIP WITH FARADAY CAGES AND INTERNAL FLEXIBILITY SIPIES

This application is a continuation of U.S. patent application Ser. No. 13/931,080 filed Jun. 28, 2013; which is a continuation of Ser. No. 13/769,292, filed Feb. 16, 2013, now abandoned; which is a continuation of U.S. patent application Ser. No. 12/292,769, filed Nov. 25, 2008, now U.S. Pat. No. 8,125,796, which is a continuation-in-part of U.S. patent application Ser. No. 12/292,553, filed Nov. 20, 2008, now U.S. Pat. No. 8,164,170, which is a non-provisional of U.S. Provisional Application No. 60/996,530, filed on Nov. 21, 2007, and U.S. Provisional Application No. 60/996,553, filed on Nov. 26, 2007. U.S. patent application Ser. No. 12/292,769, filed Nov. 25, 2008, is also a non-provisional of U.S. Provisional Application No. 60/996,553, filed on Nov. 26, 2007. The contents of all of the above-mentioned applications are hereby incorporated herein by reference.

BACKGROUND

1. Field of the Disclosure

The disclosure relates to a Faraday Cage surrounding semiconductor microchips, photovoltaic cells, and/or other micro and/or nano devices combined with the applicant's prior internal sipe inventions, including at least one compartment surrounded by at least one internal sipe, such as a slit. More specifically, a Faraday Cage coincides with an outer compartment, which forms one surface of a sipe surrounding an inner compartment including a personal computer microchip and/or a photovoltaic cell and/or a fuel cell and/or a battery.

The disclosure also relates to a semiconductor wafer, such as silicon, containing a multitude of microchips, such as with one or more core microprocessors. Instead of separating the microchips into separate dies in the conventional process, the entire semiconductor wafer is used essentially intact as a computer. More specifically, interconnects can be added to the printed circuit architecture of the wafer to connect the wafer microchips to other wafer microchips and/or other components on the wafer or external to it. Still more specifically, each microchip can be a complete system on a chip (SoC). Even more specifically, the semiconductor wafer can be used with other conventional interface devices for power and data, including wireless such as radio and/or optic, and/or wired such as fiber optic and/or electric, including for each SoC microchip on the wafer. In another specific embodiment, two or more semiconductor wafers can be stacked in vertical layers; for example, with a first wafer including microprocessors or cores; a second wafer including random access memory or RAM; and a third wafer including other components; each SoC microchip, in this example, can have one or more components on each of the three wafers.

The disclosure also relates to the semiconductor wafer including one or more of the applicant's prior internal sipe inventions, with an outer compartment having an internal sipe.

The disclosure also relates to the semiconductor wafer being surrounded by one or more Faraday Cages integrated into the internal sipe invention.

2. Brief Description of the Prior Art

Faraday Cage surrounding semiconductor microchips, photovoltaic cells, and/or other micro and/or nano devices, are described by the applicant in his U.S. application Ser.

No. 10/802,049 filed Mar. 17, 2004, and published as Pub. No. US 2004/0215931 A1 on Oct. 28, 2004.

The applicant's prior internal footwear sipe inventions, including at least one compartment surrounded by at least one internal sipe, such as a slit, are described by the applicant in his U.S. patent application Ser. No. 11/802,930, filed May 25, 2007 and published as Pub. No. US 2008/0086916 A1 on Apr. 17, 2008, as well as in several earlier U.S. applications filed by the applicant.

Existing semiconductor wafers, currently up to 300 mm in diameter, are always cut into a large number of separate dies, with one microchip formed into a package from each flaw-free die cut from the semiconductor wafer; some dies are inherently defective and are discarded. A typical semiconductor wafer is shown in the applicant's U.S. application Ser. No. 10/684,657 filed Oct. 15, 2003.

As described in Wikipedia, "wafer scale integration" (WSI) is a yet-unused system of building very-large integrated circuit networks that use an entire silicon wafer to produce a single "super-chip." Through a combination of large size and reduced packaging, WSI could lead to dramatically reduced costs for some systems, notably massively parallel supercomputers.

Many companies including TI and ITT attempted to develop "wafer scale integration" (WSI) production systems in the 1970's and '80's, but all failed and no products were released. Further attempts at WSI appear to be largely abandoned for decades.

One critical problem that has not been overcome is that of inherent flaws in semiconductor wafers. It has been an ongoing goal to develop methods to handle faulty areas of the wafers through logic, as opposed to sawing them out of the wafer. Generally, this approach uses a grid pattern of sub-circuits and "rewires" around the damaged areas using appropriate logic.

The overwhelming difficulty of this approach is illustrated by the history of famous computer pioneer Gene Amdahl, who attempted to develop WSI as a method of making a supercomputer, starting Trilogy Systems in 1980 and garnering investments from Groupe Bull, Sperry Rand, and Digital Equipment Corporation, who (along with others) provided as estimated \$230 million in financing. The design called for a 2.5" square chip with 1200 pins on the bottom. After burning through about one third of the capital with nothing to show for it, Amdahl eventually declared the idea would only work with a 99.99% yield, which would not happen for 100 years. There were several subsequent efforts in the 1980's, but none successful.

Another well-known problem is the field of wafer-scale integration technology is the impact of thermal expansion on external connectivity. More specifically, when a WSI microelectronic complex is connected to a circuit board by thousands of, for example, connectors positioned between the microelectronic complex and the circuit board, these connectors can be damaged due to the different rates of thermal expansion experience by the surfaces of the microelectronic complex and circuit board.

Taking for example a finished silicon wafer, packaged in a material such as ceramic, the wafer typically expands at a rate of 3 ppm/C. In contrast, the material of the circuit board typically expands at a rate of 20-40 ppm/C. Thus, as the two materials heat up, the two surfaces will expand at different rates, potentially damaging many of the connectors distributed between the wafer and the circuit board. See Norman, et al., U.S. Pat. No. 7,279,787.

The present disclosure solves these longstanding problems with existing technology.

SUMMARY

In the applicant's disclosure regarding Faraday Cages and siped compartments, two entirely different technologies (siped cushioning compartments and Faraday Cages) from two entirely different and unrelated fields of technology ("macro" athletic footwear and microelectronics), each with their own completely different art, have been integrated into a surprising new combination wherein the integrated structural components have simultaneous dual functions that are completely separate and unrelated.

Moreover, the integrated use of a sipe media both as a lubricant between the siped compartments and as a cooling media for a microchip (or other electronic component) to deal with the critical microprocessor heat dissipation problem is an additional surprising combination of two entirely separate technical functions that are completely different into a single new structural and material component with dual functions, again from entirely different and unrelated fields of art.

The applicant's semiconductor wafer computer disclosure solves the two longstanding problems that have heretofore made all such wafer-scale integration efforts fail.

First, the applicant's disclosure solves the longstanding inherent flaw intolerance problem discussed above by using microchips on the wafer that are personal computer systems on the chip (SoC), so that each wafer microchip can operate independently, including communicating with other wafer microchips, such as wirelessly in free space and/or waveguides by radio or optical device. In the applicant's approach, the wafer microchips that inherently are flawed during manufacture or fail in operation require no elaborate special handling like that described above. Generally, if they fail fatally, the remaining wafer microchips inherently ignore them since they do not communicate with other microchips; also, partial failure can be detected remotely and the microchip can be turned off, if necessary.

Second, the applicant's disclosure solves the longstanding unequal thermal expansion problem discussed above by reducing, even almost eliminating the need for pins, since both on wafer and off wafer communication by the independently functioning microchips (personal computer SoC's) on the wafer can be accomplished by radio or optics connections that do not require pins or require far fewer structural connections. In addition, the applicant's use of stacked semiconductor wafers, which can have very similar or identical materials and expansion rates, can obviate the need for conventional motherboards or reduce their role.

Moreover, since the personal computer microchips on the semiconductor wafer can operate independently, such as in conventional clusters of several or many or all of the personal computers on the wafer (or including personal computers off the wafer as well), the operations occurring on the wafer can be asynchronous. The applicant's disclosure thereby also solves the longstanding problem of synchronizing such wafer scale operations by obviating the need for synchronizing them, as would be necessary without the applicant's disclosure.

These and other features of the disclosure will become apparent from the detailed description that follows.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1-9 of this application are FIGS. 1-6, 52, 61, 67, 74, and 79 of both the applicant's U.S. application Ser. No. 11/802,033 filed May 18, 2007 and published on Apr. 10, 2008 as Pub. No. US 2008/0083140 A1 and PCT Applica-

tion PCT/US2007/011976 filed 18 May 2007 and published on 31 Jan. 2008 as Int. Pub. No. WO 2008/013594 A2, as well as U.S. application Ser. No. 11/802,930 filed May 25, 2007 and published on Apr. 17, 2008 as Pub. No. US 2008/0086916 A1; all three applications are incorporated herein by reference. In addition, FIGS. 1-6 (except 6D-6F) of this application are FIGS. 83-88 of both the applicant's U.S. application Ser. No. 11/282,665 filed Nov. 21, 2005 and published on Nov. 9, 2006 as Pub. No. US 2006/0248749 A1 and PCT Application PCT/US2005/042341 filed 21 Nov. 2005 and published on 1 Jun. 2006 as Int. Pub. No. WO 2006/058013 A2, both of which are incorporated herein by reference.

FIG. 1A-4A show a frontal or sagittal plane cross section view of an example of a device **510** such as a flexible insert with a siped compartment or chamber or bladder.

FIGS. 1B-6B shows a horizontal plane view of a device **510** example.

FIG. 1C is an exploded perspective view showing the relationship of an insert device **510**, which has a siped compartment or chamber or bladder, with a midsole, bottom sole, and upper of a shoe or other footwear.

FIGS. 1D-1F show an example of the **510** invention using typical commercial air bladder embodiments as compartment **501** modified with outer compartment **500**.

FIGS. 5A-6A show a frontal or sagittal plane cross section view of an example of a device **510** such as a flexible insert with two siped compartments or chambers or bladders or combination.

FIG. 7A shows a computer laptop with the **510/511/513** inventions and FIG. 7B shows a semiconductor microchip with the **510/511/513** inventions.

FIGS. 7C-7D show additional examples of the **510/511/513** inventions applied to electronic game controllers and cell phone.

FIG. 8 is a chart showing approximate comparison of hardness scales.

FIGS. 9A-9C show a large urban telephone book lying flat, solid wooden block, and an upright very thick book to illustrate basic concepts of sipe flexibility and stability.

FIGS. 10-22 of this application are FIGS. 10A-10I, 16A-16Z, 16AA, and 16AB, 17A-17D, 21B, 22A, 23A-23E, 25A-25D, 26A-26C, 27A-27H, 28, 29A, 30A-30C, and 31 of both the applicant's U.S. application Ser. No. 10/802,049 filed May 17, 2004 and published on Oct. 28, 2004 as Pub. No. US 2004/0215931 A1.

FIGS. 10A-10I are simplified diagrams of a section of a computer network, such as the Internet, showing an embodiment of a system architecture utilizing an internal firewall to separate that part of a networked PC (including a system reduced in size to a microchip) that is accessible to the network for shared processing from a part that is kept accessible only to the PC user; also showing the alternating role that each PC in the network may play as either a master or slave in a shared processing operation involving one or more slave PC's in the network; and showing a home or business network system which can be configured as an Intranet; in addition, showing PC and PC microchips controlled by a controller (including remote) with limited or no processing capability; and showing PC and PC microchips in which an internal firewall **50** can be reconfigured by a PC user.

FIGS. 11A-11K show a new hierarchical network architecture for personal computers and/or microprocessors based on subdivision of parallel processing or multi-tasking operations through a number of levels down to a processing level.

FIGS. 12A-12D show an internal firewall **50** with a dual function, including that of protecting Internet users (and/or other network users sharing use) of one or more slave personal computers PC **1** or microprocessors **40** from unauthorized surveillance or intervention by an owner/operator of those slave processors.

FIG. 13 shows that all microprocessors of a personal computer or personal computer on a microchip can have a separate input/output communication link to a digital signal processor (DSP) or other transmission/reception connection component.

FIGS. 14A-C show a PC microprocessor on a microchip similar to that of FIG. 13, except that FIGS. 14A-C show microprocessors **93** and **94** each connecting to an optical wired connection **99'** such as thin mirrored hollow wire or optical omniguide or optical fiber.

FIGS. 15A-E show multiple firewalls **50** within a personal computer **1** or PC microchip **90**.

FIGS. 16A-D show the use for security of power interruption or data overwrite of volatile memory like DRAM and non-volatile memory like Flash or MRAM (or ovonics), respectively, of the network portion of a personal computer PC1 or system on a microchip PC90.

FIGS. 17A-C show exemplary microchip and photovoltaic cell embodiments.

FIGS. 18A-H show exemplary microchip and Faraday Cage embodiments.

FIG. 19 shows a silicon wafer **500** used to make microchips.

FIG. 20A shows a top view of a microchip **501** surrounded by adjoining portions of adjoining microchips **501** in a section of the silicon wafer **500**. FIG. 20B shows a top view of the microchip **501** embodiment of FIG. 20A after the die has been separated from the silicon wafer **500** and positioned in a microchip package **503**.

FIGS. 21A-C show alternative embodiments that unite separate fabrication processes on the same microchip **501**.

FIG. 22 shows a combination of the embodiments shown in FIGS. 20 and 21.

FIGS. 23A-23H are modifications of FIGS. 18A-27H of this application, which are FIGS. 27A-27H of U.S. application Ser. No. 10/802,049 filed Mar. 17, 2004 and published on Oct. 28, 2004, as Pub. No. US 2004/0215931 A1 and U.S. application Ser. No. 10/684,657 filed Oct. 15, 2003 and published on Aug. 18, 2005, as Pub. No. US 2005/0180095 A1, both of which applications are hereby incorporated by reference herein for completeness of disclosure.

FIGS. 23A-23H are more specific example embodiments of the FIG. 7B example; they show examples of the applicant's inventions involving one or more Faraday Cages surrounding various combinations of semiconductor microchips, photovoltaic cells, and/or other micro and/or nano devices with the applicant's internal sipe inventions **510/511/513**.

FIGS. 24A and 25A-25B are modifications of FIGS. 28 and 29A-29B of U.S. application Ser. No. 10/684,657 filed Oct. 15, 2003 and published on Aug. 18, 2005, as Pub. No. US 2005/0180095 A1, which was above incorporated by reference herein for completeness of disclosure.

FIG. 24A is a top view of a semiconductor wafer **1500**, the entire semiconductor wafer **1500** being used essentially intact as a computer, which can also include a Faraday Cage and a siped compartment. FIG. 24B is a side cross section of three stacked wafers **1500**. FIG. 24C is another top view of a wafer **1500**, but subdivided into smaller components, such as half sections and quarter sections.

FIGS. 25A-25B are modifications of FIGS. 29A-29B of the '657 application incorporated by reference herein above. FIG. 25B shows a microchip **1501** as a separated die in a package **1503** including the applicant's internal sipe inventions **510/511/513** and the Faraday Cage **300**.

FIGS. 26A-26B is FIGS. 7A-7B with the addition of Faraday Cages, as well as stacked dies in FIG. 7B.

FIGS. 27A-27B are like FIGS. 28A-28B of the '033 and '930 Applications incorporated above shows in cross-section an example of a tire **535**, such as for a wheel **533** of a transportation vehicle, with a device **510**.

It is understood that the reference numbers identified in this application and in U.S. Patent Applications '665, '033, and '930 incorporated by reference herein, are used consistently throughout the application such that like reference numbers are used to describe the same or like elements.

DETAILED DESCRIPTION

FIGS. 1-9 of this application are FIGS. 1-6, 52, 61, 67, 74, and 79 of both the applicant's U.S. application Ser. No. 11/802,033 filed May 18, 2007 and published on Apr. 10, 2008 as Pub. No. US 2008/0083140 A1 and PCT Application PCT/US2007/011976 filed 18 May 2007 and published on 31 Jan. 2008 as Int. Pub. No. WO 2008/013594 A2, as well as U.S. application Ser. No. 11/802,930 filed May 25, 2007 and published on Apr. 17, 2008 as Pub. No. US 2008/0086916 A1; all three applications are incorporated herein by reference. In addition, FIGS. 1-6 (except 6D-6F) of this application are FIGS. 83-88 of both the applicant's U.S. application Ser. No. 11/282,665 filed Nov. 21, 2005 and published on Nov. 9, 2006 as Pub. No. US 2006/0248749 A1 and PCT Application PCT/US2005/042341 filed 21 Nov. 2005 and published on 1 Jun. 2006 as Int. Pub. No. WO 2006/058013 A2, both of which are incorporated herein by reference.

FIGS. 1-9 show the applicant's prior inventions incorporating forms of insertable devices with one or more internal (or mostly internal) sipes, including slits (or channels or grooves and other shape, including geometrically regular or non-regular shapes, such as anthropomorphic shapes), into a large variety of products, including footwear and orthotics, athletic, occupational and medical equipment and apparel, padding for equipment and furniture, balls, tires and any other structural or support elements in a mechanical, architectural or any other device.

New reference numerals used in the FIGS. 1-79 are further defined as follows:

Ref. No **500**: An outer compartment, such as an outer compartment **161** or chamber **188** or bladder, at least partially or mostly or entirely enclosing a space within the outer compartment/chamber/bladder **500**, which can be located anywhere in a footwear sole or upper or both or other article described in this application. Construction and materials can be, as one embodiment example, simpler in shape but otherwise similar to those used in any commercial samples of Nike Air™.

Ref. No **501**: An inner compartment, such as an inner compartment **161** or chamber **188** or bladder, is located inside the enclosed space of the outer compartment/chamber/bladder **500**. Construction and materials of the inner compartment/chamber/bladder **501** can be, as one embodiment example, like those used in any commercial samples of gas bladders in Nike Air™.

Ref. No. **502**: Structural element that is optional anywhere within either outer compartment/chamber/bladder **500** or inner compartment/chamber/bladder **501**, of which a **501**

embodiment is shown; any flexible, resilient material can be used, including structures molded into the shape of (and using the material of) the compartment/chamber/bladder **500** or **501**, as is very common in the art, such as many commercial samples of gas bladders used in Nike Air™, as well as foamed plastic or plastic composite or other materials, like Nike Shox™ or Impax™ or Reebok DMX™ or New Balance N-ergy™. In addition, other materials can be used directly within a **501/500** compartment or can be connected to or through a **501/500** compartment, as in the cushioning components of the shoe sole heel of commercial samples of Adidas 1™, including electromechanical, electronic, and other components. Some devices may benefit from the use of rigid or semi-rigid materials for part or all of a media within a compartment.

Ref. No. **503**: Attachment of two compartment/chambers/bladders **500/501**, including particularly attachment of outer **500** to inner **501**; any practical number of attachments of any form using any materials can be used, including glue.

Ref. No. **504**: Media contained within all or part of compartment/chamber/bladder **500** or **501**, particularly **501**, can be any useful material, such as gas (including, as an example, gas used in Nike Air™ or ambient air, liquid or fluid, gel, or foam (such as a plastic like PU or EVA or equivalent or rubber (natural or synthetic) or combination of two or more; encapsulation of foam is optional); material particles or coatings, such as dry coatings like Teflon™ can also be used. An optional element in an outer compartment/chamber **500** (or an inner compartment/chamber **501** that itself contains an inner compartment/chamber, as in FIG. **87**).

Ref. No. **505**: Internal sipe or slit or channel or groove for flexibility, such as between inner and outer compartment/chamber **500/501** (or bladder) surfaces, as one embodiment example; such surfaces can be substantially parallel and directly contact in one useful embodiment example, but are not attached so that at least parts of the two surfaces can move relative to each other, such as to facilitate a sliding motion between surfaces; the surfaces can be in other useful forms that allow portions of the surfaces to be proximate to each other but not contacting in an unloaded condition or in a partially loaded condition or in a maximally loaded condition.

Ref. No. **506**: Media of internal sipe **505**; media **506** can be any useful material like those used in media **504**; media **506** can be located in part or all of **505** to decrease (or increase) sliding resistance between **500/501** or **505** surfaces, for example, to lubricate the surfaces with any suitable material; silicone or Teflon™ can be used, for example; an optional element.

Ref. No. **507**: Metal particles.

Ref. No. **508**: Shock absorbing fluid containing **507**; a magnetorheological fluid.

Ref. No. **509**: Electromagnetic field-creating circuit.

Ref. No. **510**: A flexible insert or component including siped compartments **161** or chambers **188** or bladders used for example as outer and inner compartments/chambers/bladders **500/501** for footwear soles or orthotics or uppers or other uses; a useful embodiment being two or more compartment or chambers (or bladders) **161/188** (or mix) that are separated at least in part by an internal sipe **505**, including the example of at least one **501** (either **161/188** or bladder) inside at least one **500** (either **161/188** or bladder) and being separated by an internal sipe **505**.

Ref. No. **580**: Computer or computer component.

Ref. No. **581**: Micro-technology and nanotechnology devices, including a semiconductor microchip device (a part of a die or an entire die or multiple dies assembled together) microelectromechanical systems (MEMS), field-programmable gate arrays (FPGA's) and faraday cages, photovoltaic cells, fuel cells, batteries, and including devices assembled at the molecular or atomic scale.

FIGS. **1-7** show, as numeral **510**, examples of a device or flexible insert including siped compartments **161** or chambers **188** or bladders (another term used in the art) for use in any footwear soles, including conventional soles **22** or the applicant's prior inventions, including footwear/shoe soles **28** and midsole inserts **145** as described in applicant's published '087 Application and Ser. No. 11/282,665 U.S. Application published Nov. 9, 2006, as Pub. No. US 2006/0248749 A1 incorporated by reference, or for orthotics **145** as described in the applicant's published '034 U.S. Application, '869 Patent, and WO 02/09547 WIPO publication, as well as to be published by WIPO PCT Application Number PCT/US2005/042341, all incorporated by reference herein, including for uppers for footwear or orthotics (or including uppers), or for other flexibility uses in athletic equipment like helmets and apparel including protective padding and guards, as well as medical protective equipment and apparel, and other uses, such as protective flooring, improved furniture cushioning, balls and tires for wheels, and other uses.

The device or flexible insert with siped compartments or chambers **510** include embodiments like two or more of either compartments **161** or chambers **188** or bladders (or a mix including two or more of a compartment, a chamber, and a bladder) that are separated at least in part or in several parts or mostly or fully by an internal sipe **505**. The flexible insert **510** can be inserted during assembly of an article by a maker or manufacturer or is insertable by a user or wearer (into an article like a shoe, for example, as part of a removable midsole insert **145** described above), or integrated into the construction of a device as one or more components.

Siped compartments or chambers **510** include example embodiments such as FIGS. **1-7**, which generally show at least one inner compartment **161** or chamber **188** inside at least one other outer compartment **161** or chamber **161**; and the two compartments/chambers **161/188** being separated by an internal sipe **505**.

One practical example embodiment of the invention is any prior commercial embodiment of Nike Air™ gas bladder or compartment (like typical examples in FIGS. **12-16** of U.S. Pat. No. 6,846,534, which is hereby incorporated by reference) that is installed unattached, as is, located within the space enclosed partially or fully by a new, slightly larger outer compartment of one additional layer of the same or similar material, with the same or a simpler or the simplest geometric shape; that is, not necessarily following indentations or reverse curves, but rather incorporating straighter or the straightest lines, as seen in cross-section: for example, following the outermost side curvature seen in FIGS. **12-16**, but with upper and lower surfaces that are substantially flat and parallel (or curved and parallel), to facilitate ease of movement between the two surfaces of the sipe **505** formed, increasing the resulting flexibility.

The new additional, outer compartment thus thereby has created by its presence an internal sipe **505** between the two unconnected compartments. The new internal sipe **505** provides much greater flexibility to any footwear sole **22** or **28**, since it allows an inner, otherwise relatively rigid Nike Air™ compartment structure to become an inner compartment **501** (instead of typically being fixed into the other materials such

as EVA of the footwear sole) to move freely inside the new outer compartment **500**, which becomes a new compartment that is fixed to the footwear sole, rather than the conventional Nike Air™ bladder. The flexibility improvement allows the shoe sole to deform under a body weight load like a wearer's bare foot sole, so that stability is improved also, especially lateral stability.

The result is that the conventional, inner Nike Air™ compartment—now contained by a new outer compartment—can move easily within the overall footwear sole, allowing the sole to bend or flex more easily in parallel with the wearer's bare foot sole to deform to flatten under a body weight load, including during locomotion or standing, so that footwear sole stability is improved also, especially lateral stability. The extent to which the inner Nike Air™ compartment is “free-floating” within the new outer compartment can be controlled or tuned, for example, by one or more attachments (permanent or adjustable) to the outer compartment or by the media in the internal sipe.

The internal sipe **505** includes at least two surfaces that can move relative to each other to provide a flexibility increase for a footwear sole so that the shape of the footwear sole can deform under a body weight load to better parallel to the shape of the barefoot sole of a wearer under a same body weight load. The relative motion between the two internal sipe **505** surfaces increases the capability of the footwear sole to bend during locomotion under a wearer's body weight load to better parallel the shape of said wearer's bare foot sole.

Also, the sliding motion between internal support surfaces within the shoe sole **28** allowed by internal sipe **505** in response to torsional or shear forces between a wearer's foot and the ground assists in controlling and absorbing the impact of those forces, whether sudden and excessive or chronically repetitive, thereby helping to protect the wearer's joints from acute or chronic injury, especially to the ankles, knees, hips, lower back, and spine.

A benefit of the siped compartments/chambers **510** is that, as a single unitary component, it can be used in any conventional manner in constructing the footwear sole **28**, generally like that used with a conventional single layer compartment such as used in Nike Air™; i.e. the outer surface of **510** can, as a useful embodiment, adhere to the adjacent materials like plastic such as PU (polyurethane) or EVA (ethyl vinyl acetate) or other plastic or rubber of the footwear sole that contact the **510** component, just as would be the case with the outer surface of existing single compartment **161** or chamber **188** of commercial examples of Nike Air™. However, the internal sipe **505** formed by the use of an inner compartment/chamber **501** in the siped compartment/chamber **510** provides flexibility in a footwear sole **28** that is absent in the relatively rigid footwear sole **28** formed with a conventional, single layer compartment **161** or chamber **188** of the many Nike Air™ commercial examples.

The sipe surfaces can in one useful example embodiment be formed by the inner surface (or part or parts of it) of the outer compartment **500** and the outer surface (or part or parts of it) of the inner compartment **501**. Such sipe surfaces can be substantially parallel and directly contact each other in one useful embodiment example, but the two surfaces are generally not attached to each other, so that the sipe surfaces can move relative to each other to facilitate a sliding motion between the two surfaces.

The sipe surfaces can be in other useful forms that allow portions of the surfaces to be proximate to each other in an unloaded condition, rather than contacting; such surfaces

can make partial or full direct contact under a wearer's body weight load (which can vary from a fraction of a “g” to multiple “g” forces during locomotion) or remain somewhat separated; the amount of sipe surface area making direct contact can also vary with a wearer's body weight load. The sipes surfaces also may not be parallel or only partially parallel, such as the areas of direct surface contact or proximal surface contact.

To preclude the surfaces of the internal sipe **505** from directly contacting each other (whether loaded or unloaded), the sipe surfaces can include an internal sipe media **506** located between the surfaces to reduce friction by lubrication and increase relative motion and therefore flexibility. Useful example embodiments of the internal sipe media **506** include any useful material known in the art (or equivalent), such as a liquid like silicone as one example, a dry material like Teflon™ as another example, or a gas like that used in Nike Air™ as a further example. The media **506** can be located in all of the sipe **505** or only part or parts, as shown in FIGS. 1-6.

The media **506** can be used to decrease (or increase) sliding resistance between the inner surfaces of the sipe; for example, to lubricate with any suitable material known in the art. The internal sipe media **506** is an optional feature.

The attachments **503** can be simply passive (i.e. static) or actively controlled by electronic, mechanical, electromagnetic, or other useful means. The attachments **503** can, for example, be designed to break away as a failsafe feature to compensate for a predetermined extreme torsional load, for example, to reduce extreme stress on critical joints (in lieu of a wearer's cartilage, tendons, muscle, bone, or other body parts being damaged); the attachments **503** can then be reset or replaced (or, alternatively, return automatically upon relief of extreme stress to a normal position).

Example embodiments of the compartments and chambers **500/501** can include a media **504** such as a gas (like that used in Nike Air™ or ambient atmospheric air), a liquid or fluid, a gel, a foam (made of a plastic like PU or EVA, both of which are common in the footwear art, or equivalent, or of a rubber (natural or synthetic) or blown rubber or a rubber compound or equivalent or of another useful material or of a combination of two or more of the preceding foam plastic/rubber/etc.) or a useful combination of one or more gas, liquid, gel, foam, or other useful material.

FIGS. 2A, 3A, and 4A show examples of embodiments of siped compartment/chamber/bladders **510** wherein either the inner compartment/chamber/bladder **501** or the outer compartment **500** can have one or more openings, for pressure equalization, assembly facilitation, or other purposes.

FIG. 5A shows an example embodiment with an inner compartment/chamber/bladder **501**¹ having a smaller inner compartment/chamber/bladder **501**²; additional smaller inner compartments **501** are possible in a similar progression, either enclosed within the previous larger inner compartment **501** or within the same **501** or **500**.

FIG. 6A shows an example embodiment with two inner compartment/chamber/bladders **501**¹ and **501**² which are layered within outer compartment/chamber/bladder **500**; additional compartment/chamber **501** layers can be useful also.

FIG. 1B shows an example embodiment of the device **510** in a horizontal plane view of FIGS. 1A, 2A, 3A, 4A, 5A, and 6A.

FIG. 1C is an exploded perspective view showing the relationship of an insert device **510**, which has a siped compartment or chamber or bladder, with a midsole **148**, bottom sole **149**, and upper **21** of a shoe or any other

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footwear. The midsole **148** and bottom sole **149** (and any additional layers not shown) together form a shoe sole which can be any footwear sole, including any conventional sole **22** or any sole **28** according to one of the applicant's patents or applications. In the simple example shown, the device **510** can be inserted during a conventional manufacturing or assembly process or by a footwear wearer and can be located anywhere in the footwear sole and/or upper.

FIG. **1E** shows a detailed frontal plane cross section of an example conventional commercial gas bladder/compart-
ment/chamber shown as inner bladder or compartment or chamber **501** in schematic perspective in FIG. **1D**. FIG. **2F** shows the same detailed cross sectional view of the same inner bladder or compartment or chamber **501** enveloped by an outer bladder or compartment or chamber **500** forming an insertable siped compartment **510**, including wherein all or part of the sipe **505** is a slit.

FIG. **7A** shows a perspective view of a laptop personal computer **580** or any other electronic device with one of a potential multitude of potential example embodiments of the applicant's **510** or **511** or **513** inventions described above are used in two areas, but other useful embodiments with any of the previously described variations can be employed beneficially. For example, as shown, the **510/511/513** inventions can be used for cushioning selected corners of the laptop or can be integrated in the laptop frame in the same or other areas; and/or individual components of the laptop computer such as a harddrive can be protected by being mounted in a protected **510/511/513**, such as the housing of the hard-drive forming an inner compartment **501** of the **510** siped compartment invention.

FIG. **7B** shows a semiconductor microchip device **581** example including a part of a die or an entire die or multiple dies assembled together using the applicant's **510/511/513** invention; other devices **581** can include other micro-technology and nanotechnology devices, including microelectromechanical systems (MEMS), field-programmable gate arrays (FPGA's) and faraday cages, photovoltaic cells, fuel cells, batteries, and including devices assembled at the molecular or atomic scale. The FIG. **7B** example embodiment can include a single layer of die or dies or can include a stacked arrangement of two or more layers of separate dies. FIG. **7C** shows a cross section of the semiconductor microchip device **581** example of FIG. **7B**, showing a stacked arrangement of three layers of separate dies **581 #1**, **581 #2**, and **581 #3**.

FIGS. **7C** and **7D** show other examples such as electronic game controllers or any other remote controllers **580** and cell phones **580** that can beneficially incorporate any of the applicant's **510/511/513**.

FIG. **8** shows a chart displaying "Approximate Comparison of Hardness Scales". Products using the **510** or **511** or **513** inventions can be made of any material and any material hardness shown in chart, including Rockwell C and B Scales and Brinell Hardness Scale, including metallic and ceramic, generally for non-cushioning, relatively rigid structural elements into which a degree of flexibility is to be introduced using the applicant's **510/511/513** inventions, specific examples of which are shown in FIGS. **7A-7B**.

FIG. **9A** shows that, in an analogous way, especially to the thicker heel portion of a typical shoe sole, a thick urban area telephone book has in effect hundreds of mostly "internal sipes", each page being in effect separated by a horizontal internal slit from each adjacent page, each of which thereby is able to move freely relative to each other, resulting in a flexible telephone book that bends quite easily as is well known by all, while at the same time relatively stable when

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a vertical force like body weight is applied. FIG. **9B** shows, in contrast, if the same wood fiber material with the same dimensions as a thick telephone book were formed instead into a single piece of wood with no pages, like a solid particle board, it would be quite rigid with little flexibility, although it will support a body weight stably. FIG. **9C** shows if, instead, the sipes were rotated 90 degrees into vertical slits and open to the bottom, so that the spine of a much wider telephone book with shorter pages is on top (in the form of a shoe sole with deep open external sipes, those external sipes would also provide a substantial amount of flexibility like that of the human foot sole, but at the cost of excessive instability when a vertical force like body weight is applied, as is obvious from common experience, since the pages will splay out uncontrollably.

FIGS. **10-22** of this application are FIGS. **10A-10I**, **16A-16Z**, **16AA**, and **16AB**, **17A-17D**, **21B**, **22A**, **23A-23E**, **25A-25D**, **26A-26C**, **27A-27H**, **28**, **29A**, **30A-30C**, and **31** of both the applicant's U.S. application Ser. No. **10/802,049** filed May 17, 2004 and published on Oct. 28, 2004 as Pub. No. **US 2004/0215931 A1**.

As shown in FIGS. **10A-10F**, to deal with operational and security issues, it may be beneficial for individual users to have one microprocessor or equivalent device that is designated, permanently or temporarily, to be a master **30** controlling device (comprising hardware and/or software and/of firmware and/or other component) that remains inaccessible (using, for example, a hardware and/or software and/or firmware and/or other component firewall **50**) directly by the network but which controls the functions of the other slave microprocessors **40** when the network is not utilizing them.

For example, as shown in FIG. **10A**, a typical PC **1** may have four or five microprocessors (even on a single microprocessor chip), with one master **30** and three or four slaves **40**, depending on whether the master **30** is a controller exclusively (through different design of any component part), requiring four slave microprocessors **40**; or the master microprocessor **30** has the same or equivalent microprocessing capability as a slave **40** and multiprocesses in parallel with the slave microprocessors **40**, thereby requiring only three slave microprocessors **40**. The number of PC slave microprocessors **40** can be increased to virtually any other number, such as at least about eight, about 16, about 32, about 64, about 128, about 256, about 512, about 1024, and so on. These multiples are not required, and the number of PC master microprocessors **30** may be increased. Also included is an internal firewall **50** between master **30** and slave **40** microprocessors. As shown in preceding FIGS. **1-9**, the PC **1** in FIG. **10A** may be connected to a network computer **2** and to the Internet or WWW or present or future equivalent or successor **3**, like the Grid (or MetaInternet).

Other typical PC hardware components such as hard drive **61**, floppy diskette drive **62**, compact disk-read only memory (CD-ROM) **63**, digital video disk (DVD) **64**, Flash memory **65**, random access memory (RAM) **66**, video or other display **67**, graphics card **68**, and sound card **69**, as well as digital signal processor or processors, together with the software and/or firmware stored on or for them, can be located on either side of internal firewall **50**, but such devices as the display **67**, graphics card **68** and sound card **69** and those devices that both read and write and have non-volatile memory (retain data without power and generally have to be written over to erase), such as hard drive **61**, Flash memory **65**, floppy diskette drive **62**, read/write CD-ROM **63** or DVD **64** may be located on the PC user side of the internal firewall **50**, where the master microprocessor is also located, as shown in FIG. **10A**, for security reasons;

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their location can be flexible, with that capability controlled such as by password-authorized access.

Alternately, any of these devices that are duplicative (or for other exceptional needs) like a second hard drive 61', can be located on the network side of the internal firewall 50. RAM 66 or equivalent or successor memory, which typically is volatile (data is lost when power is interrupted), should generally be located on the network side of the internal firewall 50, but some can be located with the master microprocessor to facilitate its independent use.

However, read-only memory (ROM) devices including most current CD drives (CD-ROM's) 63' or DVD's (DVD-ROM) drives 64' can be safely located on the network side of the internal firewall 50, since the data on those drives cannot be altered by network users; preemptive control of use may remain with the PC user.

However, at least a portion of RAM can be kept on the Master 30 microprocessor side of the internal firewall 50, so that the PC user can retain the ability to use a core of user PC 1 processing capability entirely separate from any network processing. If this capability is not desired, then the master 30 microprocessor can be moved to the network side of the internal firewall 50 and replaced with a simpler controller on the PC 1 user side, like the master remote controller 31 discussed below and shown in FIG. 10I.

The master microprocessor 30 may also control the use of several or all other processors 60 owned or leased by the PC user, such as home entertainment digital signal processors 70, especially if the design standards of such microprocessors in the future conform to the requirements of network parallel processing as described above. In this general approach, the PC master processor uses the slave microprocessors or, if idle (or working on low priority, deferrable processing), makes them available to the network provider or others to use. Wireless connections 100, including optical wireless, are expected to be extensively used in home or business network systems, including use of a master remote controller 31 without (or with) microprocessing capability, with broad bandwidth connections such as fiber optic cable connecting directly to at least one component such as a PC 1, shown in a slave configuration, of the home or business personal network system; that connection links the home system to the network 2 such as the Internet 3, as shown in FIG. 10I. A business system may include broadband such as fiber optic or optical wireless links to most or all personal computers PC 1 and other devices with microprocessors, such as printers, copiers, scanners, fax machines, telephone and video conferencing equipment; other wired or wireless links also can be used.

A PC 1 user can remotely access his networked PC 1 by using another networked master microprocessor 30 on another PC 1 and using a password or other access control means for entry to his own PC 1 master microprocessor 30 and files, as is common now in Internet and other access. Alternately, a remote user can simply carry his own digitally stored files and his own master microprocessor or use another networked master microprocessor temporarily has his own.

In the simplest configuration, as shown in FIG. 10B, the PC 1 may have a single master microprocessor 30 and a single slave microprocessor 40, separated by an internal firewall 50, with both processors used in parallel or multitasking processing or with only the slave 40 so used, and connected with broad bandwidth such as optical fiber wire 99 to a network computer 2 and Internet 3 and successors like the Grid (or MetaInternet). Virtually any number of slave microprocessors 40 is possible. The other non-micro-

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processor components shown in FIG. 10A above may also be included in this simple FIG. 10B configuration.

As shown in FIG. 10C, microchips 90 are expected to integrate most or all of the other necessary computer components (or their present or future equivalents or successors), like a PC's volatile memory like RAM 66 (such as DRAM), graphics 82, sound 83, power management 84, network communications 85, and video processing 86, possibly including modem 87, non-volatile memory like flash (or magnetic like MRAM or ovonic unified memory) 88, system BIOS 88', digital signal processor (DSP) or processors 89, and other components or present or future equivalents or successors) and internal bus, on a single chip 90 (silicon, plastic, or other), known in the industry as "system on a chip". Such a PC microchip 90 can have the same architecture as that of the PC 1 shown above in FIG. 10A: namely, a master control and/or processing unit 93 and one or more slave processing units 94 (for parallel or multitasking processing by either the PC 1 or the Network 2), separated by an internal firewall 50 and connected by broad bandwidth wire 99 such as optical fiber cable to a network computer 3 and the Internet 3 and successors like the Grid (or MetaInternet). Alternatively, microchip 90 can be an "appliance" system on a chip.

Existing PC components with mechanical components like hard drive 61, floppy or other removable diskette 62, CD-ROM 63, and DVD 64, which are mass storage devices with mechanical features that will likely not become an integral part of a PC "system of a chip" may still be capable of connection to a single PC microchip 90 and control by a single PC master unit 93.

In the simplest multi-processor case, as shown in FIG. 10D, the chip 90 has a single master unit 93 and at least one slave unit 94 (with the master having a controlling function only or a processing function also), separated by an internal firewall 50 and connected by broad bandwidth wire 99 such as fiber optic cable to a network computer 3 and the Internet 3 (and successors like the Grid or MetaInternet). The other non-microprocessor components shown in FIG. 10A above may also be included in this simple FIG. 10D configuration.

As noted above, any computer may be both a user and provider, alternatively—a dual mode operating capability. Consequently, any PC 1 within the network 2, connected to the Internet 3 and successors like the Grid (or MetaInternet), can be temporarily a master PC 30 at one time initiating a parallel or multitasking processing request to the network 2 for execution by at least one slave PC 40, as shown in FIG. 10E. At another time the same PC 1 can become a slave PC 40 that executes a parallel or multitasking processing request by another PC 1' that has temporarily assumed the function of master 30, as shown in FIG. 10F. The simplest approach to achieving this alternation is for both master and slave versions of the parallel processing software to be loaded in each or every PC 1 that is to share in the parallel processing, so each PC 1 has the necessary software means, together with minor operational modifications, such as adding a switching means by which a signaled request for parallel processing initiated by one PC 1 user using master software is transmitted to at least a second PC 1, triggering its slave software to respond by initiating parallel processing.

As shown in FIGS. 10G and 10H, which are parallel to FIGS. 10E and 10F, the number of PC slave processors 40 can be increased to any virtually other number, such as at least about 4; as shown, the processing system is completely scalar, so that further increases can occur to, for example, about eight, about 16, about 32, about 64, about 128, about

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256, about 512, about 1024, and so on; the PC master microprocessors **30** can also be increased.

In summary, as noted above relative to FIG. **10I**, a PC **1** can function as a slave PC **40** and be controlled by a master controller **31**, which can be remote and which can have limited or no microprocessing capability, but can as well have similar or greater capability. Such a master controller **31** can be located on the PC user side of the internal firewall **50**, under the control of the PC user, while the microprocessors **40** reside on the network side of the internal firewall **50**. The master controller **31** may receive input from the PC user by local means such as keyboard, microphone, video-cam or future hardware and/or software and/or firmware or other equivalent or successor interface means (as does a master processor **40**) that provides input to a PC **1** or microprocessor **30** originating from a user's hand, voice, eye, nerve or nerves, or other body part; in addition, remote access by telephone, cable, wireless or other connection may also be enabled by a hardware and/or software and/or firmware and/or other means with suitable security such as password controlled access. Similarly, relative to a PC "system on a chip", a master controller unit (which could be capable of being accessed by the PC user through a remote controller **31**) with only a controlling capability can be located on the PC user side of the internal firewall **50**, under the control of the PC user, while the slave processor units **94** would reside on the network side of the internal firewall **50**.

FIGS. **10A-10C** show PC **1** with an internal firewall **50** that is configurable through either hardware and/or software and/or firmware and/or other means; software configuration is easiest and most typical, but active motherboard hardware configuration is possible and may present some security advantages, including a use of manual or electromechanical or other switches or locks. FIG. **10A** shows a CD-ROM **63'** that has been placed by a PC user on the network side of an internal firewall **50** from a previous position on the PC user side of an internal firewall **50**, which was shown in FIG. **10A**. The settings of an internal firewall **50** may default to those that safely protect the PC **1** from uncontrolled access by network users, but with capability for the relatively sophisticated PC user to override such default settings and yet with proper safeguards to protect the unsophisticated user from inadvertently doing so; configuration of an internal firewall **50** may also be actively controlled by a network administrator in a local network like that of a business, where a PC user may not be the owner or leaser of the PC being used, either by remote access on the network or with a remote controller **31**.

Similarly, FIGS. **10C** and **10D** show a PC "system on a chip" **90** with an internal firewall **50** that is configurable through either hardware and/or software and/or firmware and/or other means; software configuration is easiest and most typical. Active configuration of the integrated circuits of the PC microchip **90** is also possible and may present some speed and security advantages. Such direct configuration of the circuits of the microchip **90** to establish or change its internal firewall **50** could be provided by the use of field-programmable gate arrays (or FPGA's) or their future equivalents or successors; microcircuit electromechanical or other switches or locks can also be used potentially. For example, slave processing unit **94** can be moved to the PC user side of an internal firewall **50** from a network side position. The same active configuration of the chip circuit can use FPGA's for the simplest form of multiprocessing microchip **90** with a single slave unit **94**, transferring

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its position to the PC user's side of an internal firewall **50** from a network side as shown by slave processing units **94** in FIGS. **10C** and **10D**.

FIGS. **11G-11I** show the applicant's new hierarchical network structure and function applied to the design of a personal computer PC **1**, as discussed previously in FIGS. **10A** and **10B** of the '049 Application. FIG. **11G** shows the simplest general design, with a master M_1 microprocessor **30** and two slave S_{21} and S_{22} microprocessors **40**. FIG. **11H** shows the same network structure with an additional level of slave microprocessors **40**, S_{31} through S_{34} , while FIG. **11I** shows the same network structure as FIG. **11H** with an additional level of slave microprocessors **40**, S_{41} through S_{48} . As shown in these examples, this network structure is completely scalar, including any practical number of slave microprocessors **40** on any practical number of processing levels.

FIG. **11J** shows a useful embodiment in which each microprocessor **30** and **40** has, in addition to internal cache memory, its own random access memory (RAM) **66** or equivalent memory (volatile like DRAM or non-volatile like Flash memory, magnetic such as MRAM memory, or ovonic unified memory), integrated on-microchip **90** or separate off-microchip. A significant amount of such microchip RAM (volatile like DRAM or non-volatile like Flash memory, magnetic such as MRAM memory, or ovonic unified memory), significantly greater than cache memory (SRAM) and other on-chip memory used on microprocessor chips today, can be beneficial in improving the efficient operation of the microprocessor; if located off microprocessor chip, the size of such memory can substantially exceed the size of the associated microprocessor, but an on-microprocessor chip location for DRAM or Flash (or MRAM or ovonic memory), like cache (SRAM) memory, offers the best potential for improving microprocessor speed and efficiency. The design can also incorporate (or substitute) conventional shared memory or RAM **66'** (i.e. memory used by all, or some, of the microprocessors **30** or **40** (or **90**) of the personal computer PC **1**).

FIGS. **11A-11C** are parallel to FIGS. **11G-11I** above, but show PC microprocessor **90** architecture rather than macro PC **1** architecture; a PC microprocessor **90** is as earlier described in FIG. **10C**, a personal computer on a microchip.

FIG. **11D** is like FIG. **11J**, also except for showing PC microprocessor **90** architecture instead of PC **1** architecture. FIG. **11D** shows a useful embodiment in which each PC microprocessor **93** or **94** has its own integrated on-microchip (or separate off microchip) random access memory (RAM) **66** or equivalent memory (volatile like DRAM or non-volatile, like Flash memory, magnetic such as MRAM memory, or ovonic unified memory). A significant amount of such RAM or other memory, significantly greater than cache (SRAM) memory or other on-microchip memory used on microprocessor chips today, can be beneficial in improving the efficient operation of the microprocessor; if located off-microprocessor chip, the size of such memory can substantially exceed the size of the associated microprocessor, but an on-microprocessor chip **90** location for DRAM or Flash (or MRAM or ovonic memory), like cache (SRAM) memory, offers the best potential for improving microprocessor speed and efficiency. The microchip design can also incorporate (or substitute) conventional shared memory or RAM **66'** (i.e. memory used by all, or some, of the PC microprocessors **93** or **94** of the personal computer PC microprocessor **90**).

FIGS. **11A-11D** show a different and improved basic microchip architecture which can exclude or reduce the

currently used superscalar approach in microprocessors to execute multiple instructions during each clock cycle. The FIGS. 11A-11D architecture is much simpler and, by integrating memory with microprocessor, reduces memory bottlenecks. The simplicity of the FIGS. 11A-11D microchip design, which may have little or no superscalar components, compared to conventional superscalar designs (the inherent extreme complexity of which creates a very substantial memory overhead), can result in the use of a much greater proportion of independent, non-superscalar processors per microchip, exclusive of integrating memory or RAM 66 onto the microprocessor chip 90, as discussed in FIG. 11D.

FIGS. 11G-11J, by using the same architecture for PC 1 networks as FIGS. 11A-11D, import the same advantage of microchip parallel processing performance to parallel processing in PC 1 networks.

FIG. 11K shows a direct connection of optical fiber 99 from Internet 3 (or another network) to random access memory (RAM) microchip 66'. The connection may be at a central portion 140 of RAM chip 66' to provide equal access to stored data on RAM chip 66'. The direct connection can be anywhere on RAM chip 66'. Digital signal processor (DSP) 89 is on RAM chip 66' for connection with optical fiber 99. RAM chip 66' is for shared memory use among PC's 1 and for broadcast use. RAM chip 66' can include volatile or non-volatile (flash-type) memory. RAM chip 66' can have more than one DSP 89, such as shown in FIG. 20B.

All FIGS. 11A-11K, like the preceding figures of this application, show sections of a network of personal computers PC 1 (or PC microprocessors 90) or microprocessors 30 or 40 which can be parts of the WWW or Internet or Internet II or the Next Generation Internet (meaning connected to it) or Intranets or Extranets or other networks.

Also, except for FIGS. 11A-11C and 11G-11I, all of the FIG. 11 series show personal computers PC 1 and microprocessors 30 or 40 as occupying the same location. This dual representation was done for economy of presentation and to show the parallel functionality and interchangeability in conceptual terms of personal computer PC 1 and microprocessors 30 or 40 in the structure of the new network. So, taking FIG. 16A as an example, M_1 , S_{21} and S_{22} show three personal computers PC 1 or, alternatively, one microprocessor 30 and two microprocessors 40.

As noted initially in FIG. 10C, a personal computer PC 1 can be reduced in size to a PC microprocessor chip 90, so preceding Figures showing personal computer PC 1 also generally represent PC microprocessor chip 90.

Finally, FIGS. 11A-11K show a mix of electrical and optical connections, including wired 99, especially connections such as optical glass fiber or omniguides, and wireless 100, especially wireless optical (and mixtures of both in a single figure), and dense wave division multiplexing (DWDM). Generally, either 99 or 100 or a mix can be used relatively interchangeably in the network inventions shown (as well as in prior figures), though in some embodiments either highest transmission speed (i.e. broadest bandwidth) or mobility (or some other factor) may dictate a use of wired or wireless. Generally, fiber optic wire 99 and dense wave division multiplexing (DWDM) may provide the most advantageous transmission means because it has the greatest bandwidth or data transmission speed, so it may be used for connections between personal computers and microchips, including direct connections, although optical wireless 100 also offers very high bandwidth, especially with dense wave division multiplexing (DWDM). Other wireless 100 (but also including optical wireless), including with DWDM, can be used where mobility is a paramount design criteria.

The FIG. 11 embodiments can be combined with, or modified by incorporating, any other network system architectures (including client/server or peer to peer) or any other topologies (including ring, bus, and star) either well known now in the art or their future equivalents or successors.

Any of the embodiments shown in FIGS. 11A-11K can be combined with any one or more of the preceding or subsequent figures of this application to provide a useful improvement over the art.

The parallel processing network architecture shown in the preceding FIGS. 11A-11K and in earlier figures has several features unique to its basic design that provide for the security of personal computers PC 1 (or PC microprocessor 90) or microprocessor 40 that share other computers for parallel and multi-tasking processing. First, the slave personal computers PC 1 (or microprocessors 40) have only part of the operation (for large operations, only a very small part) and therefore unauthorized surveillance of a single PC 1 can provide only very limited knowledge of the entire operation, especially in only a relatively local area in which switching or routing was employed. Second, the addresses of the slave personal computers PC 1 (or microprocessors 40) are known or traceable, and therefore are not protected by anonymity (like hackers usually are) in case of unauthorized intervention. In addition, cryptography can be employed, with on microprocessor chip 30, 40, or 90 hardware 55 being used in some embodiments due to efficiency, although software and firmware can also be used, or a separate PC 1 hardware-based component 56 like an encryption microchip can be used; with either encryption component 55 or 56, micro electromechanical locks can be used to prevent access other than by the direct physical user; other MicroElectroMechanical System (MEMS) devices located on microchips like PC90 can be used for access prevention or other functions. Nonetheless, these inherent strengths can be substantially reinforced, as indicated in FIGS. 12B-12D.

FIG. 12A shows at least one internal firewall 50 performing its conventional function of keeping out intruders such as hackers from the Internet 3 from unauthorized access for either surveillance of, or intervention in, a user's personal computer PC 1 (or PC microprocessor 90) or master microprocessor 30.

FIG. 12B shows that, since Internet users can, as enabled by the applicant's network structure invention, use one or more of the slave microprocessors 40 of another's personal computer PC 1 (or PC microprocessor 90) for parallel (or multi-tasking) processing, the at least one internal firewall 50 has a dual function in also protecting Internet 3 use (or other shared use on a network) from unauthorized surveillance or intervention by a PC 1 owner/user who is providing the shared resources. To maintain the privacy necessary to operate such a cooperatively shared network arrangement, unauthorized surveillance or intervention must be carefully prevented by hardware/software/firmware or other means.

FIG. 12C therefore shows master M personal computer PC 1 (or PC microprocessor 90) using the slave S_2 microprocessor 40 of a different personal computer, PC 1', which is available for Internet 3 (or other net) shared use, while internal firewall 50' blocks unauthorized access into PC 1' by PC 1 (although PC 1' owner/user can always interrupt a shared operation and take back control and use of slave S' microprocessor 40, which then triggers off-loading action to compensate, as discussed above in FIGS. 16I-16J of the '049 Application).

FIG. 12D is similar to FIG. 12C, but shows a PC microprocessor 90 with a slave microprocessor 94 being used by Internet 3 users (or other net), so that at least one

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firewall 50 serves both to deny access such as surveillance by master M microprocessor 93 to an Internet 3 parallel processing (or multi-tasking) operation on slave S microprocessor 94 and to deny access to master M microprocessor 93 by Internet 3 (or other net) users of slave S microprocessor 94. At least one internal firewall 50 may be implemented by non-configurable hardware at the microchip level to provide protection against tampering with the internal firewall 50 by a PC 1 user, who has easier access to software or macro hardware such as PC motherboards to alter. PC 90 microchips may employ tamper-resistant construction or tamper-proof construction. As such, PC 90 microchips may be permanently locked by out-of-specification conditions or permanently destroyed by attempts at physical access.

Also, non-configurable hardware denying access from the network is the most immune to hacking from any outside source, including the Internet, and can therefore be used either for general protection or to protect an innermost kernel of the most confidential of personal files (such as passwords or financial data) and the most critical of operating system components, such as the system bios or access to file alternation.

At the same time, the FIG. 12 and earlier embodiments provide a solution to digital rights management by providing a highly safe environment for the owners of digital versions of audio, video, and software copyrighted material. Such copyrighted material as movies, television, music, and application or operating system software may be decrypted and controlled on the network user side of the PC 1 or PC 90, while the PC 1 user is denied access to the decrypted digital version of the copyrighted material. However, the network user can make the material viewable to the PC 1 user, but not copyable, via the PC 1 and PC 90 microchip architecture shown in FIGS. 10A and 10C.

Any of the embodiments shown in FIGS. 12A and 12B can be combined with one or more of any of the preceding figures of this application to provide a useful improvement over the art.

FIG. 13 is like FIG. 20B of the '049 Application (and therefore also can be combined with FIGS. 11J and 11D, respectively), but show additionally that all microprocessors 30, 40, 93, and 94 of PC 1 or PC 90₁ can have a separate input/output communication link to a digital signal processor (DSP) or other transmission/reception connection component. The additional communications linkages are shown as 141, 142, 143, and 144, which connect to M₁, S₁, S₂₁, and S₂₂, respectively, and connect to the network, including the Internet 3, the WWW, the Grid, and equivalents or successors. Like all preceding and subsequent figures, FIG. 13 is a schematic architectural plan of the new and unique components of the parallel processing system invention disclosed in this application and can represent either physical connections or virtual relationships independent of hardware. FIG. 13 shows an embodiment in which the additional linkages lead through the Internet 3 to microprocessors PC 90₂₅-90₂₈.

The additional communications linkages 141, 142, 143, and 144, as well as the original linkages 111, 112, 113, and 114 of FIGS. 20A and 20B, may have a bandwidth sufficiently broad to at least avoid constraining the processing speed of microprocessors 30, 40, 93, and 94 connected to the linkages. The ultra high bandwidth of optical connections like optical fiber or omniguides or optical wireless may provide external connections between PC 1 and PC 90₁ microprocessors that are far greater than the internal electrical connections or buses of those microprocessors, for example, by a factor of 10, or 100, or 1000, which are

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already possible with optical fiber, or 1,000,000, which is possible with optical omniguides, which are not limited to a relatively smaller band of wavelengths using DWDM like optical fiber; future increases will be substantial since the well established rate of increase for optical bandwidth is much greater than that for microprocessor speed and electrical connections. Wireless optical antennas that are positioned on the exterior of houses, buildings, or mobile reception sites, instead of inside of glass or other windows, should significantly increase the number of optical wavelengths that can be sent or received by each of the wireless optical antennas; the entire connection is freespace optical wireless, which allows for greater dense wave division multiplexing (DWDM) and thereby greater bandwidth.

A major benefit of the embodiment shown in FIG. 13 is that PC 1 and PC 90₁ can function like the FIG. 9 embodiment to efficiently perform operations that are uncoupled, so that each microprocessor M₁, S₁-S₃₄ can operate independently without microprocessors M₁, S₁, and S₂₁-S₂₂ being idled, as they may be in FIG. 13. Another benefit is that for tightly coupled parallel operations, microprocessors M₁, S₁, and S₂₁-S₂₂ can have broad bandwidth connections with microprocessors 30, 40, 93, or 94 that are not located on PC 1 or PC 90₁. Thus the embodiments shown in FIG. 13 provide an architecture that allows PC 1 or PC 90₁ the flexibility to function in parallel operations either like FIG. 13 embodiments or like the FIG. 9 embodiment of the '049 Application, depending on the type of parallel operation being performed. Studies indicate that single chip multiprocessors like PC 90₁ can also perform uniprocessor operations with a speed like that of uniprocessor architectures like wide-issue superscalar or simultaneous multithreading.

Like FIG. 20B of the '049 application, the embodiment of FIG. 13 includes broad bandwidth connection to the Internet 3 by wired means such as optical connection by fiber optic cable or omniguide or optical wireless, although other wired or non-wired means can be used with benefit, and the use of DWDM or wideband CDMA is clearly advantageous. It should be noted that the architecture of the FIGS. 20 and 21 embodiments may be particularly advantageous with ultra-wideband communication connections.

Another advantage of the embodiments shown in FIGS. 14A and 14B when functioning in the FIG. 9 (of the '049 Application) form of loosely coupled or uncoupled parallel processing or multitasking is that if PC 1 or PC 90₁ is functioning as a web server and typically uses only one microprocessor to do so, it can quickly add minor web sites using one or more additional microprocessors to meet increasing volume of visits or other use of the web site. This replication of web sites on additional microprocessors in response to increasing load can also be done using the FIG. 11 form of tightly coupled parallel processing. PC 1 and PC 90₁ or any of their microprocessors 30, 40, 93, and 94 or other components can also serve as a switch or a router, including other associated hardware/software/firmware network components.

Any of the embodiments shown in FIG. 13 can be combined with one or more of any of the preceding figures of this application to provide a useful improvement over the art.

Binary tree configurations of microprocessors shown in FIGS. 11, and 13 can be laid out in 2D using an H-tree configuration, as shown in FIG. 21C of the '049 Application, and can be combined with one or more of any of the preceding figures of this application to provide a useful improvement over the art.

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FIG. 14A shows a microprocessor PC 90₁ like that of FIG. 13, except that FIG. 14A shows the microprocessors 93 and 94 each connecting to an optical wired interconnection 99' such as thin mirrored hollow wire or omniguide or optical fiber (and other very broad bandwidth connections can be used); the interconnect can include a digital signal processor 89' employed with a microlaser 150, which can be tunable, and other components to transmit and receive digital data for microprocessors 93 and 94 into the optical wired interconnects 99' such as an omniguide using, for example, a specific wavelength of light for each separate channel of each separate microprocessor 93 and 94 utilizing dense wave division multiplexing (DWDM).

FIG. 14B shows an enlargement of the digital signal processor 89' with microlaser 150 with other transmission and reception components.

FIG. 14A shows a simple bus network connection architecture between the interconnect 99' and the microprocessors 93 and 94. However, since the interconnection 99' is optical and the bandwidth available is very broad, the optical connection 99' allows connections between microprocessors 93 and 94 in PC 90₁ that are functionally equivalent to those shown in FIG. 13 of the '049 Application, which includes a representation of physical connections. The interconnects between microprocessors 93 and 94 like FIG. 13 are shown within the omniguide 99' shown in FIG. 14A. In fact, the potential bandwidth of the optical interconnect 99' is so great that complete interconnection between all microprocessors 93 and 94 with PC 90₁ is possible, even for a much greater number of microprocessors either in a larger PC 90₁, like FIG. 11C for example, or in other PC 90s, such as PC 90₂₁-90₂₄ and 90₃₁-90₃₁₆ in FIG. 13 connected to PC 90₁ through a network such as the Internet 3, the WWW, or the Grid; consequently, any conventional network structure can be implemented. Consequently, the embodiment shown in FIG. 14A has the flexibility of those of FIG. 13 to function in parallel operations like either the FIGS. 20A-20B embodiments or like the FIG. 9 embodiment (both of the '049 Application), depending on the type of parallel operation to be performed, or the FIG. 11 embodiments.

It should be noted that the optical interconnect 99' shown in FIG. 14A can beneficially have a shape other than a thin wire or tube, such as an omniguide with any form or shape located above and connection to microlasers 150 at a suitable location such as on or near the upper surface of the microchip PC 90₁ located at least at each microprocessor 93 and 94 or connected thereto, for example; the optical interconnect 99' and microlasers 150 and associated transmission and reception components can be located elsewhere on the microchip PC 90₁ with benefit. An omniguide can take a waveform shape or rely exclusively on a mirrored (or semi-mirrored) surface or surfaces (or combination of both shape and mirrored surface) to guide lightwave signals such as propagated by a microlaser 150 substantially directly and/or by reflection. A relatively large optical interconnect 99' can enable freespace or wireless-like connections between microlasers 150; such an optical interconnect 99' can cover substantially the entire PC90 microchip or can connect multiple PC90 microchips and can connect one or more PC90 microchips to other PC components.

As shown in FIG. 14A, random access memory (RAM) 66 can be located on microchip PC 90₁ like in FIG. 11D and also can be connected directly or indirectly to the optical interconnect 99' (or use non-optical connections not shown), so that the microprocessors 93 and 94 and RAM 66 can communicate with a very broad bandwidth connection, including with RAM 66 and microprocessors 93 and 94

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located off microchip PC 90₁ on the network including the Internet 3 and WWW. Any other components of the PC 90 microchip can be connected with the optical interconnect 99' and more than one such interconnect 99' can be used on the same PC 90 or other microchip. Microlasers 150 can include, for example, 5-to-20-micron-high (or other height) vertical cavity-surface-emitting lasers (VCSELs), which can beam down waveguides built into the PC90 microchip; alternatively, freespace optics can be employed; and lenses can be employed. Radio-frequency (RF) signals can also be used for similar interconnects 99'. Micro light emitting diodes (LEDs) can substitute for one or some or all of the microlasers 150 and either can be a transceiver (transmit and receive light signals).

FIG. 14C is a side cross section of the microchip PC 90₁ shown in FIG. 14A taken at hatched line 22C (which is abbreviated). FIG. 14C shows the location of the omniguide above the surface of the microprocessors 93 and 94 and RAM 66 and connecting them while also containing two or more microlasers 150 (associated DSP and other components not shown) proximate to each to contain the optical signal generated by the microlasers 150 so that the signal can be transmitted between microprocessors 93 and 94 and RAM 66 either directly or by being reflected off the mirrored (or semi-mirrored) surface of the omniguide 99', for example. Each of the microprocessors 93 and 94 (or 30 or 40) and RAM 66 (or any other memory component such as L1 cache or L2 cache, for example, or other microchip component) can have one or more microlasers 150 and each such microlaser 150 can distinguish itself from other microlasers 150 on the microchip (or off it) that also generate wavelength signals by using, for example, a distinct wavelength of light for data transmission and/or utilizing wave or dense wave division multiplexing. FIG. 14A is a top view of the microchip PC 90₁, which is a PC system on a microchip, any of which disclosed in this application can be also more generally any microchip with multiple processors. The microlasers 150 (and associated transmission and reception components such as DSP) that are associated with RAM (or parts of it) or other memory components can either provide data in response to direct inquiries or fetches made by a microprocessor 93 or 94 or can broadcast a continual stream of current data (continually updated and repeated in continuous cycle, for example) which is used by the microprocessor as needed.

Any of the embodiments shown in FIGS. 14A, 14B and 14C can be combined with one or more of any of the preceding figures of this application to provide a useful improvement over the art.

FIG. 15A shows multiple firewalls 50, a concept indicated earlier by the at least one firewall 50 discussed in FIG. 12D. FIG. 15A shows a PC1 or microchip 90 with a primary firewall 50 and additional interior firewalls 50¹, 50², and 50³, that are within primary firewall 50. As shown, interior firewall 50³ is in the most protected position, since it is inside all the other firewalls, while the other interior firewalls 50², and 50¹ are progressively less protected, since, for example, interior firewall 50¹ is protected from the outside network only by the primary firewall 50. As shown, progressively more protected positions can be created within the PC1 or microchip 90. The interior firewalls can also be arranged in any other way within the primary firewall 50. The interior firewalls can be used to separate user files from system files, for example, or to separate various hardware components from each other. In this manner, a number of compartments can be created within the PC1 or microchip 90 to more safely protect the software, hardware, and

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firmware of the PC1 or microchip 90, just as ships have a number of separate watertight compartments to protect against flooding and avoid sinking. Any of the primary or interior (or other inner firewalls discussed below) can be hardware, software, or firmware, or a combination, and can coexist in layers, so that a firewall 50, for example, may have a hardware firewall, a software firewall, and a firmware firewall, either as independent units or as integrated components. W³ in FIG. 15A and subsequent Figures denotes the World Wide Web.

FIG. 15B shows another embodiment of compartments created by inner firewalls within a PC1 or microchip 90. Primary firewall 50 and interior firewall 50¹ are like FIG. 15A, but interior firewalls 50², 50³, and 50⁴ are shown perpendicular to firewalls 50 and 50¹ (just to illustrate in a simplified schematic way, which may be different in an actual embodiment). In this way, an upper row of compartments U¹ and U² can be used, for example, to bring from the network files which are first authenticated and then enter into the U¹ compartment, are decrypted, and undergo a security evaluation, such as by virus scan, before transfer to the most secure compartment U². Any operations could potentially occur in any compartment, depending on the level of security desired by the user (by over-ride) for example, but an advantageous default system would allow for files with the highest levels of authentication, encryption, and other security evaluations to be allowed into the most secure compartments.

Similarly, operating system files can also be authenticated and brought from the network side of the PC1 or microchip 90 into compartment O¹ for decryption and security evaluation or other use, and then finally transferred into the most secure compartment O². Again, similarly, a row of compartments can be used for separating hardware, such as a master microprocessor 30 or 93 being located in compartment M¹ and a remote controller 31, for example, located in compartment M².

Also, additional inner firewalls 50²², 50³³, and 50⁴⁴ can be located outside the primary firewall 50, but within the network portion of the PC1 or microchip 90, to separate user files in compartment U from operating system files in compartment O from hardware such as a slave microprocessor in compartment S on the network side. In the example shown, an additional row is shown for hardware, including a hard drive in a compartment HD on the network side, a hard drive in compartment HD¹ on the PC1 or microchip 90 user's side, and flash memory (such as system bios 88) in compartment F². Each microprocessor 30, 40, 93, or 94 can have its own compartment in a manner like that shown in FIG. 15B, as can associated memory or any other hardware component.

FIG. 15C shows an inner firewall 50 embodiment similar to FIG. 15B, but FIG. 15C shows that any file or set of files, such as operating files O or user data files U or application files A, can have its own inner firewall 50^O or 50^U or 50^A. Similarly, any hardware component, such as hard drive HD, also can have its own inner firewall 50^{HD}. Additionally, more than one file or set of files or hardware components can be grouped together within an inner firewall, such as 50^S shown in FIG. 15C.

FIGS. 15D and 15E show operating system files O or application files A like those shown in FIG. 15C, but organized differently in discrete layers, each separate grouping of the operating or application files having a separate firewall 50 (and optionally with as well as a PC1 or PC90 firewall shown in earlier Figures), so that the firewall structure is like that of an onion. The operating system files

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O or application files A can have a parallel structure, with an innermost kernel operating system or application file located in the center, with additional features in other files in subsequent layers, from the simplest to the most complex and from the most secure and trusted to the least secure and trusted.

Using this structure, as shown in FIG. 15D, an innermost operating system core O¹ may be firmware stored in a read-only memory (ROM), located in a microchip for quick access, so that a simplest version operating system with all core features can be protected absolutely from alteration and can be available almost immediately, without lengthy boot up procedures required by loading the operating system from a hard drive, for example. The core operating system O¹ can include a core of the system BIOS or of the operating system kernel, for example; it would be advantageous for this core to be capable of independent operation, not dependent on components in other levels to operate at the basic core level (similarly, other levels can advantageously be independent of higher levels).

A secondary operating system O² can be software located advantageously on flash or other microchip non-volatile memory such as magnetic (or less advantageously, a hard drive or other mechanical storage media) and can consist of additional features that are more optional, such as those not always used in every session, or features that require updating, changing, or improving, such features coming from trusted sources located on a network, such as the Internet or the Web; additional portions of or upgrades to the system BIOS and the operating system kernel can be located in O², for example.

A third level operating system O³ located, for example, on a hard drive, can consist of additional software features that are used only occasionally and are more optional, and can be loaded as needed by a user into DRAM or magnetic memory microchip for execution, for example. Operating systems O² and O³ can include, for example, the most recent upgrades from a known and trusted source, such as a commercial software vendor or open source software developer, that are downloaded from a network, including the Internet and the Web, or loaded from conventional memory media like CD or floppy diskette. All three levels of such operating systems O¹, O², and O³ together can constitute, for example, roughly the equivalent of a conventional PC operating system typical in the year 2000.

A fourth level operating system O⁴, for example, can consist of special use or single use operating system additions, especially software coming from untrusted or unauthenticated sources on a network, such as the Internet or the Web.

For example, the graphical interface of the operating system can be in 2D only at the O¹ level, in 3D at the O² level, rendering at the O³ level, and animation in the O⁴ level; additionally, a standard format can be maintained in the O¹ and O² levels, with user or vendor customization at the O³ level.

As shown in FIG. 15E, application files such as A¹, A², A³, and A⁴ can be structured the same way as operating system files O in FIG. 15D and with the same layered approach to firewalls 50 as in FIG. 15D. Typical application software of the year 2000 can be restructured in this manner.

The kernel operating system files O¹ and O², as well as kernel application files A¹ and A² can be located in any personal computer PC1 or PC90, including at the level of an appliance including the simplest device, advantageously in ROM and in non-volatile read/write memory such as Flash (or magnetic such as MRAM, or ovonic memory) micro-

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chips, for example, as described in FIGS. 15D and 15E above. Inclusion of wireless connection capability is advantageous, as is the use of DWDM.

An advantage of the file and firewall structures shown in FIGS. 15D and 15E is that a system crash or file corruption should never occur at the simple and unalterable level O¹ or A¹ and any level above O¹ or A¹ can be recovered at a lower level, specifically the highest level at which there is a stable system or uncorrupted data. For example, a word processing application program can have the most basic functions of a typewriter (i.e. storing alphanumeric, punctuation, spacing, and paragraph structure data) stored on a ROM microchip in A¹ and related user files (i.e. such as a word document) on U². Insertion of a digital video file into a word document can be handled at the A³ level and insertion of a downloaded special effect at the A⁴ level. In this example, a crash caused by the insertion at the least secure and most complex A⁴ level would not disturb the word document located at the U² or U³ level. Rebooting and/or recovery can be automatic when detected by the operating system or at the option of the user.

Thus, FIGS. 15A-15E illustrate embodiments wherein a PC1 or microchip 90 includes a hierarchy of firewalls. In the context of the present invention, firewalls may be structured to allow varying degrees of access from the network side of PC1 or microchip 90. As discussed above, ROM may totally deny access from the network side, effectively creating an innermost firewall. Hardware, software, firmware, or combinations thereof may be structured to deny or allow a predetermined maximum level of access from the network side, effectively creating outer firewalls. Similarly, intermediate firewalls effectively may be created.

The embodiments of FIGS. 15A-15E, as well as earlier FIGS. 12A-12D and earlier embodiments, provide a solution to digital rights management by providing a highly safe environment for the owners of digital versions of audio, video, and software copyrighted material. Such copyrighted material as movies, television, music, and application or operating system software may be decrypted and controlled on the network user side of the PC 1 or PC 90, while the PC 1 user is denied access to the decrypted digital version of the copyrighted material. However, the network user can make the material viewable to the PC 1 user, but not copyable, via the PC 1 and PC 90 microchip architecture shown in FIGS. 10A and 10C. For example, a copyrighted movie or music album may be a file that is associated with control and other software; all files located on one or more specific hardware components may be grouped together within an inner firewall, such as 50^S shown in FIG. 15C.

Additional security for copyright owners may be provided by using a digital signal processor (DSP), and/or analog and/or other components grouped within the inner firewall 50^S to convert network user selected decrypted digital files into analog files before they are transmitted off the PC 90 microchip, so that only an analog signal exits the PC 90 microchip for viewing or listening by the PC 1 user. As such, direct digital copying by the PC 1 user of copyrighted digital files provided over the Internet is prevented.

Any of the embodiments shown in FIGS. 15A-15E can be combined with one or more of any of the preceding figures of this application to provide a useful improvement over the art.

Additionally, an inner firewall can divide any hardware component into a separate network side compartment and a separate firewall protected side compartment. For example, a hard drive 61 can have a controller 61' that is divided into two compartments, HD and HD¹, as above. As shown in

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FIG. 24 of the '049 Application, the user side HD¹ compartment of the controller 61' can have a read capability controller r and a write capability controller w, while the network side HD compartment can be limited to a read capability controller r only. The user side HD¹ compartment controller can be, for example, used to control only the upper surface of the hard drive 61 platters, while the network side HD compartment controller can be used to control only the lower surface of the hard drive 61 platters, so that a single hard drive can effectively serve a dual role as both a network-accessible hard drive and a user-accessible hard drive, while maintaining a firewall 50 between them. Additionally, the network side HD controller can optionally have a write capability also, which can be preemptively turned on or off by the PC 1 or microchip 90 user. Other relative allocations between network and user of the HD 61 platters can be made and can be configurable by the user or system administrator or not configurable.

Similarly, CD drives 63 or DVD drives 64 (read only or read/write) can have a controller 63' or 64' like that of the HD controller 61' above that is divided by a firewall 50, so that some laser beams are under network control and other laser beams are under user control, like the above hard drives. Floppy disk drives, "Zip" drives, and other removable disk or diskette drives can similarly be divided by a firewall 50 so that there is a physical user portion of the disk or diskette and a physical network portion of the disk or diskette, both either fixed or configurable by a user or system administrator or other authorized source. Memory microchips such as RAM or Flash or other can also be divided into network and user sides in a similar manner.

The use of volatile memory on the network side of the PC1 or microchip 90 is particularly useful in eliminating viruses and other security problems originating from the network side, such as malicious hackers on the Internet. When the network side of the firewall 50 of the PC1 or microchip 90 is returned to its user (preemptively or otherwise), volatile memory like random access memory (RAM) such as DRAM on the network side can first be erased. For example, volatile memory can be purged by momentarily interrupting power to the network side of the PC1 or microchip 90, thereby erasing all network data so that no network data is retained when the user regains control of the network side of the PC1 or microchip 90 for the user's use, except at the user's option; other conventional means may be employed. Of course, when the user is specifically using the network side, for example, for Web browsing, the operating system or the user can selectively save network side files or transfer them to the user side.

On the network side, non-volatile memory like Flash, MRAM, and ovonic memory with network data must be overwritten to obtain the same erasure-type protection, which can be a disadvantage if it takes much more time. Moreover, for relatively large storage media, such as CD-RW or DVD-RW with write-once capability, network data writing must be tracked to be effectively erased. Any new network file on non-volatile memory with only a write-once capability can be erased by overwriting all "0's" to "1's", so that, for example, the network data written on a CD-RW or DVD-RW would be converted to all "1's" or "pits" (no unpitted writing surface within the network data sector, permanently overwriting the file); optionally, the operating system or the user can selectively save network side files or transfer them to the user side, or vice versa. There is a disadvantage to using Flash memory, since repeated overwriting will eventually degrade it.

FIGS. 16A-16D show the use for security of power interruption or data overwrite of volatile memory like DRAM and non-volatile memory like Flash or MRAM (or ovonics), respectively, of the network portion (N) of a personal computer PC1 or system on a microchip PC90; the network (N) portion being created within a PC1 or PC90 by a firewall 50 (as described above in previous figures) and including resources that, when idled by a user, can be used by the network, including the Internet (I) or the World Wide Web. Such use is to prevent the unplanned or approved mixture of user and network files by either files being retained in the "swing space" (N) during the transition from use by a network user to use by the PC1/PC90 user or vice versa.

As shown in FIG. 16A and FIG. 16C, when the network portion (N) of the PC1 personal computer or PC90 microchip is idled by a user, for example, power is interrupted to volatile memory like DRAM and/or data is overwritten to files in non-volatile memory like Flash or MRAM (or ovonics), so that no files exist in the network portion (N) after such interruption or overwriting.

After the step shown in FIGS. 16A and 16C, the network portion (N) can be used safely from a security viewpoint by a user from the network, including the Internet and the World Wide Web (and potentially including other network resources), as shown in FIG. 16B, or by the PC1/PC90 user, as shown in FIG. 16D, potentially including other resources from the user portion (U) of the PC1 or PC90. As noted earlier, the FIG. 16 approach can advantageously be used as an additional feature to other conventional security measures.

Any of the embodiments shown in FIGS. 16A-16D can be combined with one or more of any of the preceding figures of this application to provide a useful improvement over the art.

The PC 90 microchip as previously described, or a personal computer PC 1 (or any microchip, including a special or general purpose microprocessor on a microchip, alone or including one or more other system components as previously described) may include one or more photovoltaic cells 201, as are well known in the art. The photovoltaic cells 201 may be located on the PC 90 microchip or located near the PC 90 microchip, such as adjoining it or adjacent to it, or located less near, such as in the PC 90 microchip user's home, office, or vehicle, either inside or outside, or may be located more remotely.

FIG. 17A shows one or more photovoltaic cells 201 located on a PC 90 microchip. The photovoltaic cells 201 may use electromagnetic radiation, such as visible light, as a power source that is directed to the cells 201 by an optical waveguide 202, which may include a size that is sufficient to allow the cells 201 to generate electrical power at maximum output level or at a most efficient level. In addition, visible light in freespace (without a waveguide 202) may also serve as a power source and can be directed by the use of one or more lenses 204.

FIG. 17B shows a single microchip 200 including both a PC 90 and one or more photovoltaic cells 201. FIG. 17B shows a top view of a multi-layer microchip having one or more photovoltaic cells 201 on one side of a microchip 200, with a PC 90 on the other side of the microchip 200, as shown in FIG. 17C in a bottom view of the same microchip as FIG. 17B. Besides being integrated on the same microchip 200, the photovoltaic cells 201 may be located separately from the PC 90 microchip, and the two separate elements may be joined or adjoining.

A light source for the photovoltaic cells 201 can be direct or indirect and can be sunlight or artificial light, including light from a laser, or a combination, and can be optionally focused by a lens 204. The light may be coherent with one or more discrete frequencies, such as from a laser, or incoherent with many frequencies. The artificial light may be generated by well known conventional means that are conventionally powered by electricity distributed by the existing electrical power grid, as is well known in the art.

A single photovoltaic cell or a number of cells 201 may power each component on the PC 90 microchip, such as the master microprocessor 93 or slave microprocessors 94, DRAM or MRAM, Flash memory, DSP, or laser 150, or any of the other components previously described. The photovoltaic cells 201 may be connected to one or more batteries. The photovoltaic cells 201 can be located remotely as a separate unit, such as on the PC 90 microchip user's roof at home, car, or office, so that the cells 201 provide general local power or power dedicated to the PC 90 microchip and/or associated components. The PC 90 microchip may be a network server, router, or switch, so that any network component can be powered by photovoltaic cells 201, including the Internet, an Intranet, or the World Wide Web.

The FIG. 17A-17C embodiments advantageously eliminate the need for a microchip, such as the PC 90 microchip, to have a wired connection 99 that typically provides power or data or both, but which also provides a connection means for the entry of electromagnetic flux, which can impair or destroy the functioning of the PC 90 microchip. The embodiments shown rely on light, which does not transmit electromagnetic flux, for power and data.

FIG. 18A shows a single microchip 200, combining a PC 90 microchip (or any microchip, including a special or general purpose microprocessor on a microchip, alone or including one or more other system components as previously described) and one or more photovoltaic cells 201, that is substantially surrounded by a Faraday Cage 300, such as is well known in the art, that is optimized to shield against magnetic flux, including high frequency flux (and may include shielding against electric flux). Faraday Cage 300 may be constructed of a mesh structure, or may also be a continuous structure without holes, which has an advantage of preventing entry by very high frequency electromagnetic flux, and may incorporate other microchip structures, such as a heat sink 301.

FIG. 18B shows separate PC 90 microchip and one or more photovoltaic cells 201; the two separate components are connected by a wire 99, and all three components are substantially surrounded by a Faraday Cage 300, also known as a Faraday Shield or Screen.

FIG. 18C shows the same components as FIG. 18B, but shows each component substantially surrounded by a separate Faraday Cage 300, all of which may be connected. For portable handheld wireless devices, the ground for the Faraday Cage 300 may be the user's body.

As shown in FIG. 18D, the PC 90 microchip may be located in a housing for any of the PC's described previously, such as a case of a laptop personal computer 401 or a PC cell phone 402, which may also have a separate Faraday Cage 300, so that the PC 90 microchip is substantially surrounded by more than one Faraday Cage 300. The inner Faraday Cage 300 surrounding the PC 90 microchip may be optimized to shield against specific frequencies of magnetic flux, such as high frequency flux in the microwave range, which may be assisted by the relatively smaller size of the PC 90 microchip (compared to its housing). FIG. 18D

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shows an inner Faraday Cage **300** surrounding only a portion, the PC **90**, of a microchip such as the combined microchip **200**.

As shown in FIGS. **18E** and **18F**, the PC **90** microchip can be separate from the photovoltaic cell or cells **201** and can be joined by a wired connection **99**.

As shown in FIG. **18E**, an inner Faraday Cage **300** may surround only a portion of a PC **90** microchip, such as a Magnetic Random Access Memory (MRAM) component.

FIG. **18F** shows Faraday Cage **300** that surrounds only a portion of one or more photovoltaic cells **201**, such as a part conducting an electrical current flow directly to the PC **90** microchip.

The PC **90** microchip may also be powered by one or more fuel cells **211** or one or more batteries (each with one or more cells) **221** or any combination of such batteries **221**, fuel cells **211**, or photovoltaic cells **201**. As shown in FIGS. **18E** and **18F**, the PC **90** microchip is typically separate from a fuel cell or cells **211** or batteries **221** and can be joined by a wired connection **99**, as shown, as is the case with a photovoltaic cell or cells **201**. A wired connection **99** can be configured to protect the PC **90** microchip from electromagnetic flux through the use of RF traps or Ferrite grommets or beads **212** on the wire or cable connection **99**.

By providing power without an external wired connection **99**, both fuel cells **211** and batteries **221** isolate the PC **90** microchip from a power grid that can transmit electromagnetic flux, but to do so a battery or batteries **221** can be configured to provide connection to the power grid only intermittently when charging is required.

FIG. **18G** shows a microchip, such as a PC **90** microchip, surrounded by a Faraday Cage **300** but without including a photovoltaic cell **201** shown in FIGS. **18A-18F**.

FIG. **18H** shows a PC housing such as a laptop PC **401** or PC cell phone **402** including a PC **90** microchip and separate Faraday Cages **300** surrounding both the microchip and housing. Also shown is an antenna **499** (or antennas) for wireless communication that can be separated from the Faraday Cage **300** to protect the electrical components of the PC by an RF trap or Ferrite grommets or beads **212**. The antenna **499** can project externally from the PC housing or be located internally in the PC housing, such as in the screen housing of a laptop PC **401**. In an exemplary implementation, the antenna **499** is located outside of at least one Faraday Cage **300**.

Any of the embodiments shown in FIGS. **17A-17C** and **18A-18H** may be combined with one or more other embodiments shown in those figures or in preceding FIGS. **1-16** and described herein.

FIG. **19** shows a silicon wafer **500** used to make microchips. The largest wafers **500** in current use are 300 mm (12 inches) in diameter and can contain as many as 13 billion transistors. Current state of the art in microchip fabrication is 0.13 micron process technology and the next process will be measured in nanometers (90 nm). As shown in FIG. **19**, microchips **501** are separated by an edge portion **502**. A microchip **501** can be a PC **90** microchip.

FIG. **20A** shows a top view of a microchip **501** surrounded by adjoining portions of adjoining microchips **501** in a section of the silicon wafer **500**. The microchip **501** is bounded by edge portions **502**. Although the current state of the art in microchip fabrication on a silicon wafer is to use only one process on a wafer, embodiments of the invention use two or more fabrication processes on a single wafer **500**.

As shown in the example of FIG. **20A**, one process can be located on one section **511** on the microchip **501**, while a second process can be located on a second section **521** of the

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microchip **501**. A third process can be located on a third section **531** of the microchip **501**; additional processes can also be located on other sections of the microchip **501**.

The processes can be completely separate while at least sharing the common silicon wafer **500** base, and the processes can occur at different fabrication facilities, including those owned by different manufacturers. Alternatively, two or more separate processes may have common sub-processes that can be integrated, i.e., performed at the same time. Sections of the microchip **501** that are not undergoing a process can be protected from that process by a protective coating that is unaffected by that process and removed after that process. There can be one or more temporary protective coatings, which can remain on for more than one process.

The separate sections of the separate fabrication processes of the microchip **501** can be in any shape or pattern of the microchip. As shown in the FIG. **20A** example, one or more separate processes can be located on adjoining portions of adjoining microchips. For example, as shown in FIG. **20A**, section **521** is located on the lower portion of one row of microchips **501** and on the upper portion of the adjoining row of microchips **501**, which would be positioned upside down of the wafer **500**, so that the contiguous area of the section **521** process is maximized. Similarly, section **531** is shown in the example located on the lower portion of the adjoining row of microchips **501** and on the upper portion of the middle row of microchips **501**. Alternatively, all of the microchips **501** of the wafer **500** can be positioned upright on the wafer.

Embodiments of the invention include any fabrication process of a silicon wafer **500** and can include wafers made of other materials suitable for microelectronic devices, such as gallium arsenide. The fabrication processes in current widespread use are generally CMOS (complementary metal-oxide semiconductor), but can be bipolar or other. The separate processes (and separate sections **511**, **521**, and **531** shown in FIG. **29A**) can be for general purpose microprocessor (including one or more cores), memory (DRAM or non-volatile such as Flash or MRAM or ovonic), analog (including radio and/or laser), digital signal processing (DSP), micro-electromechanical system (MEMS), field programmable gate arrays (FPGA), graphic processing unit (GPU), microprocessor chipset, and others.

Embodiments of the invention facilitate a "system on a chip" (SoC), such as the earlier described PC **90** microchip, by allowing most or all of the micro components of a PC to be located on a single microchip. Even the consolidation of only two microchips into a single microchip provides a significant increase in processing speed and reduced power consumption. The silicon die becomes the motherboard for all the micro components of the PC, leaving only the macro components like battery, power supply, and input/output (I/O) connections to be located on the printed circuit motherboard. The result is ultra-large-scale-integration.

FIG. **20B** shows a top view of the microchip **501** embodiment of FIG. **20A** after the die has been separated from the silicon wafer **500** and positioned in a microchip package **503**.

The fabrication processes illustrated in FIGS. **20A-20B** can include material such as silicon germanium, gallium arsenide, indium phosphide and others used, for example, as deposits on silicon. Besides using different materials in different sections of the microchip, different size processes can be used in different microchip sections, such as a 0.13 micron process on section **511** and a 0.18 micron process on section **521** in the FIG. **29A** example. All or parts of the microchip **501** can be synchronous or asynchronous. Both

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different size and different material processes can be combined on different sections of the microchip **501**.

Although the maximum increase in speed and decrease in power consumption can be achieved by putting all micro or nano components on a single "system on a chip," such as for a PC, even a minimal combination of just two different micro or nano components of a single microchip **501** can yield a very significant increase in speed and decrease in power consumption. To take a very simple example, a silicon wafer **500** can have 256 MB of DRAM manufactured onto a section **531** of the microchips **502** located on the wafer by one factory; when that DRAM process is completed, a second factory can add a general purpose CPU like a Pentium 4 to a second section **511** of the microchips **501** on the silicon wafer **500**. Such an approach allows direct communication between microprocessor and DRAM on the microchip **501** for much greater speed and reduced power. Since 256 MB DRAM is an inexpensive commodity product currently, especially if purchased as wafers **500**, there would be little or no increase in the production time of the microprocessor.

FIG. **21A** illustrates an embodiment of the invention, which is an alternative method of uniting separate fabrication processes on the same microchip **501**. Sections **501**¹, **501**², and **501**³ of FIG. **21A** correspond to sections **511**, **521**, and **531** of FIGS. **20A** & **20B** in that both sets of sections represent three separate processes, but in FIG. **21A** each section is a separate die cut from a wafer **500** and all three sections are united in a single package **503**. The section dies **501**¹, **501**², and **501**³ can be held together by the chip package **503** or can be glued together, or a combination of the two in parts or the whole. In addition, the section dies can be assembled into a chip package **503** or the package can be assembled around the dies or a combination of both partially or completely.

The separate process dies illustrated in the FIG. **21A** example may be assembled with the surface that the process is on in each die being substantially level with each other, so that both process surfaces of the dies form a plane that is substantially flat. The edges of the dies are configured so adjoining dies fit together as closely as possible, as shown in FIG. **21A** at **502**² and **502**³.

The circuits of dies **501**¹, **501**², and **501**³ are connected at their edges **502**³ and **502**² by interconnect lines **580** that can be widened as shown in **581** of FIG. **21B**, which shows a portion of die edge **502**³ and **502**² in an enlarged view. A process can be added in the area **591** overlapping the edges of the dies at **502**³ and **502**² bounded by lines **590**; in that process interconnect lines **580** of the two separate dies can be connected by laying down connections at **582** that connect to the enlarged portions **581** of the interconnect lines **580**, as illustrated in FIG. **21B**.

FIG. **21C** shows that the die edges **502**³ and **502**² can have any shape or pattern, not just a straight line shown above in FIGS. **21A** & **21B**.

FIG. **22** shows a combination of the embodiments shown in FIGS. **20** and **21**. Microchip **501** as shown in FIG. **20** is shown assembled with separate dies **501**¹, **501**², and **501**³ into a microchip package **503**, with edges between dies at **502**¹, **502**², **502**³, and **502**²³, which could include a connection process such as the example shown in FIG. **21B**.

The microchip **501** dies shown in FIGS. **20-22** can be packaged using FCPGA (flip-chip pin grid array), FCBGA (flip-chip ball grid array), BBUL (bumpless build-up layer) or other technology.

Any of the embodiments shown in FIGS. **19**, **20A-20B**, **21A-21C**, and **22** can be combined with one or more other

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embodiments shown in those figures or in the preceding FIGS. **1-18** and described herein.

FIGS. **23A-23H** are new inventions based on FIGS. **18A-27H** of this application, which are FIGS. **27A-27H** of U.S. application Ser. No. 10/802,049 filed Mar. 17, 2004 and published on Oct. 28, 2004, as Pub. No. US 2004/0215931 A1 and U.S. application Ser. No. 10/684,657 filed Oct. 15, 2003 and published on Aug. 18, 2005, as Pub. No. US 2005/0180095 A1, both of which applications are hereby incorporated by reference herein for completeness of disclosure.

FIGS. **23A-23H** are example new embodiments based on the FIG. **7B** example; they show examples of the applicant's inventions involving one or more Faraday Cages surrounding various combinations of semiconductor microchips, photovoltaic cells, and/or other micro and/or nano devices with the applicant's internal sipe inventions **510/511/513**. In the exemplary embodiments shown, the Faraday Cages **300** coincide with the outer compartment **500**, which forms a surface of the sipe **505** (as previously defined) which can partially or completely surround the PC **90** microchip and/or photovoltaic cell **201** and/or fuel cell **211** and/or battery **221**; any one or more of which (**90/201/211/221**) can be connected by wire **99**, which can also be protected by Faraday Cage **300** so that all components are protected; and one or more Faraday Cages **300** and or internal sipes **505** can be inside a Faraday Cage **300** and/or internal sipe **505** so as to provide additional protection, including of one or more individual components, as shown in several useful examples in the FIGS. **23A-23H**.

Heat sink **301** or other microchip cooling device can be augmented or replaced by a cooling media **506**, which can for example circulate by convection alone within the sipe **505** as shown in FIG. **23G** for example or can be allowed to circulate away from the microchip or other micro or nano device to a heat sink in another location, for example, including by circulation powered by a pump, for example, or other powered means. One or more attachments **503** can provide external connection from the microchip/cell **200** or other component like fuel cell **211** or battery **221** to the outer compartment **500** and/or Faraday Cage **300** and then to other external connections, such as a wire **99** as shown in FIG. **23C** or an antenna **499** as shown in FIG. **23H**; as shown in the FIG. **23D** example, one or more attachments **503**, which can be one or more conventional pins, can connect a microchip PC **90** and/or other component with an internal Faraday Cage **300**. Advantageous combinations of the FIGS. **23A-23H** embodiments can be made with embodiments shown in FIGS. **24A-24C**, **25A-25B** and **26A-26B**, as well as with previous FIGS. **1-22**.

FIGS. **24A** and **25A-25B** are based on FIGS. **28** and **29A-29B** of U.S. application Ser. No. 10/684,657 filed Oct. 15, 2003 and published on Aug. 18, 2005, as Pub. No. US 2005/0180095 A1, which was above incorporated by reference herein for completeness of disclosure.

FIG. **24A** is a top view of a semiconductor wafer **1500**, of which 300 mm. is a current example using a 90 nanometer process, made of silicon, gallium arsenide, or any other suitable semiconductor materials in current use or future equivalents. The wafer **1500** contains a multitude of microchips **1501** with each microchip **1501** including, for example, 1 or more core microprocessors **93** or **94**, including at least 2 or 4 or 8 or 16 or 32 or 64 or 128 or 256 or 512 or 1028 or more cores. But instead of separating the microchips **1501** into separate dies in the conventional process along lines **1502**, the entire semiconductor wafer **1500** of any size (including smaller than 300 mm) is used essentially

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intact (that is, undiced into dies) as a computer (or in one example embodiment, the wafer **1500** can be virtually entire and intact, excluding deactivated or disconnected incomplete microchips **1501** located around the periphery of a circular wafer and microchips **1501** with manufacturing defects or other damage, such as failure during use, which can remain inactively on the wafer); instead of separating along lines **1502**, interconnects **1505** of any length or configuration can be added to the printed circuit architecture to connect the microchips **1501** to other microchips **1501** and/or other components on the wafer **1500** or external to it, using any interconnection means or architecture known in the art.

The microchips **1501** on the wafer **1500** can for example each be a complete personal computer (PC **90**) system on a chip (SoC), including for example microprocessors, random access memory, radio and/or optical communication components, and other operational components necessary for each microchip **1501** to be capable of functioning as a fully independent PC **90** unit on the semiconductor wafer **1500**. Each PC **90** microchip **1501** on the wafer **1500** is in effect a stand-alone unit capable of operating independent of any other PC **90** microchip **1501** on the wafer; this means that operations of PC **90** microchips **1501** occurring on the wafer **1500** can occur asynchronously, such as some or all of the PC **90** microchips **1501** on a wafer can operate together as a cluster of personal computers, as is conventional in the art, and can also include other personal computers in the cluster that are not on the wafer **1500**. The semiconductor wafer **1500** can be used with other conventional interface devices for power and data communication, including wireless such as radio and/or optic (See FIG. **14**, for example), and/or wired such as fiber optic and/or electric; such connections can reduce or eliminate the need for interconnections between the PC **90** microchips **1501** or pins connecting a wafer **1500**, for example, to a motherboard (not shown). One or more PC **90** microchips **1501** can also include these connections, and all can include wireless communication capability such a radio or optic device, so that even cluster operations between PC microchips **1501** on a wafer **1500** can occur completely on wafer **1500**, obviating the need for pins and interconnects.

Alternatively, more than one semiconductor wafer **1500** can be stacked in vertical layers, for example, with wafer **1500 #1** including microprocessors or cores; wafer **1500 #2** including random access memory or RAM such as DRAM; and wafer **1500 #3** including other components, as shown in the FIG. **24B** example, which is similar to FIG. **7B** of this application. The wafers stacked can be entire wafers as shown, or portions of wafers (such as FIG. **24C** below), or one or more entire wafers with one or more portions of wafers or dies or other components.

By reducing or eliminating the need to go “off microchip” or off wafer in this case to complete computation operations, the applicant’s invention provides huge savings in terms of speed and energy efficiency.

FIG. **24A** also shows an example of the semiconductor wafer **1500** including one or more of the applicant’s **510** or **511** or **513** internal sipe invention, with an outer compartment **500** with a sipe **505**. In addition, FIG. **24A** shows an example of the semiconductor wafer **1500** including one (or potentially more) Faraday Cage **300** surrounding the semiconductor wafer **1500**. Moreover, FIG. **24A** shows an example embodiment wherein both one or more of the internal siped inventions **510/511/513** and one or more of the Faraday Cages **300** can be used with the same semiconductor wafer **1500**. Finally, FIG. **24A** shows an example

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wherein one or more of the outer compartment **500** of the internal siped invention coincide with one or more of the Faraday Cages **300**. Each of the four example embodiments described above in this paragraph can be used independently from each other or in any combination, including all four together as shown in FIG. **24A**.

FIG. **24B** shows a side cross section of vertically stacked semiconductor wafers **1500**, which have interconnects or vias **1504** between facing proximate surfaces of the wafers. FIG. **24B** shows a three wafer example embodiment; two or more wafers can be stacked in any manner, including with each directly aligned on top of one another, as shown, or partially overlapping in any manner. Two or more of the wafers **1500** can contact directly as shown or a non-wafer layer can be between them.

As with the example embodiments shown in FIG. **24A**, FIG. **24B** shows those stacked wafers **1500** including one or more of the applicant’s **510** or **511** or **513** internal sipe invention, with an outer compartment **500** with a sipe **505**. In addition, FIG. **24A** shows an example of the stacked semiconductor wafers **1500** including one or more Faraday Cage **300** surrounding the semiconductor wafer **1500**. Moreover, FIG. **24A** shows an example embodiment wherein both one or more of the internal siped inventions **510/511/513** and one or more of the Faraday Cages **300** can be used with the same stacked semiconductor wafers **1500**. Finally, FIG. **24A** shows an example wherein one or more of the outer compartment **500** of the internal siped invention coincide with one or more of the Faraday Cages **300** surrounding the stacked wafers **1500**. Each of the four example embodiments described above in this paragraph can be used independently from each other or in any combination, including all four together as shown in FIG. **24B**.

FIG. **24C** is another top view of a circular wafer **1500**, but subdivided into smaller components, such as half sections at the upper example and quarter sections in the lower sections. Other smaller component embodiments, including other non-rectangular embodiments can be used beneficially. Advantageous combinations of the FIGS. **24A-24C** embodiments with embodiments shown in FIGS. **23A-23H** can also be made, as well as with FIGS. **25A-25B** and **26A-26B**, as well as with previous FIGS. **1-22**.

The invention examples shown in FIGS. **24A-24B** can be used as stand-alone computers or as networked computers or as components of a computer. As noted above, sipe media **506** can also be a coolant, which can circulate to a location outside the outer compartment **500** (not shown) or be enclosed within the **300/500** structure (shown in FIG. **24B** partially filling sipe **505** as one example and completely filling sipe **505** as another example (not shown).

FIGS. **25A-25B** are modifications of FIGS. **20A** and **20B** of this application (which are FIGS. **29A-29B** of the ‘657 application incorporated by reference herein above). FIG. **25A** shows a top view of PC **90** microchips **1501** on a portion of the semiconductor wafer **1500**, including interconnects **1505** between the microchips **1501**, and the microchips **1501** being bounded by edge portions **1502**. FIG. **25B** shows a microchip **1501** as a separated die in a package **1503** including the applicant’s internal sipe inventions **510/511/513** and the Faraday Cage **300**. Advantageous combinations of the FIGS. **25A-25B** embodiments can be made with embodiments shown in FIGS. **23A-23H**, **24A-24C**, and **26A-26B**, as well as with previous FIGS. **1-22**.

FIGS. **26A-26B** are like FIG. **7A-7B** with the addition of Faraday Cages, as well as the addition of stacked dies in a side view in FIG. **26B**. Advantageous combinations of the FIGS. **26A-26B** embodiments can be made with embodi-

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ments shown in FIGS. 23A-23H, 24A-24C, and 25A-25B, as well as with previous FIGS. 1-22.

By way of background, FIG. 28A of the '033 and '930 Applications incorporated by reference herein above shows in cross-section an example of a tire 535, such as for a wheel 533 of a transportation vehicle, with a device 510; the internal sipe 505 and/or inner compartment/chamber/bladder 501 can be pressurized or not (valve not shown). As shown in the example, inner compartment/chamber/bladder 501 can have one or more direct attachments 503 to the wheel and the structural elements shown can be made of any useful material as is conventional in the art, including plastic and/or plastic composite and/or carbon fiber. The outer compartment/chamber/bladder 500 can be abbreviated to cover only part of inner compartment/chamber/bladder 501, as shown in FIG. 28A, (possibly pressure-sealed to the wheel like a conventional automobile tire and wheel); the outer compartment/chamber/bladder 500 can also be abbreviated further to cover only a lesser portion, including at least a tread portion, which can include rubber (natural or synthetic, as can other or all parts of the outer compartment 500. FIG. 28B of the '033 and '930 applications shows in a side view cross-section an example of shape of structural elements 502 of the inner compartment 501 (not shown for simplicity).

FIG. 27A is new in this application and shows another tire 535 example embodiment similar to that of FIG. 28A discussed in the previous paragraph, but maximizing lateral stability by locating the structural elements 502 at the sides of the tire 535, while maximizing soft ride by locating the media such as gas in the central portion of the tire 535. New FIG. 27B is a side view of FIG. 27A and is the same as FIG. 28B discussed above.

FIGS. 23A-27 can be combined in any manner with each other and with any or all of FIGS. 1-22 of this applications, as well as with the Figures of the applicant's patents and applications incorporated by reference herein this application.

Broadly, the flexible inserts or components 510, 511, and 513 can be usefully employed anywhere that cushioning already is being used, or could be with beneficial effect, such as protective padding or cases for equipment of any sort, including portable devices like PC laptops or video players and/or games, cell phones, personal digital assistants (PDA's), and personal digital music players like Apple iPods™ and MP3 players, as examples, such as the mounting of delicate electronic (or other) components like hard-drives or for vibration dampening, such as in automobile structural and body components and connections.

The applicant's two earlier applications, U.S. application Ser. No. 11/190,087 published as Publication No. US 2005/0268487 A1 on Dec. 8, 2005 describing footwear and U.S. application Ser. No. 11/108,034 published as Publication No. US 2005/0217142 A1 on Oct. 6, 2005 describing orthotics, as well as U.S. Pat. No. 7,010,869, issued Mar. 14, 2006 (of which the '034 Application is a continuation), are hereby expressly incorporated by reference in its entirety for completeness of disclosure. The applicant's earlier application Ser. No. 11/179,887 published as Publication No. US 2005/0241183 A1 on Nov. 3, 2005 describing footwear is hereby expressly incorporated by reference in its entirety for completeness of disclosure.

The applicant's other footwear U.S. Pat. Nos. 4,989,349; 5,317,819; 5,544,429; 5,909,948; 6,115,941; 6,115,945; 6,163,982; 6,308,439; 6,314,662; 6,295,744; 6,360,453; 6,487,795; 6,584,706; 6,591,519; 6,609,312; 6,629,376; 6,662,470; 6,675,498; 6,675,499; 6,708,424; 6,729,046; 6,748,674; 6,763,616; 6,789,331; 6,810,606; 6,877,254;

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6,918,197; 7,010,869; 7,082,697; 7,093,379; 7,127,834; 7,168,185; 7,174,658; 7,234,249; 7,287,341; 7,334,350; and 7,334,356 are all hereby incorporated by reference herein in their entirety into this application for completeness of disclosure of the applicant's novel and useful combination of one or more of any of the features or components of any of the figures of this application with one or more of any of the features of any one or more of the preceding applicant's patents listed above in this paragraph.

The applicant's other footwear U.S. Applications with Publication Numbers US 20020000051; 20020007571; 20020007572; 20020014020; 20020014021; 20020023373; 20020073578; 20020116841; 20030046830; 20030070320; 20030079375; 20030131497; 20030208926; 20030217482; 20040134096; 20040250447; 20050016020; 20050086837; 20050217143; 20060032086; 20060248749; 20070240332; 20070271817; 20080000108; 20080005931; 20080022556; 20080083140; and 20080086916 are hereby incorporated by reference herein in their entirety into this application for completeness of disclosure of the applicant's novel and useful combination of one or more of any of the features or components of any of the figures of this application with one or more of any of the features of any one or more of the preceding applicant's published U.S. Applications listed above in this paragraph.

The applicant's non-footwear patents on global network computers, U.S. Pat. Nos. 6,167,428; 6,732,141; 6,725,250; 7,024,449; 7,035,906, and 7,047,275 are all hereby incorporated by reference herein in their entirety into this application for completeness of disclosure of the applicant's novel and useful combination of one or more of any of the features or components of any of the figures of this application with one or more of any of the features of any one or more of the preceding applicant's patents listed above in this paragraph.

The applicant's non-footwear applications on global network computers are U.S. application Ser. Nos. 09/085,755; 09/884,041; 09/935,779; 10/663,911; 11/196,527; 11/329,423; and 11/338,887, as well as U.S. application Ser. No. 10/802,049 with U.S. Publication Number US 2004/0215931 published Oct. 28, 2004; U.S. application Ser. No. 10/684,657 with U.S. Publication Number US 2005/0180095 published Aug. 18, 2005; U.S. application Ser. No. 11/196,527 filed Aug. 4, 2005 with U.S. Publication Number US 2006/0095497 published May 4, 2006; U.S. application Ser. No. 11/329,423 filed Jan. 11, 2006 with U.S. Publication Number US 2006/0177226 published Aug. 10, 2006; U.S. application Ser. No. 11/338,887 filed Jan. 25, 2006 with U.S. Publication Number US 2006/0190565 published Aug. 24, 2006; all of these applications are hereby incorporated by reference herein in their entirety into this application for completeness of disclosure of the applicant's novel and useful combination of one or more of any of the features or components of any of the figures of this application with one or more of any of the features of any one or more of the preceding applicant's applications, including published applications, listed above in this paragraph.

To avoid confusion, any numeral in the above listed non-footwear patents or applications relating to computers that is the same as a numeral in the above listed footwear patents or applications should have (or be presumed to have) a prefix numeral of "1" added to the numeral (the equivalent of adding "1000" to the numeral, so that all the numerals of both sets of applications and patents remain unique).

In the following claims, the term "chamber" means a compartment 161 or a chamber 188 or a bladder and the term

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“sipe” means a sipe 505 or a slit or a channel or a groove as described in the textual specification above and associated figures of this application.

The foregoing shoe designs meet the objectives of this invention as stated above. However, it will clearly be understood by those skilled in the art that the foregoing description has been made in terms of the preferred embodiments and various changes and modifications may be made without departing from the scope of the present invention which is to be defined by the appended claims.

What is claimed is:

1. A microchip comprising:

at least one outer chamber, compartment, or bladder;
at least one inner chamber, compartment, or bladder
inside said outer chamber, compartment, or bladder;
at least one internal sipe separating at least a part of said
outer chamber, compartment, or bladder and at least a
part of said inner chamber, compartment, or bladder;
at least one Faraday cage; and

said microchip being configured to connect to at least one
network of computers, said microchip comprising at
least a first internal hardware firewall configured to
deny access to at least a first protected portion of said
microchip from said network; and

wherein said at least one internal sipe is formed by at
least a portion of an inner surface of said outer
chamber, compartment, or bladder and at least a
portion of an outer surface of said inner chamber,
compartment, or bladder; and

the inner and outer surface portions forming said at
least one internal sipe oppose each other and can
move relative to each other in a sliding motion; and
wherein at least a portion of an outer surface of said
outer chamber, compartment, or bladder is proximate
to an outer surface of said microchip.

2. The microchip according to claim 1, wherein said at
least one outer chamber, compartment, or bladder includes at
least one said Faraday Cage.

3. The microchip according to claim 1, wherein said
microchip comprises at least one general purpose micropro-
cessor including at least two general purpose core or general
purpose processing units, and wherein

at least a first said core or processing unit is located within
said first protected portion of said microchip that is
protected by at least said first internal hardware fire-
wall;

at least a second said core or processing unit is located
within a second portion of said microchip that is not
protected by at least said first internal hardware fire-
wall; and

at least said second core or processing unit is separated
from said first core or processing unit by at least said
first internal hardware firewall and is located between
at least said first internal hardware firewall and said at
least one network of computers.

4. The microchip according to claim 3, wherein said
microchip further comprises a master controller of the
microprocessor located within said first protected portion of
said microchip that is protected by at least said first internal
hardware firewall, and said master controller is configured to
control said second said core or processing unit.

5. The microchip according to claim 1, wherein said
microchip further comprises a wire cable configured for
electronic devices, said wire cable comprising:

at least one wire;

at least one inner layer surrounding a portion of said at
least one wire;

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at least one outer layer surrounding a portion of said at
least one inner layer; and

at least one internal sipe of said wire cable separating at
least a part of one said outer layer and at least a part of
one said inner layer;

wherein said at least one internal sipe of said wire cable
is formed by at least a portion of an inner surface of
said outer layer and at least a portion of an outer
surface of said inner layer; and

the inner layer and outer layer surface portions of said
wire cable forming said at least one internal sipe of
said wire cable oppose each other and can move
relative to each other in a sliding motion.

6. A microchip comprising:

at least one outer chamber, compartment, or bladder;
at least one inner chamber, compartment, or bladder
inside said outer chamber, compartment, or bladder;
at least one internal sipe separating at least a part of said
outer chamber, compartment, or bladder and at least a
part of said inner chamber, compartment, or bladder;

said microchip being configured to connect to at least one
network of computers, said microchip comprising at
least four internal hardware firewalls, said at least four
internal hardware firewalls configured to deny access to
at least four protected portions of said microchip from
said network of computers;

at least one Faraday cage; and

wherein said at least one internal sipe is formed by at
least a portion of an inner surface of said outer
chamber, compartment, or bladder and at least a
portion of an outer surface of said inner chamber,
compartment, or bladder; and

the inner and outer surface portions forming said at least
one internal sipe oppose each other and can move
relative to each other in a sliding motion; and

wherein at least a portion of an outer surface of said
outer chamber, compartment, or bladder is proximate
to an outer surface of said microchip.

7. The microchip according to claim 6, wherein said at
least one outer chamber, compartment, or bladder includes at
least one said Faraday Cage.

8. The microchip according to claim 6, wherein said
microchip comprises at least one general purpose micropro-
cessor including at least two general purpose core or general
purpose processing units, and wherein

at least a first said core or processing unit is located within
said first protected portion of said microchip that is
protected by at least said first internal hardware fire-
wall;

at least a second said core or processing unit is located
within a second portion of said microchip that is not
protected by at least said first internal hardware fire-
wall; and

at least said second core or processing unit is separated
from said first core or processing unit by at least said
first internal hardware firewall and is located between
at least said first internal hardware firewall and said at
least one network of computers.

9. The microchip according to claim 8, wherein said
microchip further comprises a master controller of the
microprocessor located within said first protected portion of
said microchip that is protected by at least said first internal
hardware firewall, and said master controller is configured to
control said second said core or processing unit.

10. The microchip according to claim 6, wherein said
microchip further comprises a wire cable configured for
electronic devices, said wire cable comprising:

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at least one wire;
 at least one inner layer surrounding a portion of said at
 least one wire;
 at least one outer layer surrounding a portion of said at
 least one inner layer; and
 at least one internal sipe of said wire cable separating at
 least a part of one said outer layer and at least a part of
 one said inner layer;
 wherein said at least one internal sipe of said wire cable
 is formed by at least a portion of an inner surface of
 said outer layer and at least a portion of an outer
 surface of said inner layer; and
 the inner layer and outer layer surface portions of said
 wire cable forming said at least one internal sipe of
 said wire cable oppose each other and can move
 relative to each other in a sliding motion.

11. The microchip according to claim 6, wherein said
 microchip comprises at least one general purpose micropro-
 cessor with at least four general purpose cores or general
 purpose processing units, each protected by at least one of
 said at least four internal hardware firewalls.

12. A microchip comprising:
 at least one outer chamber, compartment, or bladder;
 at least one inner chamber, compartment, or bladder
 inside said outer chamber, compartment, or bladder;
 at least one internal sipe separating at least a part of said
 outer chamber, compartment, or bladder and at least a
 part of said inner chamber, compartment, or bladder;
 at least one Faraday cage; and
 said microchip being configured to connect to at least one
 network of computers, said microchip comprising at
 least a first internal hardware firewalls configured to
 deny access to at least a first protected portion of said
 microchip from said network; and
 the microchip further including at least one general
 purpose microprocessor with at least a general pur-
 pose core or processing unit including a master
 controller of the microprocessor and at least one
 general purpose core or general purpose processing
 unit; and
 wherein said at least one internal sipe is formed by at
 least a portion of an inner surface of said outer
 chamber, compartment, or bladder and at least a
 portion of an outer surface of said inner chamber,
 compartment, or bladder; and
 the inner and outer surface portions forming said at
 least one internal sipe oppose each other and can
 move relative to each other in a sliding motion; and
 wherein at least a portion of an outer surface of said
 outer chamber, compartment, or bladder is proximate
 to an outer surface of said microchip.

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13. The microchip according to claim 12, wherein said at
 least one outer chamber, compartment, or bladder includes at
 least one said Faraday Cage.

14. The microchip according to claim 12, wherein said at
 least one general purpose microprocessor includes at least
 two general purpose cores or general purpose processing
 units, and wherein

at least a first said core or processing unit is located within
 said first protected portion of said microchip that is
 protected by at least said first internal hardware fire-
 wall;

at least a second said core or processing unit is located
 within a second portion of said microchip that is not
 protected by at least said first internal hardware fire-
 wall; and

at least said second core or processing unit is separated
 from said first core or processing unit by at least said
 first internal hardware firewall and is located between
 at least said first internal hardware firewall and said at
 least one network of computers.

15. The microchip according to claim 14, wherein said
 master controller of the microprocessor is located within
 said first protected portion of said microchip that is protected
 by at least said first internal hardware firewall, and said
 master controller is configured to control said second said
 core or processing unit.

16. The microchip according to claim 12, wherein the
 master controller of the microprocessor is also protected by
 a second said internal hardware firewall.

17. The microchip according to claim 12, wherein said
 microchip further comprises a wire cable configured for
 electronic devices, said wire cable comprising:

at least one wire;
 at least one inner layer surrounding a portion of said at
 least one wire;

at least one outer layer surrounding a portion of said at
 least one inner layer; and

at least one internal sipe of said wire cable separating at
 least a part of one said outer layer and at least a part of
 one said inner layer;

wherein said at least one internal sipe of said wire cable
 is formed by at least a portion of an inner surface of
 said outer layer and at least a portion of an outer
 surface of said inner layer; and

the inner layer and outer layer surface portions of said
 wire cable forming said at least one internal sipe of
 said wire cable oppose each other and can move
 relative to each other in a sliding motion.

* * * * *